

PACKING $k^2 - c$ UNIT SQUARES: AN UPPER BOUND OF ORDER $k^{3/8}$ FOR THE DEFICIENCY

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ABSTRACT. For $n \geq 1$ let $s(n)$ be the side of the smallest square that contains n unit squares with pairwise disjoint interiors, and for an integer $k \geq 2$ let $c^*(k)$ be the largest integer c with $s(k^2 - c) = k$. We prove a tiered bound $c^*(k) < C_i k^{3/8} + a_i$ for every integer $k \geq k_i$, for example $c^*(k) < 48.38 k^{3/8} + 0.017$ ($k \geq 6.1 \cdot 10^{14}$) and $c^*(k) < 40.62 k^{3/8} + 1.4 \cdot 10^{-5}$ ($k \geq 3.4 \cdot 10^{26}$), and, with two further lemmas on the wall filler, $c^*(k) < 45.08 k^{3/8} + 0.026$ for every integer $k \geq 6.4 \cdot 10^{13}$; with much better constants for moderate k we prove $c^*(k) < 20.668 k^{2/5} + 0.623$ for every integer $k \geq 1.6 \cdot 10^7$. The tiered bound is smaller than $20.668 k^{2/5} + 0.623$ for every $k \geq 6.1 \cdot 10^{14}$, with two further lemmas for every $k \geq 5.07 \cdot 10^{13}$ (tier B1s), and with a sawtooth accounting of the wall rows and a choice of the lifts (tiers ‡, for example $c^*(k) < 41.19 k^{3/8} + 0.04$ for $k \geq 5.22 \cdot 10^{12}$) for every $k \geq 2.48 \cdot 10^{12}$ (tier C3). The frame is an L-shaped arrangement in the spirit of Erdős and Graham and of Chung and Graham, in which two bands of width slightly less than an integer b are filled with rows of b unit squares tilted by slightly more than $2 \arctan(1/b)$; this frame alone gives $c^*(k) \leq 8 \lceil \sqrt{k-4} \rceil - 1$ for $k \geq 6$, which we also prove. The rows are lifted, and the four ends of the bands are filled by a “staircase”: columns of squares with one common small tilt, whose lowest squares are replaced by horizontal layers so that the columns follow the rising end of the band. This gives an uncovered area $O(b^{2/3})$ per end and the exponent $2/5$. Filling the two walls of each end by chains of staircases whose tilt is lowered from block to block gives $O(b^{3/5})$ per end and the exponent $3/8$. These ideas are due to Bui; our proofs are self-contained. The constants of the $3/8$ bounds are certified by interval arithmetic; their range of validity cannot be tested by explicit computation, and the explicit checks test the construction at smaller sizes. Version 1.1 replaces the single bound $43.06 k^{3/8} + 2 \cdot 10^{-5}$ ($k \geq 4 \cdot 10^{26}$) of version 1.0 by the tiers. Explicit packings, checked in exact rational arithmetic, also by checkers written independently of the generating program, give for example $c^*(10^5) \leq 1583$ and $c^*(10^8) \leq 24790$. Together with the lower bound of order $k^{1/3}$ of [T] (a preprint, not refereed), $k^{1/3} \ll c^*(k) \ll k^{3/8}$; the true exponent is open. The paper has not been refereed.

1. INTRODUCTION AND RESULTS

A *packing* of n unit squares in a closed square K is a family of n closed unit squares, in any orientation, contained in K and with pairwise disjoint interiors. For $n \geq 1$ let $s(n)$ be the infimum of the sides of the squares in which n unit squares can be packed. Removing squares from a packing leaves a packing, so s is non-decreasing. By area $s(k^2) \geq k$, and the grid gives $s(k^2) \leq k$; hence $s(k^2 - c) \leq k$ for $0 \leq c < k^2$. For an integer $k \geq 2$ let

$$c^*(k) := \max\{c \in \mathbb{Z} : 0 \leq c \leq k^2 - 1, s(k^2 - c) = k\},$$

the *deficiency*: removing up to $c^*(k)$ squares from the $k \times k$ grid does not allow a smaller container, while removing $c^*(k) + 1$ squares does. The wasted area of Erdős and Graham [EG] is $W(x) := x^2 - M(x)$, where $M(x)$ is the largest number of unit squares that can be packed in a square of side x . In Section 2 we record the elementary relation

$$(1.1) \quad c^*(k) = \lim_{x \rightarrow k^-} W(x) - 1,$$

so $c^*(k)$ measures the wasted area just below an integer.

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Lower bounds. The author showed $c^*(k) \rightarrow \infty$ in [R], with a bound of order $\log k$; lower bounds of order $k^{1/4}$ and $k^{1/3}$ followed in [Q] and [T]. These are preprints that have not been refereed; see Section 11.

Upper bounds. From the literature on $W(x)$ one gets $c^*(k) = O(k^{3/5})$ (preprints [Bui], [McC]); the refereed literature gives $O(k^{7/11})$ [EG] and $O(k^{(3+\sqrt{2})/7} \log k)$ [CG], and Arslanov, Mustafin and Shangitbayev [AMS] proved $c^*(k) \leq k - 1$ for $k > 12$. See Section 10.

1.1. Results. The main result is the following.

Theorem 1.1 (computer-assisted constants). *For each tier i of Table 1 and every integer $k \geq k_i$,*

$$c^*(k) < C_i k^{3/8} + a_i.$$

Here $C_i \geq \frac{32}{5}(\frac{5}{3})^{3/8} C_{E,i}^{5/8}$, $k_i \geq \frac{3}{5} C_{E,i} b_{0,i}^{8/5}$ and $a_i \geq \frac{12}{5} C_{E,i} b_{0,i}^{-2/5}$, where $b_{0,i}$ and $C_{E,i}$ are the threshold and the end constant of Corollary 8.10 for the parameters of tier i (Table 3). The tiers marked $\dagger$ use, in addition to the lemmas used by the other tiers, the construction ZC' and Lemmas 8.2–8.5 of Section 8.2. The tiers marked $\ddagger$ use, in addition to these, Lemmas 8.6, 8.7, 8.8 and 8.11 and Corollary 8.12 (Sections 8.3 and 8.6); for them $b_{0,i}$ and $C_{E,i}$ are those of Corollary 8.12 (Table 4).

tier i	$b_{0,i}$	$C_{E,i}$	C_i	k_i	a_i	better than Thm. 1.2 for $k \geq$
B1 $\dagger$	$10^{7.7}$	17.461	46.31	$2.2 \cdot 10^{13}$	0.035	$1.04 \cdot 10^{14}$
B1s $\dagger$	$10^{7.9}$	16.968	45.49	$4.5 \cdot 10^{13}$	0.029	$5.07 \cdot 10^{13}$
B2 $\dagger$	10^8	16.721	45.08	$6.4 \cdot 10^{13}$	0.026	$6.4 \cdot 10^{13}$
A1	$10^{8.5}$	18.961	48.76	$4.6 \cdot 10^{14}$	0.019	$8.14 \cdot 10^{14}$
A1s	$10^{8.58}$	18.722	48.38	$6.1 \cdot 10^{14}$	0.017	$6.1 \cdot 10^{14}$
A2	10^9	17.523	46.42	$2.7 \cdot 10^{15}$	0.011	$2.7 \cdot 10^{15}$
A3	10^{10}	15.859	43.61	$9.6 \cdot 10^{16}$	0.0039	$9.6 \cdot 10^{16}$
A4	10^{12}	14.778	41.73	$1.5 \cdot 10^{20}$	$5.7 \cdot 10^{-4}$	$1.5 \cdot 10^{20}$
A5	10^{16}	14.153	40.62	$3.4 \cdot 10^{26}$	$1.4 \cdot 10^{-5}$	$3.4 \cdot 10^{26}$
C1 $\ddagger$	$10^{7.3617}$	14.473	41.19	$5.22 \cdot 10^{12}$	0.04	$5.22 \cdot 10^{12}$
C2 $\ddagger$	$10^{7.2}$	14.805	41.78	$2.95 \cdot 10^{12}$	0.047	$2.95 \cdot 10^{12}$
C3 $\ddagger$	$10^{7.15}$	14.983	42.09	$2.48 \cdot 10^{12}$	0.05	$2.48 \cdot 10^{12}$

TABLE 1. The tiers of Theorem 1.1. The constants are rounded up. The last column is $\max(k_i, k_{x,i})$, where $k_{x,i} := (C_i/20.668)^{40}$ (rounded up to three digits) is the point from which $C_i k^{3/8} < 20.668 k^{2/5}$; as $a_i < 0.623$, from there on the tier gives a smaller bound than Theorem 1.2. Tiers marked $\dagger$ use ZC' and Lemmas 8.2–8.5 (Section 8.2). Tiers marked $\ddagger$ use, in addition, Lemmas 8.6–8.8 and 8.11 (Sections 8.3 and 8.6); their parameters are in Table 4.

The proof is in Section 8. Its constants are certified by interval arithmetic (Corollaries 8.10 and 8.12). The range $k \geq k_i$, where one end of a band has more than 10^7 pieces, has not been tested by explicit computation, except for 15 end packings of the tiers $\ddagger$ near b_0 (“Verification of version 1.1”); the explicit checks of Remark 8.13 test the validity of the construction and of its lemmas at smaller sizes.

Version 1.0. Version 1.0 of this paper [U] stated $c^*(k) < 43.06 k^{3/8} + 2 \cdot 10^{-5}$ for every integer $k \geq 4 \cdot 10^{26}$ (with $C_E = 15.54$ and $b_0 = 10^{16}$). That statement remains proved (it is the case $(b_0, c_y, c_D, c_R, a_w, \varpi) = (10^{16}, 1/12, 4, 4, 3/2, 0.65)$ of Section 8, see Section 8.7), and it is superseded by the tier A5: $c^*(k) < 40.62 k^{3/8} + 1.4 \cdot 10^{-5}$ for $k \geq 3.4 \cdot 10^{26}$.

For moderate k the staircase alone gives a smaller bound with much better constants:

Theorem 1.2. *For every integer $k \geq k_0 := 1.6 \cdot 10^7$,*

$$c^*(k) < \frac{20}{3} \left(\frac{3}{2}\right)^{2/5} 5.03^{3/5} k^{2/5} + \frac{8}{3} \cdot 5.03 \cdot 10^{-4/3} < 20.668 k^{2/5} + 0.623.$$

The proof (Section 7) gives $c^*(k) \leq (18.89 + o(1)) k^{2/5}$ as $k \rightarrow \infty$. At $k = k_0$ the bound is 15738.5; the bound of Theorem 1.4 below is 31999 there.

Which bound to use. Of the two bounds, only Theorem 1.2 is proved for $1.6 \cdot 10^7 \leq k < 2.48 \cdot 10^{12}$ ($k < 2.2 \cdot 10^{13}$ without Sections 8.3 and 8.6, and $k < 4.6 \cdot 10^{14}$ without Section 8.2 either). From there on both theorems apply, and for each k one may take the smallest of the bounds that apply. With the tiers that use only Lemmas 8.1 and 8.9 (no †), Theorem 1.1 gives a smaller bound than $20.668 k^{2/5} + 0.623$ (Theorem 1.2) for every $k \geq 6.1 \cdot 10^{14}$ (tier A1s). With the tiers marked † as well, it does so for every $k \geq 5.07 \cdot 10^{13}$ (tier B1s). With the tiers marked ‡ as well, it does so for every $k \geq 2.48 \cdot 10^{12}$ (tier C3). Below that point Theorem 1.2 is the better bound (or the only one), and it also comes with explicit packings for moderate k (Theorem 1.6 below); Theorem 1.1 gives the exponent $3/8$. Both bounds are far above the values of explicit packings, and neither is claimed to be close to $c^*(k)$.

The frame of the proof is an L-shaped packing. For an integer $b \geq 2$ put $D := b^2 + 1$,

$$\gamma_0 := \frac{b^2 - 1}{D}, \quad h_0 := \frac{3b^2 - 1}{D}, \quad R(L) := \max(0, \lceil (L - h_0)\gamma_0 \rceil) \quad (L \in \mathbb{Z}),$$

and for integers $2 \leq b \leq k$

$$N(k, b) := (k - b)^2 + b(R(k) + R(k - b)).$$

Theorem 1.3. *Let $k \geq 2$ and $2 \leq b \leq k$ be integers with $(k, b) \neq (2, 2)$. Then $N(k, b)$ unit squares can be packed in a square of side less than k . Consequently, for all $2 \leq b \leq k$,*

$$c^*(k) \leq k^2 - N(k, b) - 1 < 6b + \frac{4k}{b} - 3.$$

Theorem 1.4. *For every integer $k \geq 6$,*

$$c^*(k) \leq 8\lceil\sqrt{k-4}\rceil - 1 < 8\sqrt{k-4} + 7.$$

The bound of Theorem 1.4 is the best that the packings of Theorem 1.3 give, up to an additive constant:

Proposition 1.5. *For all integers $2 \leq b \leq k$ one has $k^2 - N(k, b) > 8\sqrt{k} - 14$.*

The packings of Theorem 1.3 lose area at the four ends of the two tilted bands, about b at each end (Section 5). In Section 6 we lift the rows of each band and show that the deficiency is exactly the body cost of the bands, about $4k/b$, plus the uncovered areas of four end regions (Proposition 6.3). Section 7 fills an end region by a staircase of columns with uncovered area $O(b^{2/3})$; with $b \asymp k^{3/5}$ this gives Theorem 1.2. Section 8 replaces the two walls of the staircase by chains of staircases with slowly decreasing tilt; the uncovered area becomes $O(b^{3/5})$, and with $b \asymp k^{5/8}$ this gives Theorem 1.1. With tuned parameters the staircase construction gives explicit packings for moderate k :

Theorem 1.6 (computer-assisted). *For each row of Table 2, $k^2 - c$ unit squares can be packed in a square of side less than k . Hence $c^*(k) \leq c - 1$, as listed.*

For comparison, the explicit bound of Wang, Dong and Li [WDL] would give about $8.1 \cdot 10^4$ at $k = 10^5$ (Section 10).

To the best of our knowledge, no upper bound of order $o(k^{3/5})$ for $c^*(k)$, or for $W(x)$ just below an integer, has been stated in the sources we searched (listed in Section 10). The ingredients of Theorem 1.3 are standard, and the staircase and the chained wall stairs are due to Bui (Section 1.2); what seems not to have been written down is their use just below an integer, with bands of width $b - \varepsilon$.

k	b	y_0	$c = k^2 - N$	bound	Thm. 1.2	$8\lceil\sqrt{k-4}\rceil - 1$
10^5	623	257/4	1584	$c^*(k) \leq 1583$	2067	2535
10^6	2481	653/4	3973	$c^*(k) \leq 3972$	5192	7999
10^7	9875	1663/4	10040	$c^*(k) \leq 10039$	13041	25303
$3.825 \cdot 10^7$	22086	710	16989	$c^*(k) \leq 16988$	22302	49479
10^8	39314	1041	24791	$c^*(k) \leq 24790$	32757	79999

TABLE 2. Certified packings (Theorem 1.6). The band width is $b - \varepsilon$ with $\tan(\theta/2) = b/(b^2 - 1)$, and y_0 is the lift of the first row of each band. The column “Thm. 1.2” gives the integer part of $20.668 k^{2/5} + 0.623$ (Theorem 1.2 itself applies only for $k \geq 1.6 \cdot 10^7$), and the last column is the bound of Theorem 1.4. These staircase certificates are the best values we have at these k . The construction of Section 8 gives larger values here (for example $c^*(10^8) \leq 44299$, Remark 8.13); it wins only for very large k .

The order of $c^*(k)$. Theorem 1.1 (any one tier) and the lower bound of [T] give

$$k^{1/3} \ll c^*(k) \ll k^{3/8}.$$

By a theorem of Roth and Vaughan, a bound $W(x) \leq Cx^\alpha$ valid for all large real x must have $\alpha \geq 1/2$, so by (1.1) such bounds cannot give $c^*(k) = o(\sqrt{k})$ (Section 10.3); Theorems 1.1 and 1.2 use that the band width is just below an integer. These are the bounds proved here and in [T]; the true exponent of $c^*(k)$ is open: it lies between $1/3$ and $3/8$ (Section 12).

1.2. Credits. The skeleton of the packing of Theorem 1.3 (an axis-parallel square, and two bands along two of its sides filled with tilted blocks that touch both sides of the band) is the construction of Erdős and Graham [EG, Figs. 1–2]; the same L-shaped skeleton is the Type 1 packing of Chung and Graham [CG]. The block (a $1 \times b$ rectangle of b unit squares tilted by about $2\arctan(1/b)$ so that it fits in a band of width slightly less than b) is used by Arslanov, Mustafin and Shangitbayev [AMS] for small b and in a lemma for even b ; they do not chain the blocks into long bands. The staircase of Section 7, in which tilted columns of one fixed tilt keep only their upper squares and their lower squares are replaced by horizontal axis-parallel layers, so that one tilt can follow a rising ceiling, is the “primitive” construction of H. D. Bui [Bui, Section 3]. The wall filler of Section 8, in which a thin trapezoid is filled by a chain of such staircases whose tilt is lowered from block to block so that the layers close in phase, follows the ideas of [Bui, Sections 3–4] (arXiv:2508.04603v2). Our proofs of the end bounds are self-contained and do not use any argument of [Bui]; the write-up and the constants are ours.

1.3. Status. Theorems 1.3 and 1.4 and Proposition 1.5 are proved in full in Sections 2–5, Theorem 1.2 in Sections 6 and 7, and Theorem 1.1 in Sections 6 and 8. Theorem 1.2 relies on one evaluation in interval arithmetic, $\Phi(10^{-4/3}) \leq 5.0163$ (Corollary 7.4), done by the author’s program and by three independent computations; the other numerical constants were certified in the same way. The constants of Theorem 1.1 (Corollaries 8.10 and 8.12) are certified, tier by tier, by interval arithmetic on covers of the whole range $b \geq b_{0,i}$, by the author’s covering and by three coverings written independently from the text: two by verification agents (`mpmath.iv` and `python-flint arb`) and one by a reviewer (`mpmath.iv`), and all three implement the same normalised inequalities (the lists of Appendices A and B), so they check the arithmetic and the coverage but are only a weak check of the transcription of these inequalities from the lemmas; the point checks of the unnormalised formulas (at 90, 120 and 5850 points) partly address that. All programs were written by AI systems of the same family (see “Verification of version 1.1” at the end of the paper). The range of validity cannot be tested by explicit computation, and the explicit checks test the construction

at smaller b (Remark 8.13). The changes of version 1.1 in Section 8 (the parameters, the tiers and Section 8.2) received two independent blank-slate AI reviews, which found them correct with minor fixes; the fixes are included. Sections 8.3 and 8.6 (the tiers marked ‡) received two further blank-slate AI reviews: one found Lemmas 8.6, 8.7, 8.8 and 8.11 correct and Corollary 8.12 and the tiers ‡ correct after fixes (included), the other found all of it correct. Theorem 1.6 consists of the structural part of Section 6 and a finite part (explicit end packings), checked in exact rational arithmetic by the program that produced it and by checkers written independently from a description of the construction (Section 9). The proofs were written and checked with AI assistance and have not been reviewed by a human expert. The paper has not been refereed.

Organization. Section 2 proves (1.1) and the reduction used throughout. Section 3 treats the tilted block, Section 4 the L-shaped packing, and Section 5 the count and the proofs of Theorems 1.3, 1.4 and Proposition 1.5. Section 6 lifts the rows and proves the waste identity, Section 7 constructs the staircase end and proves Theorem 1.2, Section 8 constructs the chained wall fillers and proves Theorem 1.1, and Section 9 proves Theorem 1.6. Sections 10 and 11 compare with the literature, and Section 12 collects remarks on other end constructions and open problems.

2. THE REDUCTION AND THE WASTED AREA

For $x \geq 1$ let $M(x)$ be the largest number of unit squares that can be packed in a square of side x , and $W(x) := x^2 - M(x)$. A packing in a square of side x is also a packing in a larger square containing it, so M is non-decreasing; it is integer-valued and $M(x) \leq x^2$ by area.

Lemma 2.1. *Let $k \geq 2$ be an integer and $m(k) := \max_{1 \leq x < k} M(x)$. Then $c^*(k) = k^2 - m(k) - 1$, and $W(x) \rightarrow k^2 - m(k)$ as $x \rightarrow k^-$. In particular (1.1) holds.*

Proof. The maximum exists because M is non-decreasing, integer-valued and bounded by k^2 on $[1, k]$; and $M(x) = m(k)$ on some interval $[x_1, k]$, so $W(x) = x^2 - m(k) \rightarrow k^2 - m(k)$. Let $1 \leq n \leq k^2$. If $s(n) < k$, then by the definition of the infimum n unit squares can be packed in a square of some side $x < k$, so $n \leq M(x) \leq m(k)$. Conversely, if $n \leq m(k) = M(x)$ with $x < k$, then $s(n) \leq x < k$. Since $s(n) \leq k$ for $n \leq k^2$, we get, for $0 \leq c \leq k^2 - 1$,

$$s(k^2 - c) = k \iff k^2 - c > m(k) \iff c \leq k^2 - m(k) - 1.$$

As $1 \leq m(k) < k^2$ (one square fits in a square of side $1 < k$, and $M(x) \leq x^2 < k^2$), the largest such c is $k^2 - m(k) - 1$. $\square$

Lemma 2.2 (reduction). *If $N \geq 1$ unit squares can be packed in a square of side $S < k$, then $c^*(k) \leq k^2 - N - 1$.*

Proof. By assumption $N \leq M(S) \leq m(k)$, and Lemma 2.1 gives $c^*(k) = k^2 - m(k) - 1 \leq k^2 - N - 1$. $\square$

This is the source of the “ -1 ” in Theorems 1.3 and 1.6: a packing of N squares in a container of side less than k shows $c^*(k) \leq c - 1$ with $c := k^2 - N$. In the notation of [Q] and [T], $m(k)$ is also the largest number of unit squares in $[0, k]^2$ that are pairwise disjoint as closed sets, and $c^*(k) = W_{\min}(k) - 1$ with $W_{\min}(k) := k^2 - m(k)$ ([Q, Section 1]).

3. THE TILTED BLOCK

Fix an integer $b \geq 2$ and an angle $\theta \in (0, \pi/2)$, and write $\gamma := \cos \theta$, $\sigma := \sin \theta$, $u := (\gamma, \sigma)$ and $v := (-\sigma, \gamma)$. For $P \in \mathbb{R}^2$ and $0 \leq i \leq b-1$ let

$$Q_i(P) := \{P + \alpha u + \beta v : \alpha \in [i, i+1], \beta \in [0, 1]\}, \quad B(P) := \bigcup_{i=0}^{b-1} Q_i(P).$$

Each $Q_i(P)$ is a unit square, and $B(P)$ is a $b \times 1$ rectangle tilted by θ , with lowest vertex P . Put

$$f(\theta) := b\gamma + \sigma \quad (\text{width}), \quad h(\theta) := b\sigma + \gamma \quad (\text{height}), \quad d(\theta) := \sec \theta \quad (\text{row spacing}).$$

Lemma 3.1 (projections). *The projection of $B(P)$ to the x -axis is $[P_x - \sigma, P_x + b\gamma]$, of length $f(\theta)$, and its projection to the y -axis is $[P_y, P_y + b\sigma + \gamma]$, of length $h(\theta)$.*

Proof. The coordinates $x = P_x + \alpha\gamma - \beta\sigma$ and $y = P_y + \alpha\sigma + \beta\gamma$ are linear on $[0, b] \times [0, 1] \ni (\alpha, \beta)$, and $\gamma, \sigma > 0$. So x is smallest at $(0, 1)$ and largest at $(b, 0)$, and y is smallest at $(0, 0)$ and largest at $(b, 1)$. $\square$

Lemma 3.2 (width). *Let $\theta_0 := 2 \arctan(1/b)$. Then $0 < \theta_0 < \pi/2$,*

$$\cos \theta_0 = \gamma_0 = \frac{b^2 - 1}{D}, \quad \sin \theta_0 = \frac{2b}{D}, \quad f(\theta_0) = b, \quad h(\theta_0) = h_0 = \frac{3b^2 - 1}{D}, \quad d(\theta_0) = d_0 := \frac{D}{b^2 - 1},$$

with $D = b^2 + 1$, and $f(\theta) < b$ for every $\theta \in (\theta_0, \pi/2]$.

Proof. With $t = \tan(\theta_0/2) = 1/b$ the half-angle formulas give $\cos \theta_0 = (1 - t^2)/(1 + t^2) = (b^2 - 1)/D$ and $\sin \theta_0 = 2t/(1 + t^2) = 2b/D$; hence $f(\theta_0) = (b(b^2 - 1) + 2b)/D = b$ and $h(\theta_0) = (2b^2 + b^2 - 1)/D$. As $b \geq 2$, $\arctan(1/b) < \pi/4$, so $\theta_0 < \pi/2$. With $\varphi := \arctan(1/b)$ we have $f(\theta) = \sqrt{b^2 + 1} \cos(\theta - \varphi)$. For $\theta \in [\theta_0, \pi/2]$, $\theta - \varphi \in [\varphi, \pi/2 - \varphi] \subset [0, \pi]$, where $\cos$ is strictly decreasing; so $f(\theta) < f(\theta_0) = b$ for $\theta > \theta_0$. $\square$

Lemma 3.3 (stacking). *Let $P_0 \in \mathbb{R}^2$ and $P_j := P_0 + (0, j \sec \theta)$ for $j \in \mathbb{Z}$. If $(i, j) \neq (i', j')$, then $Q_i(P_j)$ and $Q_{i'}(P_{j'})$ have disjoint interiors. More precisely, in the coordinates $\alpha' = u \cdot (z - P_0)$, $\beta' = v \cdot (z - P_0)$,*

$$Q_i(P_j) = [i + j \tan \theta, i + 1 + j \tan \theta] \times [j, j + 1].$$

Proof. The map $z \mapsto (u \cdot (z - P_0), v \cdot (z - P_0))$ is a rigid motion. Since $v \cdot (0, j \sec \theta) = j$ and $u \cdot (0, j \sec \theta) = j \tan \theta$, the square $Q_i(P_j)$ has the stated form. If $j \neq j'$ the β' -intervals have disjoint interiors; if $j = j'$ and $i \neq i'$ the α' -intervals do. $\square$

Thus consecutive rows touch along their long sides: the v -component of the shift $(0, \sec \theta)$ is exactly 1.

Lemma 3.4 (one band). *Let $a \in \mathbb{R}$, $L \geq 0$, $\theta \in (0, \pi/2)$ and an integer $R \geq 0$ with $R = 0$ or $(R - 1)d(\theta) + h(\theta) \leq L$. Then the bR squares $Q_i(P_j)$, $0 \leq i < b$, $0 \leq j < R$, with $P_j = (a + \sigma, j \sec \theta)$, have pairwise disjoint interiors and lie in $[a, a + f(\theta)] \times [0, L]$.*

Proof. By Lemma 3.1 the row $B(P_j)$ has x -projection $[a, a + b\gamma + \sigma]$ and y -projection $[jd, jd + h] \subset [0, (R - 1)d + h] \subset [0, L]$. Disjointness is Lemma 3.3. $\square$

4. THE L-SHAPED PACKING

Let $k \geq 2$ and $2 \leq b \leq k$ be integers and $\theta \in (\theta_0, \pi/2)$. By Lemma 3.2,

$$\varepsilon := b - f(\theta) > 0, \quad S := k - \varepsilon = k - b + f(\theta) < k.$$

Let $R_1, R_2 \geq 0$ be integers satisfying the *length conditions*

$$(L1) \quad R_1 = 0 \quad \text{or} \quad (R_1 - 1)d(\theta) + h(\theta) \leq S, \quad (L2) \quad R_2 = 0 \quad \text{or} \quad (R_2 - 1)d(\theta) + h(\theta) \leq k - b.$$

The packing consists of three parts (Figure 1 of [EG] and the Type 1 packing of [CG] have the same skeleton).

- (1) The *grid* $\mathcal{G}$: the $(k - b)^2$ squares $[i, i + 1] \times [j, j + 1]$, $0 \leq i, j \leq k - b - 1$, in $G := [0, k - b]^2$.
- (2) The *right band* $\mathcal{A}$: the bR_1 squares of Lemma 3.4 with $a = k - b$, $L = S$, $R = R_1$; they lie in $A := [k - b, S] \times [0, S]$, since $a + f(\theta) = S$.

- (3) The *top band* $\mathcal{T}$: the image under the reflection $\iota(x, y) := (y, x)$ of the bR_2 squares of Lemma 3.4 with $a = k - b$, $L = k - b$, $R = R_2$. Before reflection they lie in $[k - b, S] \times [0, k - b]$, so $\mathcal{T}$ lies in $T := [0, k - b] \times [k - b, S]$.

Lemma 4.1. *The $N = (k - b)^2 + b(R_1 + R_2)$ squares of $\mathcal{G} \cup \mathcal{A} \cup \mathcal{T}$ form a packing in $[0, S]^2$, and $S < k$.*

Proof. Since $f(\theta) > 0$ we have $k - b < S$, so $G, A, T \subset [0, S]^2$; and $S < k$. The interiors $(0, k - b)^2$, $(k - b, S) \times (0, S)$ and $(0, k - b) \times (k - b, S)$ of G, A, T are pairwise disjoint (G and A are separated by $x = k - b$, G and T by $y = k - b$, and A and T by $x = k - b$); the corner $[k - b, S]^2$ belongs to A only. The interior of a square contained in one of these regions lies in the interior of that region, so squares from different parts have disjoint interiors. Inside $\mathcal{G}$ this is clear, inside $\mathcal{A}$ it is Lemma 3.4, and inside $\mathcal{T}$ it is Lemma 3.4 together with the fact that ι is an isometry. $\square$

If $k = b$, the grid is empty, T is degenerate and $R_2 = 0$; the lemma remains true.

4.1. An explicit angle. For an integer L let $R(L)$ be as in Section 1. Then $R(L)$ is the number of integers $j \geq 0$ with $jd_0 + h_0 < L$: if $L \leq h_0$ both are 0; otherwise the condition is $j < (L - h_0)/d_0 = (L - h_0)\gamma_0$, and the number of such $j \geq 0$ is $\lceil (L - h_0)\gamma_0 \rceil$.

Lemma 4.2. *Let $M := 14kb^2 + 1$, $p := M$, $q := bM - 1$ and $\theta := 2 \arctan(p/q)$. Then $\theta \in (\theta_0, \pi/2)$, and $R_1 := R(k)$, $R_2 := R(k - b)$ satisfy (L1) and (L2). Moreover $\cos \theta = (q^2 - p^2)/(q^2 + p^2)$ and $\sin \theta = 2pq/(p^2 + q^2)$ are rational, so all vertices of the packing have rational coordinates.*

Proof. (i) We have $p/q > 1/b$ (as $bM > bM - 1$) and $p/q < 1$ (as $(b - 1)M > 1$), so $\theta_0 < \theta < \pi/2$. Since the derivative of $\arctan$ is at most 1,

$$\theta - \theta_0 = 2 \left(\arctan \frac{p}{q} - \arctan \frac{1}{b} \right) \leq 2 \left(\frac{M}{bM - 1} - \frac{1}{b} \right) = \frac{2}{b(bM - 1)} \leq \frac{1}{7kb^4} =: \tau,$$

because $bM - 1 = 14kb^3 + b - 1 \geq 14kb^3$.

(ii) *Slack.* Let L be an integer with $R := R(L) \geq 1$, and $g(L) := L - h_0 - (R - 1)d_0$. Then $g(L) > 0$ by the characterization of $R(L)$ above, and

$$(b^2 - 1)Dg(L) = (b^4 - 1)L - (b^2 - 1)(3b^2 - 1) - (R - 1)D^2$$

is an integer; hence $g(L) \geq 1/(b^4 - 1)$.

(iii) *Derivatives.* For $b \geq 2$, $\gamma_0 \geq 3/5$, and $\tau \leq 1/224$; so $\cos \geq 3/5 - \tau \geq 1/2$ on $[\theta_0, \theta_0 + \tau]$. There $d' = \sin / \cos^2 \leq 4$, $h' = b \cos - \sin \leq b$ and $|f'| = |\cos - b \sin| \leq \sqrt{b^2 + 1} \leq b + 1$.

(iv) If $R_1 = R(k) \geq 1$ (otherwise there is nothing to prove), then by the mean value theorem, using $f(\theta_0) = b$,

$$\begin{aligned} (R_1 - 1)d(\theta) + h(\theta) - S &= [(R_1 - 1)d_0 + h_0 - k] + (R_1 - 1)(d(\theta) - d_0) \\ &\quad + (h(\theta) - h_0) + (f(\theta_0) - f(\theta)) \\ &\leq -g(k) + (\theta - \theta_0)(4(R_1 - 1) + 2b + 1). \end{aligned}$$

Here $R_1 - 1 < (k - h_0)\gamma_0 < k$ and $b \leq k$, so the last factor is at most $7k$, and the right side is at most $-1/(b^4 - 1) + 7k\tau = -1/(b^4 - 1) + 1/b^4 < 0$. This is (L1). Similarly, if $R_2 \geq 1$, $(R_2 - 1)d(\theta) + h(\theta) - (k - b) \leq -g(k - b) + \tau(4k + b) \leq -1/(b^4 - 1) + 5/(7b^4) < 0$, which is (L2). $\square$

Lemma 4.3 (maximality within the family). *If $\theta \in (\theta_0, \arctan b)$, then every pair R_1, R_2 satisfying (L1) and (L2) has $R_1 \leq R(k)$ and $R_2 \leq R(k - b)$. For $\theta \in (\theta_0, \theta_0 + \tau]$ equality is possible.*

Proof. Note $\tan \theta_0 = 2b/(b^2 - 1) < b$, so the interval is non-empty. On it $d' > 0$ and $h' = b \cos \theta - \sin \theta > 0$, so d and h are strictly increasing. If $R_1 \geq R(k) + 1$, then $(R_1 - 1)d_0 + h_0 \geq k$ by the characterization of $R(k)$, hence $(R_1 - 1)d(\theta) + h(\theta) > k > S$, contradicting (L1). The top band is the same with $k - b$ in place of S and k . The proof of Lemma 4.2 works for every $\theta \in (\theta_0, \theta_0 + \tau]$. $\square$

So $N(k, b)$ is the exact number of squares of the packings of this section with θ close to θ_0 and with the rows of each band starting at the bottom of the band. Remark 5.3 shows that the family can be enlarged.

5. THE COUNT: PROOFS OF THE BOUNDS OF ORDER $\sqrt{k}$

Proof of Theorem 1.3. By Lemmas 4.2 and 4.1, $N(k, b)$ unit squares can be packed in a square of side $S < k$. If $k > b$ then $N(k, b) \geq (k - b)^2 \geq 1$; if $k = b$ then $N(k, b) = bR(k)$, and $R(k) = 0$ only if $k \leq h_0 < 3$, that is $k = b = 2$. So $N(k, b) \geq 1$ for $(k, b) \neq (2, 2)$, and Lemma 2.2 gives $c^*(k) \leq k^2 - N(k, b) - 1$; for $(k, b) = (2, 2)$ this bound is $c^*(2) \leq 3$, which holds by definition.

For the second inequality we compute the count exactly. Put

$$\eta := \frac{2k - 10}{D} + \frac{8}{D^2}, \quad \eta' := \eta - \frac{2b}{D}.$$

We claim

$$(5.1) \quad k^2 - N(k, b) \leq 6b + b(\lfloor \eta \rfloor + \lfloor \eta' \rfloor), \quad \text{with equality if } k - b \geq 2.$$

For an integer L , using $h_0 = 3 - 4/D$ and $\gamma_0 = 1 - 2/D$,

$$(L - h_0)\gamma_0 = L - 3 + \frac{4}{D} - \frac{2L}{D} + \frac{6}{D} - \frac{8}{D^2} = L - 3 - \eta_L, \quad \eta_L := \frac{2L - 10}{D} + \frac{8}{D^2}.$$

Since L is an integer, $\lceil L - 3 - \eta_L \rceil = L - 3 - \lfloor \eta_L \rfloor$, so $R(L) = \max(0, L - 3 - \lfloor \eta_L \rfloor) \geq L - 3 - \lfloor \eta_L \rfloor$. Here $\eta_k = \eta$ and $\eta_{k-b} = \eta'$. Hence

$$k^2 - N(k, b) = 2kb - b^2 - b(R(k) + R(k - b)) \leq 2kb - b^2 - b(2k - b - 6 - \lfloor \eta \rfloor - \lfloor \eta' \rfloor) = 6b + b(\lfloor \eta \rfloor + \lfloor \eta' \rfloor).$$

If $L \geq 2$, then $L - h_0 > -1$ and $0 < \gamma_0 < 1$, so $(L - h_0)\gamma_0 > -1$ and the ceiling is at least 0; the maximum is then not active. This applies to $L = k$ always and to $L = k - b$ when $k - b \geq 2$, which gives the equality case.

Now $\lfloor \eta \rfloor + \lfloor \eta' \rfloor \leq 2\eta - 2b/D$, and therefore

$$k^2 - N(k, b) \leq 6b + \frac{(4k - 20)b}{D} + \frac{16b}{D^2} - \frac{2b^2}{D} = 6b + \frac{4kb}{D} - 2 + \frac{2 - 20b + 16b/D}{D}.$$

Here $4kb/D < 4k/b$, and for $b \geq 2$ we have $16b/D \leq 32/5 < 7$, so $2 - 20b + 16b/D < 9 - 20b < 0$. Hence $k^2 - N(k, b) < 6b + 4k/b - 2$, and $c^*(k) \leq k^2 - N(k, b) - 1 < 6b + 4k/b - 3$. $\square$

Corollary 5.1. *For every integer $k \geq 6$, $c^*(k) < 4\sqrt{6}\sqrt{k} \approx 9.798\sqrt{k}$.*

Proof. The function $g(x) = 6x + 4k/x$ has its minimum $g(x^*) = 4\sqrt{6}\sqrt{k}$ at $x^* = \sqrt{2k/3}$. Let $b = \lceil x^* \rceil = x^* + \delta$ with $0 \leq \delta < 1$. Using $4k/x^{*2} = 6$,

$$g(b) - g(x^*) = 6\delta - \frac{4k\delta}{x^*(x^* + \delta)} = \frac{6\delta^2}{x^* + \delta} < \frac{6}{x^*} \leq 3,$$

because $x^* \geq 2$ for $k \geq 6$. Also $2 \leq b \leq x^* + 1 \leq k$. Theorem 1.3 gives $c^*(k) < g(b) - 3 < 4\sqrt{6}\sqrt{k}$. $\square$

Proof of Theorem 1.4. Let $b := \lceil \sqrt{k - 4} \rceil$. Then $b \geq \lceil \sqrt{2} \rceil = 2$ and $b \leq \sqrt{k - 4} + 1 \leq k$. Moreover $D = b^2 + 1 \geq k - 3 > 0$ and $2k - 10 > 0$, so

$$\eta \leq \frac{2k - 10}{k - 3} + \frac{8}{(k - 3)^2} = 2 - \frac{4(k - 3) - 8}{(k - 3)^2} < 2,$$

since $k - 3 > 2$. Hence $\lfloor \eta \rfloor \leq 1$ and $\lfloor \eta' \rfloor \leq \lfloor \eta \rfloor \leq 1$, and (5.1) gives $k^2 - N(k, b) \leq 8b$. Since $(k, b) \neq (2, 2)$, Theorem 1.3 gives $c^*(k) \leq 8b - 1 < 8\sqrt{k - 4} + 7$. $\square$

For $k > 15$ Theorem 1.4 is stronger than Corollary 5.1, since $8\sqrt{k} + 7 < 4\sqrt{6}\sqrt{k}$ for $\sqrt{k} > 7/(4\sqrt{6} - 8) \approx 3.89$.

Proof of Proposition 1.5. If $k - b \geq 2$, equality holds in (5.1), and $\lfloor z \rfloor > z - 1$ gives

$$k^2 - N(k, b) > 6b + b\left(2\eta - \frac{2b}{D} - 2\right) = 4b + \frac{(4k - 20)b}{D} + \frac{16b}{D^2} - \frac{2b^2}{D} \geq 4b + \frac{4kb}{D} - 12,$$

using $16b/D^2 \geq 0$, $2b^2/D \leq 2$ and $20b/D \leq 10$. With $x := b + 1/b$ we have $kb/D = k/x$ and $4b = 4x - 4/b \geq 4x - 2$, so $k^2 - N(k, b) > 4x + 4k/x - 14 \geq 8\sqrt{k} - 14$ by the inequality of arithmetic and geometric means. If $b \in \{k - 1, k\}$, then $R(k - b) \in \{R(0), R(1)\} = \{0\}$ (as $h_0 \geq 11/5$) and $R(k) \leq \lceil k - h_0 \rceil \leq k - 2$, so $N(k, b) \leq 1 + k(k - 2)$ and $k^2 - N(k, b) \geq 2k - 1$; and $2k - 1 - (8\sqrt{k} - 14) = 2(\sqrt{k} - 2)^2 + 5 > 0$. $\square$

Hence the best bound obtainable from Theorem 1.3 satisfies $8\sqrt{k} - 15 < \min_b(k^2 - N(k, b) - 1) \leq 8\lceil\sqrt{k} - 4\rceil - 1$.

Remark 5.2 (where $8\sqrt{k}$ comes from). The exact count (5.1) is $6b + b(\lfloor \eta \rfloor + \lfloor \eta' \rfloor)$ with $\eta \approx 2k/b^2$. The estimate $\lfloor z \rfloor \leq z$ gives $6b + 4k/b$, minimized at about $9.80\sqrt{k}$; but for $b = \lceil\sqrt{k} - 4\rceil$ and $k \geq 23$ both floors equal 1 while η is just below 2, and the count is $8b$. (For $k \geq 31$: $\eta < 2$ by the proof of Theorem 1.4, and with $s := \sqrt{k} - 4$, $b < s + 1$ gives $b^2 + 2b < k - 1 + 4s$, so $D\eta' > 2k - 10 - 2b > b^2 + 1 = D$ as soon as $k - 10 \geq 4s$, which holds for $k \geq 31$; hence $1 < \eta' \leq \eta < 2$. For $23 \leq k \leq 30$ this was checked exactly, as was the whole range $23 \leq k \leq 2 \cdot 10^5$; $k = 22$ fails. Nothing else in the paper uses this.) Geometrically (Section 6), each band loses about $2L/b$ along its length L , because the row spacing $\sec \theta$ exceeds 1 by about $2/b^2$, and about b at each of its two ends, where the first and the last row cross the band with a rise of about 2. The total $4k/b + 4b$ is at least $8\sqrt{k}$.

Remark 5.3 (a larger family). Lemma 4.3 concerns the family in which the top band stays in $x \leq k - b$. One more row in the top band protrudes into the corner $[k - b, S]^2$, but often fits into the empty wedge above the last row of the right band. An exact computer check (all $6 \leq k \leq 30$ with $k - b \geq 2$) found 285 of 375 such enlarged packings valid; for example $k = 12$, $b = 3$ gives 126 squares instead of $N(12, 3) = 123$. We have no general proof, and this remark is not used.

6. LIFTED ROWS AND THE WASTE IDENTITY

6.1. Lifted rows and end regions. Throughout this section $b \geq 2$, $\theta \in (\theta_0, \pi/2)$, $\gamma = \cos \theta$, $\sigma = \sin \theta$, and $w := f(\theta) = b\gamma + \sigma < b$, $h := h(\theta)$, $d := \sec \theta$. A band is described in its own *strip frame*: the strip is $\Sigma := [0, w] \times [0, L_b]$, where $L_b > 0$ is the length of the band. Given a *lift* $y_0 \geq 0$ and an integer $R \geq 1$, the rows are $B(P_j)$, $0 \leq j < R$, with

$$P_j := (\sigma, y_0 + jd), \quad \text{and we put} \quad y_1 := L_b - h - (R - 1)d - y_0.$$

We assume $y_1 \geq 0$. For $y \geq 0$ the *end region* is

$$\text{Reg}(y) := \{(x', y') : 0 \leq x' \leq w, \ 0 \leq y' \leq y + (x' - \sigma) \tan \theta\},$$

and $\rho(x', y') := (w - x', L_b - y')$ is the rotation by π about the centre of Σ . The upper side of $\text{Reg}(y_0)$ is the line through the lower side of the first row $B(P_0)$; on $[0, \sigma]$ the first row is higher than this line, so $\text{Reg}(y_0)$ is slightly smaller than the part of Σ below the first row.

Lemma 6.1. *Let $\mathcal{C}_0$ and $\mathcal{C}_1$ be finite families of unit squares with pairwise disjoint interiors, contained in $\text{Reg}(y_0)$ and $\text{Reg}(y_1)$ respectively. Then the bR squares $Q_i(P_j)$ ($0 \leq i < b$, $0 \leq j < R$), the squares of $\mathcal{C}_0$ and the squares of $\rho(\mathcal{C}_1)$ form a packing in Σ .*

Proof. Containment. By Lemma 3.1 the row $B(P_j)$ has x -projection $[0, w]$ and y -projection $[y_0 + jd, y_0 + jd + h] \subset [y_0, L_b - y_1] \subset [0, L_b]$. On $\text{Reg}(y)$ we have $y' \leq y + (w - \sigma) \tan \theta = y + b\sigma \leq y + h$; since $L_b = y_0 + y_1 + (R - 1)d + h$ is at least $y_0 + h$ and at least $y_1 + h$, both end regions lie in Σ , and $\rho(\Sigma) = \Sigma$.

Separation. Let $\beta'(z) := v \cdot (z - P_0)$. By Lemma 3.3 (with this P_0), every row square lies in $\{0 \leq \beta' \leq R\}$. The upper side of $\text{Reg}(y_0)$ is the line through P_0 with direction u , and $\beta'(x', y') = -\sigma(x' - \sigma) + \gamma(y' - y_0) \leq 0$ is equivalent to $y' \leq y_0 + (x' - \sigma) \tan \theta$; so $\text{Reg}(y_0) \subset \{\beta' \leq 0\}$. Next, the highest vertex of the last row is $P_{R-1} + bu + v = (b\gamma, y_0 + (R-1)d + h)$, and

$$(6.1) \quad \rho(P_{R-1} + bu + v) = (w - b\gamma, L_b - y_0 - (R-1)d - h) = (\sigma, y_1) =: P'_0.$$

As above, $\text{Reg}(y_1) \subset \{z : v \cdot (z - P'_0) \leq 0\}$. The linear part of ρ is $-\text{id}$, so for $z \in \text{Reg}(y_1)$

$$\beta'(\rho(z)) = v \cdot (\rho(P'_0) - P_0) - v \cdot (z - P'_0) = v \cdot (P_{R-1} + bu + v - P_0) - v \cdot (z - P'_0) \geq (R-1) + 1 = R,$$

using $v \cdot (P_{R-1} - P_0) = R-1$, $v \cdot u = 0$ and $v \cdot v = 1$. Hence $\rho(\text{Reg}(y_1)) \subset \{\beta' \geq R\}$. The three sets $\{\beta' \leq 0\}$, $\{0 \leq \beta' \leq R\}$ and $\{\beta' \geq R\}$ have pairwise disjoint interiors; inside each group the interiors are disjoint by Lemma 3.3 and by assumption (ρ is an isometry). $\square$

Corollary 6.2. *Let $k > b$ be an integer, $S := k - b + w$, and $y_0 \geq 0$. For the right band put $L_b := S$ and for the top band $L_b := k - b$, and suppose that for each band an integer $R \geq 1$ is given with $y_1 \geq 0$. Let $\mathcal{C}_0, \mathcal{C}_1$ be as in Lemma 6.1 for each band. Map the right band into the plane by $(x', y') \mapsto (k - b + x', y')$, and the top band by $(x', y') \mapsto \iota(k - b + x', y') = (y', k - b + x')$. Together with the grid $\mathcal{G}$ these squares form a packing in $[0, S]^2$, with $S < k$ and*

$$N = (k - b)^2 + b(R_{\text{right}} + R_{\text{top}}) + \sum_{\text{4 ends}} |\mathcal{C}|.$$

Proof. The strips are mapped onto A and T of Section 4. Lemma 6.1 gives a packing inside each of A and T , and the separation of G, A, T is as in the proof of Lemma 4.1. $\square$

6.2. The waste identity. For $y \geq 0$ let

$$A_0(y) := yw + \frac{1}{2}\gamma\sigma(1 + b^2).$$

If $y = y_0$, this is the area of the part of Σ below the lower boundary of the first row: that boundary is the segment from $(0, y_0 + \gamma)$ to (σ, y_0) followed by the segment from (σ, y_0) to $(w, y_0 + b\sigma)$, and

$$\int_0^\sigma \left(y_0 + \gamma - \frac{\gamma}{\sigma}x\right)dx + \int_\sigma^w (y_0 + (x - \sigma) \tan \theta)dx = y_0\sigma + \frac{\gamma\sigma}{2} + y_0b\gamma + \frac{b^2\gamma\sigma}{2} = A_0(y_0).$$

For a finite family $\mathcal{C}$ in $\text{Reg}(y)$ we write $E(y) := A_0(y) - |\mathcal{C}|$, the *end cost*. It is the uncovered area of the part of Σ below the first row (for $y = y_0$), resp. above the last row (for $y = y_1$, after applying ρ).

Proposition 6.3 (waste identity). *In the situation of Lemma 6.1, with $E(y_0) = A_0(y_0) - |\mathcal{C}_0|$ and $E(y_1) = A_0(y_1) - |\mathcal{C}_1|$,*

$$wL_b - bR - |\mathcal{C}_0| - |\mathcal{C}_1| = (R-1)\tan \theta + E(y_0) + E(y_1).$$

In the situation of Corollary 6.2,

$$k^2 - N = (k^2 - S^2) + \sum_{\text{2 bands}} [(R-1)\tan \theta + E(y_0) + E(y_1)].$$

Proof. Split Σ into the part below the lower boundary of $B(P_0)$ (area $A_0(y_0)$), the part between the lower boundary of $B(P_0)$ and the upper boundary of $B(P_{R-1})$, and the part above the upper boundary of $B(P_{R-1})$. The lower boundary of $B(P_j)$ is that of $B(P_0)$ shifted up by jd , and each row has x -projection $[0, w]$; so the middle part has area $(R-1)dw + b$, where b is the area of $B(P_{R-1})$. By (6.1), ρ maps $B(P_{R-1})$ onto the rectangle $B(P'_0)$ (a rotation by π maps a rectangle with sides along u and v onto such a rectangle, and the highest vertex onto the lowest one), so the top part has area $A_0(y_1)$. Hence $wL_b = A_0(y_0) + A_0(y_1) + (R-1)dw + b$, and since $dw = b + \tan \theta$,

$$wL_b - bR - |\mathcal{C}_0| - |\mathcal{C}_1| = E(y_0) + E(y_1) + (R-1)(dw - b) = E(y_0) + E(y_1) + (R-1)\tan \theta.$$

For the second identity, $[0, S]^2$ is the union of G , A and T with disjoint interiors, G is covered by the grid, and A , T are the images of the two strips. $\square$

The term $(R - 1) \tan \theta$ is the *body cost* of a band. Since $\tan \theta \approx 2/b$ and $R \approx L_b$, the two bands cost about $4k/b$; the four ends cost $4E$, and $k^2 - S^2 = (b - w)(k + S)$ is tiny when θ is close to θ_0 . For the packing of Theorem 1.3 ($y_0 = 0$, no end squares) $E(0) = \frac{1}{2}\gamma\sigma(1 + b^2) \approx b$; this is the end cost $\approx b$ of Remark 5.2. Any packing of $\text{Reg}(y)$ with end cost $E = o(b)$, for lifts y in an interval of length $\sec \theta$, makes the total cost $4k/b + o(b)$; Section 7 constructs such packings with $E = O(b^{2/3})$.

7. THE STAIRCASE END AND THE EXPONENT $2/5$

In this section we prove Theorem 1.2. Section 7.1 packs a general trapezoid of the form $\text{Reg}(y)$ by a staircase of columns with one common tilt (Lemma 7.3); Section 7.2 specializes to the band ends, and Section 7.3 assembles the packing. The idea of replacing the lower squares of tilted columns by horizontal layers is that of [Bui, Section 3] (Section 1.2).

7.1. Columns, layers and the end lemma. Fix reals $w > 0$, $\mu > 0$, $\sigma_0 \geq 0$ and y with $g_0 := y - \sigma_0\mu > 1$, and put

$$g(X) := y + (X - \sigma_0)\mu, \quad \text{Reg} := \{(X, Y) : 0 \leq X \leq w, 0 \leq Y \leq g(X)\},$$

$\bar{g} := g(w)$ and $\Delta := \bar{g} - g_0 = w\mu$. Reg is a convex trapezoid of area $\int_0^w g$, and g is increasing with $g(0) = g_0$. For a packing in Reg let U be the area of Reg minus the number of unit squares of the packing (the *uncovered area*).

Tilt notation. For $t \in (0, 1)$ let $\alpha := 2 \arctan t$, so that

$$\cos \alpha = \frac{1 - t^2}{1 + t^2}, \quad \sin \alpha = \frac{2t}{1 + t^2}, \quad \tan \alpha = \frac{2t}{1 - t^2}, \quad \sec \alpha = \frac{1 + t^2}{1 - t^2}$$

are rational if t is. Put $u_\alpha := (\cos \alpha, \sin \alpha)$, $v_\alpha := (-\sin \alpha, \cos \alpha)$ and $\lambda(X, Y) := X + Y \tan \alpha$ (the *slab coordinate*). We use $\sec \alpha \cos \alpha = 1$, $\sec \alpha \sin \alpha = \tan \alpha$, $\sin \alpha \leq \tan \alpha$, $\sin \alpha \leq 2t$, $1 - \cos \alpha = 2t^2/(1 + t^2) \leq 2t^2$, and that $\tan \alpha$ and $\sec \alpha$ are increasing in t .

Columns. For a real ξ and integers $J \geq 0$, $n \geq 1$ put $P := (\xi - J \tan \alpha, J)$ and

$$K(\xi, J, n) := \{P + a u_\alpha + e v_\alpha : a \in [0, 1], e \in [0, n]\},$$

the union of the n unit squares $\{P + a u_\alpha + e v_\alpha : a \in [0, 1], e \in [i, i + 1]\}$, $0 \leq i < n$: a $1 \times n$ rectangle tilted by α and leaning to the left, with lowest vertex P .

Lemma 7.1 (columns). *Every point (X, Y) of $K(\xi, J, n)$ satisfies*

- (i) $\xi \leq \lambda(X, Y) \leq \xi + \sec \alpha$;
- (ii) $J \leq Y \leq J + n \cos \alpha + \sin \alpha$;
- (iii) $X \geq \xi - J \tan \alpha - n \sin \alpha \geq \xi - (J + n) \tan \alpha$;
- (iv) $X \leq \xi - J \tan \alpha + \cos \alpha \leq \xi + \sec \alpha$.

Proof. For $z = P + a u_\alpha + e v_\alpha$ we have $X = \xi - J \tan \alpha + a \cos \alpha - e \sin \alpha$ and $Y = J + a \sin \alpha + e \cos \alpha$. Hence $\lambda = \xi - J \tan \alpha + a \cos \alpha - e \sin \alpha + J \tan \alpha + a \sin \alpha \tan \alpha + e \cos \alpha \tan \alpha = \xi + a(\cos \alpha + \sin \alpha^2 / \cos \alpha) = \xi + a \sec \alpha$, which is (i). (ii), (iii) and (iv) follow from $a \in [0, 1]$, $e \in [0, n]$, $\sin \alpha, \cos \alpha \geq 0$, $\sin \alpha \leq \tan \alpha$ and $\cos \alpha \leq 1 \leq \sec \alpha$. $\square$

Lemma 7.2 (the tilt). *Let m be an integer and g_0 a real number with $1 < g_0 < m$. Put $D_m := m + g_0$, $\delta := m - g_0 > 0$, $F(t) := D_m t^2 - 2t - \delta$ and*

$$t^* := \frac{1 + \sqrt{1 + D_m \delta}}{D_m}.$$

Then $t^ \in (0, 1)$, and for $t \in (0, 1)$: $m \cos \alpha + \sin \alpha \leq g_0$ if and only if $F(t) \geq 0$, if and only if $t \geq t^*$. Moreover, if $g_0 \geq \nu > 0$ and $\delta \leq \bar{\delta}$, then $t^* \leq (1 + \sqrt{1 + 2\nu\bar{\delta}})/(2\nu)$.*

Proof. With $\cos \alpha, \sin \alpha$ as above, $(1+t^2)(m \cos \alpha + \sin \alpha - g_0) = m(1-t^2) + 2t - g_0(1+t^2) = -F(t)$. F is a convex quadratic with $F(0) = -\delta < 0$, so $F < 0$ on $[0, t^*)$ and $F > 0$ on (t^*, ∞) , where t^* is its positive root; $F(1) = 2g_0 - 2 > 0$ gives $t^* < 1$. For the bound, $f(D', e) := (1 + \sqrt{1 + D'e})/D'$ is increasing in $e \geq 0$ and decreasing in $D' > 0$, because

$$\frac{\partial f}{\partial D'} = \frac{D'e/(2\sqrt{1+D'e}) - 1 - \sqrt{1+D'e}}{D'^2} \quad \text{and} \quad \frac{D'e}{2\sqrt{1+D'e}} < \frac{\sqrt{1+D'e}}{2}.$$

Since $D_m = m + g_0 > 2g_0 \geq 2\nu$ and $\delta \leq \bar{\delta}$, $t^* = f(D_m, \delta) \leq f(2\nu, \bar{\delta})$. $\square$

The construction. Let $Q \geq 1$ be an integer (the tilt grid).

- (S1) *Column length:* $m := \lfloor \bar{g} \rfloor + 1$. So $m > \bar{g} \geq g_0$, and $\delta := m - g_0 \in (\Delta, \Delta + 1]$.
- (S2) *Tilt:* $t := \lceil t^* Q \rceil / Q$ with $t^* = t^*(m, g_0)$ of Lemma 7.2, so $t \in [t^*, t^* + 1/Q]$; we require $t < 1$.
Let $\alpha := 2 \arctan t$; this one tilt is used for all columns.
- (S3) *Slabs:* $\xi_i := m \tan \alpha + i \sec \alpha$ for $i = 0, 1, 2, \dots$, and $I := \#\{i \geq 0 : \xi_i + \sec \alpha \leq w\}$; we require $I \geq 1$, that is $m \tan \alpha + \sec \alpha \leq w$. Then $w - \sec \alpha < \xi_I \leq w$.
- (S4) *Staircase:* for $0 \leq i < I$ let $\Gamma_i := g(\xi_i - m \tan \alpha)$, $H(J) := J + (m - J) \cos \alpha + \sin \alpha$,

$$J_i := \left\lfloor \frac{\Gamma_i - m \cos \alpha - \sin \alpha}{1 - \cos \alpha} \right\rfloor, \quad n_i := m - J_i,$$

and the column $K_i := K(\xi_i, J_i, n_i)$ (n_i squares with base at height J_i).

- (S5) *Layers:* for integers $j \geq 0$ let $p(j) := \#\{i < I : J_i \leq j\}$ and $a_j := \xi_{p(j)} - j \tan \alpha$, and let j_1 be the largest $j \in \{-1, 0, 1, \dots, \lfloor \bar{g} \rfloor\}$ such that $g(a_{j'}) \geq j' + 1$ for all integers $0 \leq j' \leq j$. For $0 \leq j \leq j_1$ put $L_j := \lfloor w - a_j \rfloor$ and, if $L_j \geq 1$, the layer $\Lambda_j := [a_j, a_j + L_j] \times [j, j + 1]$ (L_j axis-parallel unit squares in a row).
- (S6) *Left wedge:* for $0 \leq r \leq \lfloor g_0 \rfloor - 1$ put $l_r := \lfloor (m - r - 1) \tan \alpha \rfloor$ and, if $l_r \geq 1$, the row $\Omega_r := [0, l_r] \times [r, r + 1]$ (l_r axis-parallel unit squares).

All columns have the same tilt and the same total length m : column i keeps its top n_i squares tilted, and its J_i lowest squares are replaced by the layers $j < J_i$. Each layer raises the top of a column by exactly $1 - \cos \alpha$, which is how a fixed tilt follows the rising ceiling ($J_0 \leq J_1 \leq \dots$, a staircase). The left wall is handled by the rows Ω_r in the wedge left of the first slab, the right wall by the layers, which run from the slab boundary to the wall $X = w$.

Lemma 7.3 (end lemma). *Assume $g_0 > 1$, $t < 1$ and $m \tan \alpha + \sec \alpha \leq w$. Then the columns $K_0, \dots, K_{I-1}$, the layers Λ_j and the wedge rows Ω_r form a packing in Reg , and*

$$U \leq B^* := w \sin \alpha + w(1 - \cos \alpha) + \Delta(\sec \alpha + m \tan \alpha) + \lfloor \bar{g} \rfloor (1 + \tfrac{1}{2} \tan \alpha) + (1 + \mu \omega) \omega \\ + \lfloor g_0 \rfloor (1 + \tfrac{1}{2} \tan \alpha) + (\delta + 1) \tan \alpha (1 + \mu(\delta + 1) \tan \alpha), \quad \omega := \sec \alpha + \bar{g} \tan \alpha.$$

The seven terms are: the triangles beside the columns; the gaps above the columns (the staircase quantum $1 - \cos \alpha$); the slope of the ceiling across one column; the ends and fractional parts of the layers; the corner at the right wall; the wedge rows; and the corner at the left wall. No small-angle expansion is used.

Proof. (a) *The staircase exists up to the right wall.* For $0 \leq i < I$ we have $0 \leq \xi_i - m \tan \alpha = i \sec \alpha \leq \xi_i \leq w - \sec \alpha$, so $g_0 \leq \Gamma_i \leq \bar{g}$, and Γ_i is non-decreasing in i . H is affine in J with slope $1 - \cos \alpha > 0$ (note $t \geq t^* > 0$), so $J_i = \max\{J \in \mathbb{Z} : H(J) \leq \Gamma_i\}$. Moreover $J_i \geq 0$, because $H(0) = m \cos \alpha + \sin \alpha \leq g_0 \leq \Gamma_i$ by Lemma 7.2 ($t \geq t^*$ and $1 < g_0 < m$); $J_i \leq m - 1$, that is $n_i \geq 1$, because $H(m) = m + \sin \alpha > m > \bar{g} \geq \Gamma_i$; the gap $\Gamma_i - H(J_i)$ lies in $[0, 1 - \cos \alpha)$, because $H(J_i + 1) = H(J_i) + 1 - \cos \alpha > \Gamma_i$; and $J_0 \leq J_1 \leq \dots \leq J_{I-1}$. So every slab up to the right wall receives a column; the columns stop only because the width runs out, and $w - \xi_I < \sec \alpha$.

(b) *Columns lie in Reg.* By Lemma 7.1(iii),(iv) and $J_i + n_i = m$, every point of K_i has $X \geq \xi_i - m \tan \alpha \geq 0$ and $X \leq \xi_i + \sec \alpha \leq w$, and by (ii) and (a), $J_i \leq Y \leq H(J_i) \leq \Gamma_i = g(\xi_i - m \tan \alpha) \leq g(X)$.

(c) *Layers lie in Reg.* For $0 \leq j \leq \lfloor \bar{g} \rfloor$ we have $a_j \leq \xi_{p(j)} \leq \xi_I \leq w$ and $a_j \geq \xi_0 - j \tan \alpha = (m - j) \tan \alpha > 0$. If $j \leq j_1$, then $g(a_j) \geq j + 1$, so every point of Λ_j has $0 \leq a_j \leq X \leq a_j + L_j \leq w$ and $0 \leq Y \leq j + 1 \leq g(a_j) \leq g(X)$. Also $j_1 \leq \lfloor \bar{g} \rfloor - 1$: for $j = \lfloor \bar{g} \rfloor$, $g(a_j) \leq g(w) = \bar{g} < j + 1$.

(d) *Wedge rows lie in Reg.* For $r \leq \lfloor g_0 \rfloor - 1$, $m - r - 1 \geq m - \lfloor g_0 \rfloor > 0$, so $0 \leq l_r \leq (m - 1) \tan \alpha \leq w$, and $Y \leq r + 1 \leq \lfloor g_0 \rfloor \leq g_0 = g(0) \leq g(X)$ on Ω_r .

(e) *Disjointness.* Two closed convex sets on the two sides of a line have disjoint interiors; all separations below are of this kind.

- Columns: for $i < i'$, $K_i \subset \{\lambda \leq \xi_i + \sec \alpha\} = \{\lambda \leq \xi_{i+1}\} \subset \{\lambda \leq \xi_{i'}\}$ and $K_{i'} \subset \{\lambda \geq \xi_{i'}\}$ (Lemma 7.1(i)). No gap is needed between slabs, because the tilt never changes.
- Wedge rows against columns and layers: $\Omega_r \subset \{\lambda \leq l_r + (r + 1) \tan \alpha \leq (m - r - 1) \tan \alpha + (r + 1) \tan \alpha = \xi_0\}$, while the columns lie in $\{\lambda \geq \xi_0\}$ and $\Lambda_j \subset \{\lambda \geq a_j + j \tan \alpha = \xi_{p(j)} \geq \xi_0\}$ (as $X \geq a_j$, $Y \geq j$ and $\tan \alpha > 0$ on Λ_j).
- Layer Λ_j against column K_i : if $i < p(j)$, then $K_i \subset \{\lambda \leq \xi_{i+1}\} \subset \{\lambda \leq \xi_{p(j)}\}$ and $\Lambda_j \subset \{\lambda \geq \xi_{p(j)}\}$. If $i \geq p(j)$, then $J_i \geq j + 1$ (the J_i are non-decreasing, so $\{i : J_i \leq j\} = \{0, \dots, p(j) - 1\}$), hence $K_i \subset \{Y \geq j + 1\}$ and $\Lambda_j \subset \{Y \leq j + 1\}$.
- Layers among themselves, and wedge rows among themselves, lie in different rows $[j, j + 1]$.

(f) *A partition of Reg.* Put

$$\text{Wd} := \text{Reg} \cap \{\lambda < \xi_0\}, \quad \mathcal{S}_i := \text{Reg} \cap \{\xi_i \leq \lambda < \xi_{i+1}\} \quad (0 \leq i < I), \quad \text{Rt} := \text{Reg} \cap \{\lambda \geq \xi_I\},$$

$\mathcal{S}_i^+ := \mathcal{S}_i \cap \{Y \geq J_i\}$, $\mathcal{S}_i^- := \mathcal{S}_i \cap \{Y < J_i\}$ and $\text{Lay} := \bigcup_i \mathcal{S}_i^- \cup \text{Rt}$. Then Wd, the $\mathcal{S}_i^+$ and Lay partition Reg (as $\xi_{i+1} = \xi_i + \sec \alpha$, the slabs tile $[\xi_0, \xi_I]$). The interior of each placed piece lies in one part: $\text{int } K_i \subset \mathcal{S}_i^+$ by Lemma 7.1(i),(ii); $\text{int } \Omega_r \subset \text{Wd}$ by (e); and $\text{int } \Lambda_j \subset \text{Lay}$, because a point of Reg with $\lambda \geq \xi_{p(j)}$ and $Y < j + 1$ lies either in Rt or in some $\mathcal{S}_i$ with $i \geq p(j)$, where $J_i \geq j + 1 > Y$. Boundaries have measure 0 and the pieces have disjoint interiors, so U is the sum over the parts of the area of the part minus the number of unit squares in the pieces assigned to it.

(g) *The column parts.* Fix i and write $J = J_i$, $n = n_i$, $H_i = H(J)$. The parallelogram $\Pi_i := \{\xi_i \leq \lambda \leq \xi_i + \sec \alpha, J \leq Y \leq H_i\}$ lies in Reg: its points have $X \geq \xi_i - H_i \tan \alpha \geq \xi_i - m \tan \alpha \geq 0$ (as $H_i \leq \Gamma_i \leq \bar{g} < m$), $X \leq \xi_i + \sec \alpha \leq w$, and $Y \leq H_i \leq \Gamma_i = g(\xi_i - m \tan \alpha) \leq g(X)$. Its horizontal sections have length $\sec \alpha$, so $\text{area}(\Pi_i) = \sec \alpha(H_i - J) = \sec \alpha(n \cos \alpha + \sin \alpha) = n + \tan \alpha$. The rest of $\mathcal{S}_i^+$ lies in $\{Y > H_i\}$; there $X \leq \lambda < \xi_i + \sec \alpha$, so $Y \leq g(\xi_i + \sec \alpha) = \Gamma_i + \mu(\sec \alpha + m \tan \alpha)$, and the sections still have length at most $\sec \alpha$. Hence, by (a),

$$\begin{aligned} \text{area}(\mathcal{S}_i^+) - n_i &\leq \tan \alpha + \sec \alpha(\Gamma_i + \mu(\sec \alpha + m \tan \alpha) - H_i) \\ &< \tan \alpha + \sec \alpha(1 - \cos \alpha) + \sec \alpha \mu(\sec \alpha + m \tan \alpha). \end{aligned}$$

Summing, with $I \sec \alpha = \xi_I - \xi_0 \leq w$, $I \tan \alpha = I \sec \alpha \sin \alpha \leq w \sin \alpha$ and $w\mu = \Delta$,

$$\sum_i [\text{area}(\mathcal{S}_i^+) - n_i] \leq w \sin \alpha + w(1 - \cos \alpha) + \Delta(\sec \alpha + m \tan \alpha).$$

(h) *The layer part.* For an integer j let $\text{Lay}_j := \text{Lay} \cap \{j \leq Y < j + 1\}$. By the argument at the end of (f), Lay_j contains $\text{Reg} \cap \{j \leq Y < j + 1, \lambda \geq \xi_{p(j)}\}$; conversely, a point of $\mathcal{S}_i^-$ at height $Y \in [j, j + 1)$ has $J_i > Y \geq j$, hence $J_i \geq j + 1$, that is $i \geq p(j)$ and $\lambda \geq \xi_{p(j)}$; and the points of Rt have $\lambda \geq \xi_I \geq \xi_{p(j)}$. So $\text{Lay}_j = \text{Reg} \cap \{j \leq Y < j + 1, \lambda \geq \xi_{p(j)}\}$.

Rows $0 \leq j \leq j_1$. The section of Lay_j at height Y lies in $[\xi_{p(j)} - Y \tan \alpha, w]$, so $\text{area}(\text{Lay}_j) \leq \int_j^{j+1} (w - a_j + (Y - j) \tan \alpha) dY = w - a_j + \frac{1}{2} \tan \alpha$, while Λ_j covers $L_j > w - a_j - 1$ (if $L_j = 0$, nothing is placed and $w - a_j < 1$). So $\text{area}(\text{Lay}_j) - L_j < 1 + \frac{1}{2} \tan \alpha$, and there are $j_1 + 1 \leq \lfloor \bar{g} \rfloor$ such rows by (c).

The rest, $\text{Lay} \cap \{Y \geq j_1 + 1\}$. First, $j_1 \geq J_{I-1} - 1$: if $0 \leq j \leq J_{I-1} - 1$ (note $J_{I-1} - 1 \leq m - 2 = \lfloor \bar{g} \rfloor - 1$), then $p := p(j) \leq I - 1$ and $J_p \geq j + 1$, so $\Gamma_p \geq H(J_p) \geq J_p \geq j + 1$ (as $(m - J) \cos \alpha + \sin \alpha \geq 0$); and $a_j = \xi_p - j \tan \alpha \geq \xi_p - m \tan \alpha$, so $g(a_j) \geq \Gamma_p \geq j + 1$. Hence the points of $\mathcal{S}_i^-$, which have $Y < J_i \leq J_{I-1} \leq j_1 + 1$, are not in $\{Y \geq j_1 + 1\}$, and $\text{Lay} \cap \{Y \geq j_1 + 1\} = \text{Rt} \cap \{Y \geq j_1 + 1\}$. Next, $j_1 + 1 \leq \lfloor \bar{g} \rfloor$ by (c) and $j_1 + 1 \geq J_{I-1}$, so $p(j_1 + 1) = I$ and, by the maximality of j_1 , $g(a_{j_1+1}) < j_1 + 2$, where $a_{j_1+1} = \xi_I - (j_1 + 1) \tan \alpha \geq \xi_I - \bar{g} \tan \alpha > w - \sec \alpha - \bar{g} \tan \alpha = w - \omega$. Hence $\bar{g} - \mu\omega = g(w - \omega) < g(a_{j_1+1}) < j_1 + 2$, that is $\bar{g} - (j_1 + 1) < 1 + \mu\omega$. The sections of Rt at heights $Y \leq \bar{g}$ lie in $[\xi_I - Y \tan \alpha, w]$ and have length less than $\sec \alpha + \bar{g} \tan \alpha = \omega$, and Rt lies below $Y = \bar{g}$. So $\text{area}(\text{Rt} \cap \{Y \geq j_1 + 1\}) < \omega(1 + \mu\omega)$. In total,

$$\text{area}(\text{Lay}) - \sum_j L_j < \lfloor \bar{g} \rfloor (1 + \frac{1}{2} \tan \alpha) + (1 + \mu\omega) \omega.$$

(i) *The wedge part.* For $0 \leq r \leq \lfloor g_0 \rfloor - 1$, the section of Wd at height $Y \in [r, r + 1]$ lies in $[0, \xi_0 - Y \tan \alpha]$ (note $\xi_0 - Y \tan \alpha = (m - Y) \tan \alpha > 0$), so $\text{area}(\text{Wd} \cap \{r \leq Y < r + 1\}) \leq \xi_0 - (r + \frac{1}{2}) \tan \alpha$, while Ω_r covers $l_r > (m - r - 1) \tan \alpha - 1 = \xi_0 - (r + 1) \tan \alpha - 1$ (or nothing, when $(m - r - 1) \tan \alpha < 1$): the waste is less than $1 + \frac{1}{2} \tan \alpha$ per row, in $\lfloor g_0 \rfloor$ rows. The points of Wd with $Y \geq \lfloor g_0 \rfloor$ have $0 \leq X < \xi_0 - Y \tan \alpha \leq (m - \lfloor g_0 \rfloor) \tan \alpha < (\delta + 1) \tan \alpha$ and $Y \leq g(X) \leq g_0 + \mu(\delta + 1) \tan \alpha < \lfloor g_0 \rfloor + 1 + \mu(\delta + 1) \tan \alpha$, so their area is less than $(\delta + 1) \tan \alpha (1 + \mu(\delta + 1) \tan \alpha)$.

Adding (g), (h) and (i) gives $U \leq B^*$. $\square$

The bound holds for every y with $g_0 > 1$: no tuning of y , m or α is needed, and by (a) every slab up to the right wall gets a column.

7.2. The band ends.

Corollary 7.4. *Let $b \geq 10^4$ be an integer, $x := b^{-1/3}$, $\kappa := 3/4$, $\tan(\theta/2) = b/(b^2 - 1)$, and w, γ, σ as in Section 6. Let y be any real number with*

$$\kappa b^{2/3} \leq y \leq \kappa b^{2/3} + 2.$$

Apply the construction of Section 7.1 with $\sigma_0 = \sigma$ and $\mu = \tan \theta$ (so that $\text{Reg} = \text{Reg}(y)$ of Section 6), and with any integer $Q \geq 16b$. Then the construction gives a packing in $\text{Reg}(y)$ with

$$U \leq \Phi(x) b^{2/3} \leq \Phi(10^{-4/3}) b^{2/3} \leq 5.03 b^{2/3},$$

where Φ , defined in the proof, is non-decreasing on $(0, 10^{-4/3}]$, $\Phi(10^{-4/3}) \leq 5.0163$ and $\Phi(x) \rightarrow \sqrt{A/\kappa} + 3/2 \approx 4.3284$ as $x \rightarrow 0$, with $A := 6.000002$. Consequently the end cost of Proposition 6.3 satisfies $E(y) = U + \frac{1}{2} \tan \theta \leq 5.03 b^{2/3} + 1.0001/b$.

Proof. (i) *Band facts.* Write $t_\theta := b/(b^2 - 1)$ and $u := b^{1/3} = 1/x$. Then $t_\theta \leq 1.00000002/b$ (since $bt_\theta = b^2/(b^2 - 1)$), $\sigma < 2t_\theta \leq 2.0000001/b$ and $\mu = 2t_\theta/(1 - t_\theta^2) \leq 2.000001/b$; so $\Delta = w\mu < b\mu \leq 2.000001$ and $\sigma\mu \leq 4.1 \cdot 10^{-8}$. Also $w < b$ by Lemma 3.2, as $t_\theta > 1/b$, and $w \geq b\gamma = b(1 - 2t_\theta^2/(1 + t_\theta^2)) \geq b - 2bt_\theta^2 \geq b - 2.1/b$.

(ii) *Ranges.* $g_0 = y - \sigma\mu \in [\kappa u^2 - 4.1 \cdot 10^{-8}, \kappa u^2 + 2]$. Put $\nu := \kappa u^2 - 5 \cdot 10^{-8} = p(x)u^2$ with $p(x) := \kappa - 5 \cdot 10^{-8}x^2$; then $g_0 \geq \nu > 1$. Next $\bar{g} = y + (w - \sigma)\mu \leq y + \Delta \leq \kappa u^2 + 4.000001$, $m = \lfloor \bar{g} \rfloor + 1 \leq \kappa u^2 + 5.000001$, $\lfloor g_0 \rfloor \leq \kappa u^2 + 2$, and $\delta = m - g_0 \in (\Delta, \Delta + 1]$, so $\delta \leq \bar{\delta} := 3.000001$ and $\delta + 1 \leq 4.000001$.

(iii) *Tilt.* By Lemma 7.2 with $g_0 \geq \nu$, $\delta \leq \bar{\delta}$ and $A = 2\bar{\delta}$,

$$ut^* \leq u \frac{1 + \sqrt{1 + 2\nu\bar{\delta}}}{2\nu} = \frac{x + \sqrt{x^2 + Ap(x)}}{2p(x)} \leq \frac{x + \sqrt{x^2 + A\kappa}}{2p(x)},$$

and $u/Q \leq u/(16u^3) = x^2/16$. So $t \leq t^* + 1/Q \leq x\vartheta(x) =: \bar{t}(x)$ with

$$\vartheta(x) := \frac{x + \sqrt{x^2 + A\kappa}}{2p(x)} + \frac{x^2}{16}.$$

Put $\bar{\tau} := 2\bar{t}/(1 - \bar{t}^2)$ and $\bar{\varsigma} := (1 + \bar{t}^2)/(1 - \bar{t}^2)$. Then $\tan \alpha \leq \bar{\tau}$, $\sec \alpha \leq \bar{\varsigma}$, $\sin \alpha \leq 2\bar{t} \leq 2\bar{t}$ and $1 - \cos \alpha \leq 2\bar{t}^2 \leq 2\bar{t}^2$. At $x_0 := 10^{-4/3}$ (that is, $b = 10^4$), $\bar{t} \leq 0.0672$ and $\bar{\tau} \leq 0.1349$.

(iv) *The hypotheses of Lemma 7.3.* $g_0 > 1$ by (ii); $t \leq \bar{t}(x) \leq \bar{t}(x_0) \leq 0.0672 < 1$ by the monotonicity in (v); and $m \tan \alpha + \sec \alpha \leq (0.75 b^{2/3} + 5.000001) \cdot 0.1349 + 1.01 < b - 2.1/b \leq w$, since $b^{2/3} \leq b/21.5$ for $b \geq 10^4$. Lemma 7.3 gives the packing and $U \leq B^*$. We divide each term of B^* by $u^2 = b^{2/3}$, using $w < b = u^3$, $\Delta \leq 2.000001$, $\mu \leq 2.000001 x^3$ and $\omega = \sec \alpha + \bar{g} \tan \alpha$:

$$\begin{aligned} T_1 &= w \sin \alpha / u^2 \leq 2ut \leq 2\vartheta(x), & T_2 &= w(1 - \cos \alpha) / u^2 \leq 2ut^2 \leq 2x\vartheta(x)^2, \\ T_3 &= \Delta(\sec \alpha + m \tan \alpha) / u^2 \leq 2.000001[x^2\bar{\varsigma} + (\kappa + 5.000001 x^2)\bar{\tau}], \\ T_4 &= [\bar{g}](1 + \tfrac{1}{2} \tan \alpha) / u^2 \leq (\kappa + 4.000001 x^2)(1 + \tfrac{1}{2}\bar{\tau}), \\ T_5 &= (1 + \mu\omega) \omega / u^2 \leq (1 + 2.000001 x \bar{\omega}) \bar{\omega}, & \bar{\omega} &:= x^2\bar{\varsigma} + (\kappa + 4.000001 x^2)\bar{\tau}, \\ T_6 &= [g_0](1 + \tfrac{1}{2} \tan \alpha) / u^2 \leq (\kappa + 2x^2)(1 + \tfrac{1}{2}\bar{\tau}), \\ T_7 &= (\delta + 1) \tan \alpha (1 + \mu(\delta + 1) \tan \alpha) / u^2 \leq 4.000001 \bar{\tau} x^2 (1 + 2.000001 \cdot 4.000001 x^3 \bar{\tau}). \end{aligned}$$

For T_5 we used $\omega/u^2 \leq \bar{\omega}$ and $\mu\omega \leq 2.000001 x^3 u^2 \bar{\omega} = 2.000001 x \bar{\omega}$. Let $\Phi(x)$ be the sum of the seven right-hand sides.

(v) *Monotonicity.* On $0 < x \leq x_0$, $p(x)$ is positive and decreasing and $x + \sqrt{x^2 + A\kappa}$ is increasing, so ϑ is increasing; $\bar{t} = x\vartheta$ is increasing and less than 1, so $\bar{\tau}$ and $\bar{\varsigma}$ are increasing. Every right-hand side in (iv) is a sum of products of non-negative non-decreasing functions of x . So Φ is non-decreasing in x , that is non-increasing in b , and $\Phi(x) \leq \Phi(x_0)$ for all $b \geq 10^4$.

(vi) *Values.* In interval arithmetic, $\Phi(x_0) \leq 5.01625$ (the terms are about 2.89126, 0.19400, 0.20946, 0.80975, 0.10545, 0.80515 and 0.00116). As $x \rightarrow 0$, $T_1 \rightarrow \sqrt{A\kappa}/\kappa = \sqrt{A/\kappa}$, $T_4, T_6 \rightarrow \kappa$ and the other terms tend to 0, so $\Phi \rightarrow \sqrt{A/\kappa} + 3/2$. Finally $A_0(y)$ exceeds the area of $\text{Reg}(y)$ by $\frac{1}{2} \tan \theta$ (the triangle with vertices $(0, y - \sigma\mu)$, (σ, y) , $(0, y + \gamma)$), and $\tan \theta \leq 2.000001/b$. $\square$

The values in (iii) and (vi) were certified in interval arithmetic (`mpmath.iv`, 40 digits, with x_0 enclosed in an interval) by the author's program and by three independent computations (two reviews and an independent reimplementations), which all give $\Phi(x_0) \in [5.0162487, 5.0162488]$; one of them also bounds Φ on all of $(0, x_0]$ by a subdivision into 4000 intervals, without using (v), and finds the maximum 5.0163 at x_0 . For larger b the constant improves: $\Phi(10^{-5/3}) \leq 4.6351$, $\Phi(10^{-2}) \leq 4.4685$, $\Phi(10^{-8/3}) \leq 4.3583$. The choice $\kappa = 3/4$ is close to the minimizer $(\sqrt{6}/4)^{2/3} \approx 0.721$ of the limit $\sqrt{6/\kappa} + 2\kappa$. The window has length $2 > \sec \theta$, which is what Section 7.3 needs: the bound holds for every y in the window, not only at sampled points.

7.3. Proof of Theorem 1.2. Let $C_E := 5.03$, $k \geq k_0 = 1.6 \cdot 10^7$, $b_* := (3k/(2C_E))^{3/5}$ (so $k = \frac{2}{3}C_E b_*^{5/3}$) and $b := \lceil b_* \rceil$. Since $k_0 \geq \frac{2}{3}C_E \cdot 10^{20/3} \approx 1.5565 \cdot 10^7$, we have $b \geq b_* \geq 10^4$; since $3/(2C_E) < 1$, $b \leq k^{3/5} + 1$. Take $\tan(\theta/2) = b/(b^2 - 1) > 1/b$; then $w < b$ and $S := k - b + w < k$. Use the packing of Corollary 6.2 with

$$y_0 := \tfrac{1}{4} \lceil 3b^{2/3} \rceil \in [\kappa b^{2/3}, \kappa b^{2/3} + \tfrac{1}{4}]$$

and, in each band, $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$. Here $h = b\sigma + \gamma < 3$ (as $b\sigma = 2b^2(b^2 - 1)/((b^2 - 1)^2 + b^2) < 2$) and $y_0 \leq 0.75(k^{3/5} + 1)^{2/3} + \frac{1}{4} \leq 0.75 k^{2/5} + 1$, so $h + 2y_0 < 1.5 k^{2/5} + 5 < k - b \leq L_b$; hence $R \geq 1$ and $y_1 = L_b - h - (R - 1)d - y_0 \in [y_0, y_0 + d)$. As $d < 1.0001$, both y_0 and y_1 lie in the window of Corollary 7.4, whatever value y_1 takes. Fill the four end regions $\text{Reg}(y_0)$, $\text{Reg}(y_1)$ of the two bands by Corollary 7.4 with Q the least power of 2 that is at least $16b$. By Corollary 6.2 this is a packing in $[0, S]^2$, and by Proposition 6.3

$$k^2 - N = (k^2 - S^2) + \sum_{2 \text{ bands}} [(R - 1) \tan \theta + E(y_0) + E(y_1)].$$

- $(R-1)\tan\theta = (R-1)d\sin\theta = (L_b - h - y_0 - y_1)\sin\theta \leq L_b\sigma < 2L_bt_\theta$, so the two bands give less than $(S+k-b)2t_\theta < 4kb/(b^2-1) = 4k/b + 4k/(b(b^2-1))$.
- $b-w = 2t_\theta(bt_\theta-1)/(1+t_\theta^2) = 2t_\theta/((b^2-1)(1+t_\theta^2)) < 2b/(b^2-1)^2$, so $k^2 - S^2 < 2k(b-w) < 4kb/(b^2-1)^2$.
- $\sum E \leq 4(C_E b^{2/3} + \frac{1}{2}\tan\theta)$ by Corollary 7.4.

Hence $k^2 - N < 4k/b + 4C_E b^{2/3} + \lambda'$ with $\lambda' := 4k/(b(b^2-1)) + 4kb/(b^2-1)^2 + 2\tan\theta \leq 8.001k/b^3 + 4.1/b$. Since $k \leq \frac{2}{3}C_E b^{5/3}$, $8.001k/b^3 \leq 8.001 \cdot \frac{2}{3}C_E b^{-4/3} \leq 1.25 \cdot 10^{-4}$, and $4.1/b \leq 4.1 \cdot 10^{-4}$; so $\lambda' < 1$, and Lemma 2.2 gives $c^*(k) \leq k^2 - N - 1 < 4k/b + 4C_E b^{2/3}$. Finally $4k/b \leq 4k/b_* = \frac{8}{3}C_E b_*^{2/3}$ and $b^{2/3} \leq (b_* + 1)^{2/3} \leq b_*^{2/3} + \frac{2}{3}b_*^{-1/3} \leq b_*^{2/3} + \frac{2}{3} \cdot 10^{-4/3}$, so

$$c^*(k) < \frac{20}{3}C_E b_*^{2/3} + \frac{8}{3}C_E \cdot 10^{-4/3} = \frac{20}{3}\left(\frac{3}{2}\right)^{2/5} C_E^{3/5} k^{2/5} + \frac{8}{3}C_E \cdot 10^{-4/3}.$$

In interval arithmetic,

$$\frac{20}{3}\left(\frac{3}{2}\right)^{2/5} 5.03^{3/5} \in [20.66740, 20.66741] \quad \text{and} \quad \frac{8}{3} \cdot 5.03 \cdot 10^{-4/3} \in [0.62259, 0.62260],$$

which gives Theorem 1.2. For every $C > \sqrt{A/\kappa} + 3/2 \approx 4.3284$ we have $\Phi(b^{-1/3}) \leq C$ for all large b , and the same argument with C in place of C_E gives $c^*(k) \leq (\frac{20}{3}(\frac{3}{2})^{2/5} C^{3/5} + o(1))k^{2/5}$; letting C tend to the limit gives $c^*(k) \leq (18.89 + o(1))k^{2/5}$. $\square$

Remark 7.5 (comparison and exact checks). The construction of Section 7.1 was implemented in exact rational arithmetic, with Q the least power of 2 that is at least $16b$. For $b = 400, 1600, 6400, 25600, 102400$ and five lifts each in the window of Corollary 7.4, all 25 end packings were checked by an exact checker that does not use Lemma 7.3 (shapes, containment, separating axis tests for all pairs of pieces with overlapping bounding boxes; for $b = 102400$ also by a check of the separating lines of the proof); all pass, $U \leq B^*$ holds in each, and $U/b^{2/3}$ stays between 3.26 and 4.36. Deliberately corrupted versions (a column shifted by 2^{-30} , a layer lengthened, a column given one more square, and so on) were rejected. The full packings of Section 7.3, counted exactly (the four ends checked by exact containment and by the separating lines of the proof; the rows not checked, their correctness rests on Lemmas 3.3 and 6.1; the identity of Proposition 6.3 checked exactly), give:

k	b	y_0	$c^*(k) \leq$	$20.668 k^{2/5} + 0.623$
10^5	484	185/4	1794	$(k < k_0)$
10^6	1927	465/4	4467	$(k < k_0)$
10^7	7669	1167/4	10579	$(k < k_0)$
$1.6 \cdot 10^7$	10167	352	12707	15738.5
$3.825 \cdot 10^7$	17152	499	17865	22303.0
10^8	30530	2931/4	26998	32757.2

For $k < k_0$ these packings are still valid (Lemma 7.3 does not need $b \geq 10^4$); only the closed form of Theorem 1.2 is restricted to $k \geq k_0$. An independent reimplementaion, written from the text of the proof only (with its own admissible choices of Q), checked the end packings exactly for b up to $4 \cdot 10^5$ and many lifts in the window, and the full packings for $k = 1.6 \cdot 10^7, 3.825 \cdot 10^7, 10^8, 10^9$ and $3 \cdot 10^9$: no overlaps, no containment errors, $U \leq B^*$ in every case, and $c^*(10^9) \leq 67065$, $c^*(3 \cdot 10^9) \leq 99640$ (against 82281.4 and 127687.7 from Theorem 1.2); 8 of 8 deliberately corrupted versions were rejected. It relies on Lemmas 3.3 and 6.1 for the rows. The tuned packings of Theorem 1.6 are 4.9–11.8% better than these (Section 9).

8. CHAINED WALL FILLERS AND THE EXPONENT 3/8

In the end lemma of Section 7 the columns cost about $b/\sqrt{m}$ and the two walls about m each. In this section the walls are filled more efficiently: each wall is a thin trapezoid, and it is filled by a

chain of staircases (*blocks*) whose tilt is lowered slightly from block to block, so that the horizontal layers opened in one block close just before the next block. The wall cost becomes $O(m^{3/4})$, the best column length is $m \asymp b^{4/5}$, the end cost is $O(b^{3/5})$, and Proposition 6.3 gives Theorem 1.1. The ideas (horizontal layers under tilted columns, and filling the wall trapezoids by stairs whose tilt is lowered from block to block) are those of Bui [Bui, Sections 3–4]; the construction and the proof below are self-contained and do not use any argument of [Bui].

The construction has six real parameters: the threshold b_0 , the window coefficient c_y of the lifts, the wall widths c_D , c_R , the wall tilt factor a_w and the proof parameter ϖ of Lemma 8.1. Version 1.0 of this paper used the single choice $(10^{16}, 1/12, 4, 4, 3/2, 0.65)$. Version 1.1 keeps all lemmas of version 1.0 and needs no new lemma for the parameters; it states them with the parameters, adds two short arguments that version 1.0 did not need (the error of the wall tilts in (E6) for all tilts up to $1/10$, and the room condition in Section 8.7), and certifies the constants for each row of Table 3 (Section 8.5); this gives the tiers of Theorem 1.1. Section 8.2 adds a variant ZC' of the wall filler and two lemmas (Lemmas 8.3 and 8.4) with a smaller bound; only the tiers marked $\dagger$ use them. Sections 8.3 and 8.6 add a sawtooth accounting of two groups of wall rows (Lemmas 8.6–8.8) and a choice of the lift of each band (Lemma 8.11); only the tiers marked $\ddagger$ use them.

Throughout this section we use the tilt notation of Section 7.1, the columns $K(\xi, J, n)$ and Lemmas 7.1 and 7.2, and we put

$$q := 1 - \cos \alpha = \frac{\tan \alpha^2}{\sec \alpha (\sec \alpha + 1)} \leq \frac{\tan \alpha^2}{2}, \quad \text{so that} \quad \sec \alpha - 1 = q \sec \alpha, \quad \sec \alpha \sin \alpha = \tan \alpha.$$

q is increasing in t . On $t \in (0, 1/10]$ we have $\tan \alpha \leq 2.1t$, and $t \mapsto \tan \alpha$ has derivative $2(1+t^2)/(1-t^2)^2 \leq 2.1$, so neighbouring points of a grid $(1/Q^*)\mathbb{Z}$ differ in $\tan \alpha$ by at most $\delta_Q := 2.1/Q^*$. A tilt with index 0 (as t_0, α_0) is the *initial tilt* of a wall. Two closed convex sets that lie in the two closed half-planes of one line have disjoint interiors; every disjointness statement below is of this kind.

Grids. Q is the grid of the main tilt: any power of 2 with $Q \geq 16b$. The wall tilts and the cut tilt use the grid $Q^* := \max(2^{40}, Q)$, and the phases the grid $Q_f := \max(2^{48}, Q)$.

8.1. The wall filler. For reals $\chi \geq 4$, $\zeta > 0$ and $L \geq 0$ let $r(x) := \chi + \zeta x$ and

$$Z = Z(\chi, \zeta, L) := \{(x, y) : 0 \leq x \leq L, 0 \leq y \leq r(x)\},$$

a right trapezoid of area $L\chi + \zeta L^2/2$. The input is a grid tilt $t_0 \in (0, 1/10] \cap (1/Q^*)\mathbb{Z}$; put $N_0 := \lfloor \chi \rfloor - 1 \geq 3$. For a block β with tilt t_β we write $\lambda_\beta(x, y) := x + y \tan \alpha_\beta$.

Construction ZC.

(Z1) *Block 0:* tilt t_0 , base $F_0 := \lfloor \chi - N_0 \cos \alpha_0 - \sin \alpha_0 \rfloor \geq 0$, total length $M_0 := F_0 + N_0$, offset $s_0 := M_0 \tan \alpha_0$.

(Z2) *Block β* has tilt t_β (with $\cos \alpha, \sin \alpha, \tan \alpha, \sec \alpha, q$), offset s_β , base F_β and total length $M_\beta = F_\beta + N_0$. If $s_\beta + \sec \alpha > L$ the chain stops (block β does not exist). Otherwise, for $i = 0, 1, 2, \dots$ let $\xi_i := s_\beta + i \sec \alpha$; stop (reason ‘ L ’) if $\xi_i + \sec \alpha > L$; else let $\Gamma_i := r(\xi_i - M_\beta \tan \alpha)$ and $\tilde{J}_i := \lfloor (\Gamma_i - M_\beta \cos \alpha - \sin \alpha)/q \rfloor$; stop (reason ‘saturation’) if $\tilde{J}_i > M_\beta - 1$; else place the column $K(\xi_i, J_i, n_i)$ with $J_i := \tilde{J}_i$, $n_i := M_\beta - J_i$. Let P_β be the number of columns and $\hat{\xi}_\beta := s_\beta + P_\beta \sec \alpha$.

(Z3) *Next block.* If block β stopped by saturation (it is *complete*), let $\kappa_\beta := q^2/\zeta$, let $t_{\beta+1}$ be the largest grid point $t' \leq t_\beta$ with $\tan \alpha' \leq \tan \alpha - \kappa_\beta$, and put $\hat{\kappa}_\beta := \tan \alpha - \tan \alpha_{\beta+1} \geq \kappa_\beta$ and $\eta_\beta := \hat{\kappa}_\beta - \kappa_\beta \in [0, \delta_Q)$. *Phase:* let $J_{l_0} := J_0$ and $J_L := J_{P_\beta-1}$ (indices of block β). If $J_L > J_{l_0}$, put $A := (\Gamma_0 - M_\beta \cos \alpha - \sin \alpha)/q$, $B := q/(\zeta \sec \alpha)$, $p := \#\{i : J_i \leq J_{l_0}\}$, $f := p - B(J_{l_0} + 1 - A)$,

$$\psi := \hat{\xi}_\beta - \tan \alpha_{\beta+1} - (s_\beta + p \sec \alpha) + J_{l_0} \hat{\kappa}_\beta, \quad \omega_0 := \text{frac}(-(\psi + f(\sec \alpha - 1) - (\sec \alpha - 1))),$$

and $\omega := \lceil \omega_0 Q_f \rceil / Q_f \in [0, 1]$; otherwise $\omega := 0$. Then $s_{\beta+1} := \hat{\xi}_\beta + \omega$ and, with primes for the tilt $t_{\beta+1}$,

$$\begin{aligned} F_{\beta+1} &:= \max\{J \in \mathbb{Z} : J + N_0 \cos \alpha' + \sin \alpha' \leq r(s_{\beta+1} - (J + N_0) \tan \alpha')\} \\ &= \left\lfloor \frac{\chi + \zeta s_{\beta+1} - N_0(\cos \alpha' + \zeta \tan \alpha') - \sin \alpha'}{1 + \zeta \tan \alpha'} \right\rfloor, \end{aligned}$$

$$M_{\beta+1} := F_{\beta+1} + N_0.$$

- (Z4) *Rows.* Block β blocks row j if $J_{\beta,0} \leq j$ ($J_{\beta,i}$ is J_i of block β). For each integer $0 \leq j < \lfloor r(L) \rfloor$, walk from $x = 0$ to $x = L$: a *stretch* starts at $x = 0$ or just after a blocking block β , at the abscissa $\xi_{\beta,p} - j \tan \alpha_\beta$ with $p := \#\{i : J_{\beta,i} \leq j\}$ ($p = P_\beta$ allowed, $\xi_{\beta,P_\beta} = \hat{\xi}_\beta$), and ends at the next blocking block β_2 , at the abscissa $s_{\beta_2} - (j+1) \tan \alpha_{\beta_2}$, or at $x = L$. For a stretch from a to a^+ with $a^+ - a \geq 1$, $a \geq 0$ and $j+1 \leq r(a)$, place the row $[a, a + \lfloor a^+ - a \rfloor] \times [j, j+1]$.

Each block is one staircase of Section 7.1 (one tilt, total length M_β , J non-decreasing); a block ends when its staircase is used up, and the next block starts after a phase gap $\omega < 1$ with the slightly smaller tilt, chosen so that every layer opened in the block ends just before the next block with a small residual (see (R) below).

Parameters. Let $\varpi \in (0, 1)$ (a free proof parameter; the tiers of Section 8.5 use $\varpi = 0.65$ or $\varpi = 0.681818$), let $q_{\min}$ be the value of q at the slope $\varpi \tan \alpha_0$, that is $q_{\min} := (\varpi \tan \alpha_0)^2 / (s_m(s_m + 1))$ with $s_m := \sqrt{1 + \varpi^2 \tan^2 \alpha_0}$, and put $M_{\max} := r(L) + N_0 q_0$ (q_0 is q at t_0),

$$\Lambda := \frac{q_{\min} N_0 - 1 - \zeta \tan \alpha_0}{\zeta}, \quad \bar{K} := \frac{L}{\Lambda}, \quad \varepsilon_W := (\sec \alpha_0 - 1) + N_0 \delta_Q + \frac{1}{Q_f}.$$

Hypotheses.

- (W1) $\Lambda \geq N_0 \tan \alpha_0$ (in particular $\Lambda > 0$);
- (W2) $\Delta_{\text{dr}} := \frac{q_0}{N_0} \left(L + \bar{K} \frac{1 + \zeta \tan \alpha_0}{\zeta} \right) + \bar{K} \delta_Q < (1 - \varpi) \tan \alpha_0$;
- (W3) $N_0 \cos \alpha_0 \geq \frac{\zeta \sec \alpha_0}{q_{\min}} + \tan \alpha_0 + \zeta(2 + \sec \alpha_0 + M_{\max} \tan \alpha_0) + \frac{1 + \zeta \tan \alpha_0}{q_{\min}}$;
- (W4) $\varepsilon_W < 1$;
- (W5) $\delta_Q \leq \varpi \tan \alpha_0$ and $t_0 \leq 1/10$.

Lemma 8.1 (wall filler). *Under (W1)–(W5) the construction ZC does not fail, its pieces form a packing in Z , and the uncovered area $U_Z := \text{area}(Z) - \# \text{squares}$ satisfies*

$$\begin{aligned} U_Z \leq B_W &:= L c_{\text{col}} + \bar{K} \left[(N_0 - 1)(\varepsilon_W + \tan \alpha_0) + \frac{1 + \zeta \tan \alpha_0}{q_{\min}} (1 + \tan \alpha_0) + A_T \right] \\ &\quad + \lfloor \chi \rfloor (1 + \tfrac{1}{2} \tan \alpha_0) + w_0 (1 + \zeta w_0) + r(L) (1 + \tfrac{1}{2} \tan \alpha_0) + w_e (4 + 2\zeta w_e), \end{aligned}$$

where $c_{\text{col}} := \tan \alpha_0 + \sec \alpha_0 q_0 + \sec \alpha_0 \zeta (\sec \alpha_0 + N_0 q_0 \tan \alpha_0)$, $A_T := h_T w_T$,

$$h_T := 1 + \frac{\zeta \sec \alpha_0}{q_{\min}} + \tan \alpha_0 + \zeta(1 + \sec \alpha_0 + M_{\max} \tan \alpha_0), \quad w_T := 1 + (r(L) + 1) \left(\frac{q_0^2}{\zeta} + \delta_Q \right),$$

$w_0 := (1 + N_0 q_0) \tan \alpha_0$ and $w_e := 1 + \sec \alpha_0 + (r(L) + N_0 q_0) \tan \alpha_0$.

The terms are: column triangles, gaps and ceiling slope; aligned closings; unaligned closings; transition slivers; the left end; the right end.

Proof. (a) *One block.* Fix an existing block β with tilt t ($\cos \alpha, \sin \alpha, \tan \alpha, \sec \alpha, q$), offset s , base F and total $M = F + N_0$, and assume the invariants

- (I1) $\tan \alpha \geq \varpi \tan \alpha_0$ (hence $q \geq q_{\min}$);
- (I2) $s \geq M \tan \alpha$;
- (I3) $F \geq 0$ and $F + N_0 \cos \alpha + \sin \alpha \leq r(s - M \tan \alpha) < F + 1 + N_0 \cos \alpha + \sin \alpha + \zeta \tan \alpha$.

(I3) holds for $\beta = 0$ ($\Gamma_0 = r(0) = \chi$ and the definition of F_0) and for $\beta + 1$ by the definition of $F_{\beta+1}$ as a maximum ($J = F_{\beta+1} + 1$ violates the defining inequality, and $r(s' - (F' + 1 + N_0) \tan \alpha') = r(s' - M' \tan \alpha') - \zeta \tan \alpha')$. $\Gamma_i = r(s - M \tan \alpha) + i\zeta \sec \alpha$ is affine in i , so $\tilde{J}_i = \lfloor A + i/B \rfloor$, and the J_i are non-decreasing. $H(J) := J + (M - J) \cos \alpha + \sin \alpha$ is affine in J with slope q , so $J_i = \max\{J : H(J) \leq \Gamma_i\}$ and the gap $\Gamma_i - H(J_i)$ lies in $[0, q]$. $J_0 \geq F$ because $H(F) = F + N_0 \cos \alpha + \sin \alpha \leq \Gamma_0$ by (I3); so $J_i \geq F$, $n_i \leq N_0$, and $n_i \geq 1$ (no saturation). By (I3), $A < F + (1 + \zeta \tan \alpha)/q$, so $J_0 < F + (1 + \zeta \tan \alpha)/q$. The block is not empty: it exists, so $s + \sec \alpha \leq L$, and $J_0 < F + (1 + \zeta \tan \alpha_0)/q_{\min} < F + N_0$ by (W1); so $\tilde{J}_0 \leq M - 1$ and $P \geq 1$. *Containment of columns* (Lemma 7.1, $J_i + n_i = M$): $X \geq \xi_i - M \tan \alpha \geq s - M \tan \alpha \geq 0$ by (I2), $X \leq \xi_i + \sec \alpha \leq L$, and $J_i \leq Y \leq H(J_i) \leq \Gamma_i = r(\xi_i - M \tan \alpha) \leq r(X)$. *Length of a complete block*. Saturation at $i = P$ means $A + P/B \geq M$, so $P > B(N_0 - (1 + \zeta \tan \alpha)/q)$ and

$$(8.1) \quad \Lambda_\beta := P \sec \alpha > \frac{qN_0 - 1 - \zeta \tan \alpha}{\zeta} \geq \Lambda, \quad \text{equivalently} \quad q \leq \frac{\zeta \Lambda_\beta + 1 + \zeta \tan \alpha}{N_0}$$

(the middle expression is increasing in q and decreasing in $\tan \alpha$; use $q \geq q_{\min}$, $\tan \alpha \leq \tan \alpha_0$). *Last column of a complete block*: $J_L := J_{P-1} \geq \tilde{J}_P - 1/B - 1 \geq M - 1/B - 1$, so $n_L := M - J_L \leq 1 + 1/B = 1 + \zeta \sec \alpha/q$.

(b) *Passing to block $\beta + 1$* (β complete; primes refer to $t_{\beta+1}$). *Tilt drift*. Let C_β be the sum of $\kappa_\gamma + \delta_Q$ over the complete blocks $\gamma \leq \beta$, and suppose (I1) for the blocks $0, \dots, \beta$. By (8.1) these blocks have total length at least $(\beta + 1)\Lambda$ inside $[0, L]$, so $\beta + 1 \leq \bar{K}$; and by (8.1), $\kappa_\gamma = q_\gamma^2/\zeta \leq (q_0/N_0)(\Lambda_\gamma + (1 + \zeta \tan \alpha_0)/\zeta)$. Summing ($\sum \Lambda_\gamma \leq L$) gives $C_\beta \leq \Delta_{\text{dr}} < (1 - \varpi) \tan \alpha_0$ by (W2). By induction $\tan \alpha_\beta \geq \tan \alpha_0 - C_{\beta-1}$, so $\tan \alpha - \kappa_\beta \geq \tan \alpha_0 - C_\beta + \delta_Q > \varpi \tan \alpha_0 + \delta_Q \geq \tan \alpha(1/Q^*)$ by (W5): a positive grid point t' exists, and $\tan \alpha' > \tan \alpha - \kappa_\beta - \delta_Q \geq \tan \alpha_0 - C_\beta > \varpi \tan \alpha_0$. So (I1) holds for $\beta + 1$, and $t' \leq t$, $\cos \alpha' \geq \cos \alpha$. $F' \geq F$. $J = F$ is admissible in the definition of F' : $r(s' - M \tan \alpha') \geq r(\hat{\xi} - M \tan \alpha) = \Gamma_P$, saturation gives $\Gamma_P \geq M \cos \alpha + \sin \alpha + qM = M + \sin \alpha$, and $F + N_0 \cos \alpha' + \sin \alpha' \leq M + \sin \alpha' \leq M + \sin \alpha$. $J'_0 \leq J_L$. By maximality of J_L , $\Gamma_{P-1} < H(J_L + 1) \leq J_L + n_L + \sin \alpha$. As $s' \leq \hat{\xi} + 1 = \xi_{P-1} + \sec \alpha + 1$, $r(s') \leq \Gamma_{P-1} + \zeta(1 + \sec \alpha + M \tan \alpha)$. By (I3) for $\beta + 1$, $F' + N_0 \cos \alpha' + \sin \alpha' \leq r(s' - M' \tan \alpha') \leq r(s')$, so $F' < J_L + n_L + \sin \alpha - N_0 \cos \alpha' + \zeta(1 + \sec \alpha + M \tan \alpha)$; and by (a) for block $\beta + 1$, $J'_0 < F' + (1 + \zeta \tan \alpha')/q'$. With $n_L \leq 1 + \zeta \sec \alpha/q$, $M \leq M_{\max}$ (from $F + N_0 \cos \alpha + \sin \alpha \leq r(L)$), $\cos \alpha' \geq \cos \alpha_0$, $q, q' \geq q_{\min}$, $\sec \alpha \leq \sec \alpha_0$, $\sin \alpha \leq \tan \alpha_0$, hypothesis (W3) gives $J'_0 < J_L + 1$. (I2) for $\beta + 1$: $M' = F' + N_0 \leq J'_0 + N_0 \leq J_L + N_0 \leq M - 1 + N_0$, so $M' \tan \alpha' \leq M \tan \alpha + N_0 \tan \alpha_0 \leq s + \Lambda_\beta = \hat{\xi} \leq s'$ by (I2), (8.1) and (W1). *Aligned closings*. For $J_{l_0} \leq j < J_L$ put $p(j) := \#\{i : J_i \leq j\}$; then $1 \leq p(j) \leq P - 1$ and, since $j \geq \lfloor A \rfloor = J_{l_0}$, $p(j) = \lceil B(j+1-A) \rceil = B(j+1-A) + f_j$ with $f_j \in [0, 1)$. The stretch of row j from $a_j = \xi_{p(j)} - j \tan \alpha$ to $s' - (j+1) \tan \alpha'$ has length $\ell_j = s' - \tan \alpha' - s - p(j) \sec \alpha + j\hat{\kappa}$. Modulo 1 (as $p(j)$ is an integer and $\sec \alpha = 1 + (\sec \alpha - 1)$, and $B(\sec \alpha - 1) = q(\sec \alpha - 1)/(\zeta \sec \alpha) = q^2/\zeta = \kappa$),

$$\ell_j \equiv \omega + C + j\eta - f_j(\sec \alpha - 1), \quad C := \hat{\xi} - \tan \alpha' - s - B(1 - A)(\sec \alpha - 1).$$

By (Z3), $\psi + f(\sec \alpha - 1) \equiv C + J_{l_0}\eta$, so $\omega_0 + C + J_{l_0}\eta - (\sec \alpha - 1)$ is an integer and

$$\ell_j \equiv (\omega - \omega_0) + (j - J_{l_0})\eta + (1 - f_j)(\sec \alpha - 1) \pmod{1},$$

where $0 \leq \omega - \omega_0 < 1/Q_f$, $0 \leq (j - J_{l_0})\eta < N_0\delta_Q$ (as $J_L - J_{l_0} \leq N_0 - 1$) and $0 < (1 - f_j)(\sec \alpha - 1) \leq \sec \alpha_0 - 1$. The right-hand side lies in $(0, \varepsilon_W]$ with $\varepsilon_W < 1$ (W4). Also $\ell_j \geq \hat{\xi} - \xi_{p(j)} - \tan \alpha' \geq \sec \alpha - \tan \alpha' > 0$. Hence

$$(R) \quad \ell_j \geq 0 \quad \text{and} \quad \text{frac}(\ell_j) \leq \varepsilon_W.$$

(c) *Global structure*. By (a), (b) and induction every block satisfies (I1)–(I3), ZC never fails, the bases F_β are non-decreasing, the tilts non-increasing, $s_{\beta+1} \geq \hat{\xi}_\beta$, $J_{\beta+1,0} \leq J_{\beta,L}$, and there are at most $\bar{K}$ complete blocks. For $Y \geq 0$ and $\beta < \beta''$ we have $\lambda_{\beta''} \leq \lambda_\beta$. A stretch of row j runs from just after a blocking block β (left line $\lambda_\beta = \xi_{\beta,p(j)}$) to the next blocking block $\beta_2 > \beta$ (line

$\lambda_{\beta_2} = s_{\beta_2}$), passing below all blocks in between, whose columns have $J \geq J_{\beta'',0} \geq j+1$. If β is not the last block and $\beta+1$ does not block row j , then $j < J_{\beta+1,0} \leq J_{\beta,L}$, so $p(j) \leq P_\beta - 1$. Hence $p(j) = P_\beta$ (left end on $\lambda_\beta = \hat{\xi}_\beta$) occurs only for stretches that end at the next block's line (*transition rows*, $j \geq J_{\beta,L}$) or for the last block (*end rows*).

(d) *Containment of rows*. A placed row has $a \geq 0$, $a + \lfloor a^+ - a \rfloor \leq a^+ \leq L$ (as a^+ is L or $s_\beta - (j+1)\tan\alpha_\beta \leq s_\beta \leq L$), and $j+1 \leq r(a) \leq r(X)$ on it. So every piece lies in Z .

(e) *Disjointness*. Columns of one block lie in the slabs $\xi_i \leq \lambda_\beta \leq \xi_{i+1}$ (Lemma 7.1(i)). For $\beta < \beta''$, a column of β lies in $\{\lambda_\beta \leq \hat{\xi}_\beta\} \subset \{\lambda_{\beta''} \leq \hat{\xi}_\beta\} \subset \{\lambda_{\beta''} \leq s_{\beta''}\}$ (as $Y \geq 0$), and a column of β'' in $\{\lambda_{\beta''} \geq s_{\beta''}\}$. A row piece of a stretch of row j from block β to block β_2 (or $x = L$): all columns of blocks $\gamma < \beta$ and the columns $i < p(j)$ of β lie in $\{\lambda_\beta \leq \xi_{\beta,p(j)}\}$, and the row piece in $\{\lambda_\beta \geq \xi_{\beta,p(j)}\}$ (its lower left corner is on that line and $\tan\alpha_\beta > 0$); the columns $i \geq p(j)$ of β and all columns of the blocks strictly between β and β_2 have $J \geq j+1$, so they lie in $\{Y \geq j+1\}$ and the row piece in $\{Y \leq j+1\}$; the columns of β_2 and of later blocks lie in $\{\lambda_{\beta_2} \geq s_{\beta_2}\}$, and the row piece in $\{\lambda_{\beta_2} \leq s_{\beta_2}\}$ (its top right corner is on or left of that line). For the first stretch of a row (left end $x = 0$) only the last two cases occur. Different rows are separated by the lines $Y = j+1$, and the stretches of one row by the lines above. So ZC is a packing in Z .

(f) *Accounting*. For each column $K_{\beta,i}$ let its *cell* be $\{z \in Z : \xi_i \leq \lambda_\beta < \xi_{i+1}, Y \geq J_i\}$. The cells are pairwise disjoint: slabs of one block are disjoint, and with $E_\beta := \{z \in Z : \lambda_\beta(z) < s_\beta\}$ we have $E_0 \subset E_1 \subset \dots$, and the cells of block β lie in $E_{\beta+1} \setminus E_\beta$. For $Y \in [j, j+1)$ and a slab of a blocking block, $Y \geq J_i$ iff $J_i \leq j$; so the complement of the cells in $Z \cap \{j \leq Y < j+1\}$ is the disjoint union of the *stretch regions* $\{j \leq Y < j+1, \lambda_\beta \geq \xi_{\beta,p(j)} \text{ (or } X \geq 0), \lambda_{\beta_2} < s_{\beta_2} \text{ (or } X \leq L)\} \cap Z$. The interior of every placed piece lies in its cell or stretch region, and boundaries have measure 0; so U_Z is the sum over the cells of (area $- n$) plus the sum over the stretch regions of (area $-$ placed squares).

(f1) *Cells*. The parallelogram $\{\xi_i \leq \lambda_\beta \leq \xi_i + \sec\alpha, J_i \leq Y \leq H(J_i)\}$ has area $\sec\alpha(n \cos\alpha + \sin\alpha) = n + \tan\alpha$, so the part of the cell in $\{Y \leq H(J_i)\}$ has area at most $n + \tan\alpha$. The rest of the cell lies in $\{Y > H(J_i)\}$, where $X \leq \xi_i + \sec\alpha - H(J_i)\tan\alpha$, so $Y \leq r(\xi_i + \sec\alpha - H(J_i)\tan\alpha) = \Gamma_i + \zeta(\sec\alpha + (M - H(J_i))\tan\alpha)$ with $M - H(J_i) = nq - \sin\alpha \leq N_0q$; its sections have length at most $\sec\alpha$. So (area of the cell) $- n \leq \tan\alpha + \sec\alpha q + \sec\alpha \zeta(\sec\alpha + N_0q \tan\alpha) \leq c_{\text{col}}$ (all terms increase with $t \leq t_0$, and the gap is less than q). The slab of column (β, i) meets the line $Y = 0$ in $[\xi_i, \xi_i + \sec\alpha] \subset [0, L]$; these intervals have disjoint interiors (consecutive inside a block, and $s_{\beta+1} \geq \hat{\xi}_\beta$) and length $\sec\alpha \geq 1$, so there are at most L columns, and the cells cost at most Lc_{col} .

(f2) *Stretch areas*. A stretch region between the lines $\lambda_\beta = c$ and $\lambda_{\beta_2} = s'$ has horizontal sections of length $(s' - Y \tan\alpha_{\beta_2}) - (c - Y \tan\alpha_\beta)$, affine in Y with slope $\tan\alpha_\beta - \tan\alpha_{\beta_2} \geq 0$; so its area is at most $\ell + \tan\alpha_0$, where $\ell := s' - (j+1)\tan\alpha_{\beta_2} - (c - j \tan\alpha_\beta)$ is the stretch length of (Z4) (with $\frac{1}{2}\tan\alpha_0$ for an end at $x = 0$ or $x = L$). A placed piece leaves at most $\text{frac}(\ell) + \tan\alpha_0$; if $0 \leq \ell < 1$ the waste is at most $\ell + \tan\alpha_0$ as well.

(f3) *The stretches*, each counted by its right end.

- Right end at $s_{\beta+1}$, left end in block β with $J_{\beta,0} \leq j < J_{\beta,L}$ (*aligned closings*): by (R), $\ell \geq 0$ and $\text{frac}(\ell) \leq \varepsilon_W$; the left end has $a_j \geq \xi_p - M \tan\alpha \geq 0$ and $j+1 \leq J_{p(j)} \leq H(J_{p(j)}) \leq \Gamma_{p(j)} \leq r(a_j)$, so the piece is placed when $\ell \geq 1$. Waste at most $\varepsilon_W + \tan\alpha_0$, in at most $N_0 - 1$ rows per complete block.
- Right end at $s_{\beta+1}$, left end before block β ($j < J_{\beta,0}$; *unaligned closings*): these rows have $J_{\beta+1,0} \leq j < J_{\beta,0}$, at most $J_{\beta,0} - F_\beta < (1 + \zeta \tan\alpha_0)/q_{\min}$ per complete block; the left end is $x = 0$ or a slab line with $p \leq P - 1$, and the waste is at most $1 + \tan\alpha_0$.
- Right end at $s_{\beta+1}$, left end on $\hat{\xi}_\beta$ ($j \geq J_{\beta,L}$; *transition rows*): their union lies in $\{\lambda_\beta \geq \hat{\xi}_\beta, \lambda_{\beta+1} < s_{\beta+1}, Y \geq J_{\beta,L}\}$. Its points have $X < s_{\beta+1}$, so $Y \leq r(s_{\beta+1}) < J_L + n_L + \sin\alpha +$

- $\zeta(1 + \sec \alpha + M \tan \alpha)$ (see (b)): height at most h_T . At height Y its section has length $\omega + Y\hat{\kappa} \leq w_T$ (as $\hat{\kappa} \leq q_0^2/\zeta + \delta_Q$). Area at most A_T per complete block.
- Right end at s_0 , left end $x = 0$: for $j \leq \lfloor \chi \rfloor - 1$ the piece is placed ($j + 1 \leq \chi = r(0)$), waste at most $1 + \frac{1}{2} \tan \alpha_0$, in at most $\lfloor \chi \rfloor$ rows. The part of E_0 with $Y \geq \lfloor \chi \rfloor$ has $X < s_0 - \lfloor \chi \rfloor \tan \alpha_0 \leq (M_0 - \lfloor \chi \rfloor) \tan \alpha_0 < w_0$ and $Y \leq r(w_0) \leq \lfloor \chi \rfloor + 1 + \zeta w_0$: area at most $w_0(1 + \zeta w_0)$.
 - Right end at $x = L$ (*final stretches*), at most one per row $j < \lfloor r(L) \rfloor$. If the left end is $x = 0$ or a slab line with $p \leq P - 1$, the piece is placed when $\ell \geq 1$ and the waste is at most $1 + \frac{1}{2} \tan \alpha_0$. Otherwise, by (c), the left end is on $\hat{\xi}$ of the last block, and these end rows lie in $E_{\text{end}} := \{\lambda_{\text{last}} \geq \hat{\xi}_{\text{last}}\}$, where $L - \hat{\xi}_{\text{last}} < 1 + \sec \alpha_0$ (the last block stopped with reason ‘ L ’, or the next block could not start). An end row with $a_j = \hat{\xi} - j \tan \alpha \geq 0$ and $j + 1 \leq r(a_j)$ is placed with waste at most $1 + \frac{1}{2} \tan \alpha_0$ (counted in $r(L)(1 + \frac{1}{2} \tan \alpha_0)$). The other points of E_{end} : (i) rows with $a_j \geq 0$ and $j + 1 > r(a_j)$; as $a_j \geq L - w_e$, these rows have $j > r(L) - \zeta w_e - 1$, and their points have $X \in [L - w_e, L]$, $Y \leq r(L)$: area at most $w_e(1 + \zeta w_e)$; (ii) rows with $a_j < 0$: then $j > \hat{\xi}/\tan \alpha \geq (s + \sec \alpha)/\tan \alpha \geq M + 1/\tan \alpha > N_0$ by (I2), so there are at most $r(L) - \lfloor \chi \rfloor + 1 \leq \zeta L + 2$ such rows, and $\hat{\xi} < j \tan \alpha \leq r(L) \tan \alpha_0$ gives $L < w_e$, so each such row meets Z in area at most $L < w_e$: area at most $(\zeta w_e + 2)w_e$. In total at most $w_e(4 + 2\zeta w_e)$.

(f4) Summing (the cells; per complete block, of which there are at most $\bar{K}$, the aligned, unaligned and transition rows; the start; the final stretches) gives $U_Z \leq B_W$. If there is no block at all ($s_0 + \sec \alpha_0 > L$), all rows run from 0 to L , rows $j \leq \lfloor \chi \rfloor - 1$ are placed with waste at most 1, and the rest has area at most $L(1 + \zeta L) \leq w_e(1 + \zeta w_e)$, as $L < s_0 + \sec \alpha_0 \leq w_e$. $\square$

8.2. The variant ZC' , transition rows and the end region. This subsection is new in version 1.1. Only the tiers marked $\dagger$ in Tables 1 and 3 use it. It changes one rule of ZC and replaces two crude terms of B_W , the transition term A_T and the end term $w_e(4 + 2\zeta w_e)$.

Construction ZC' . It is identical to ZC except for the last sentence of (Z4), which becomes:

(Z4') For a stretch of row j from a to a^+ put

$$a^* := \max(a, 0, (j + 1 - \chi)/\zeta)$$

(the least $x \geq \max(a, 0)$ with $r(x) \geq j + 1$); if $a^+ - a^* \geq 1$, place the row $[a^*, a^* + \lfloor a^+ - a^* \rfloor] \times [j, j + 1]$.

(Z1)–(Z3) (blocks, columns, tilts, phases) and the stretches (their left and right ends) are unchanged.

Observation (O). If $a \geq 0$ and $j + 1 \leq r(a)$, then $a^* = a$, and ZC' places exactly the piece that ZC places (if $a^+ - a < 1$, neither places a piece). So ZC' differs from ZC only on stretches where ZC places nothing because $a < 0$ or $j + 1 > r(a)$; there ZC' may place a row that starts further to the right.

Lemma 8.2. *Under (W1)–(W5) the construction ZC' does not fail, its pieces form a packing in Z , and $U_Z \leq B_W$.*

Proof. (Z1)–(Z3) are those of ZC , so parts (a)–(c) of the proof of Lemma 8.1 apply verbatim (they do not involve the rows). *Containment:* a placed row of ZC' has $a^* \geq 0$, $a^* + \lfloor a^+ - a^* \rfloor \leq a^+ \leq L$, and $r(X) \geq r(a^*) \geq j + 1$ on it, as r is increasing and $a^* \geq (j + 1 - \chi)/\zeta$. *Disjointness:* the piece lies in $[a, a^+] \times [j, j + 1]$. Every separation statement of Lemma 8.1(e) about a row piece uses only that (1) its lower left corner (a', j) satisfies $\lambda_\beta(a', j) \geq \xi_{\beta, p(j)}$ (or $a' \geq 0$ for a first stretch), (2) its top right corner $(a'', j + 1)$ satisfies $\lambda_{\beta_2}(a'', j + 1) \leq s_{\beta_2}$ (or $a'' \leq L$ for a last stretch), and (3) the piece lies in $\{j \leq Y \leq j + 1\}$. As $a^* \geq a$, $a^* + \lfloor a^+ - a^* \rfloor \leq a^+$ and $\lambda_\beta, \lambda_{\beta_2}$ are increasing in X , (1)–(3) hold for the piece of ZC' because they hold for the segment $[a, a^+] \times [j, j + 1]$; so (e) holds verbatim. *Accounting:* the partition of (f) into cells and stretch regions does not depend on the placement rule, and the pieces that ZC' places in a stretch lie in its region. Each bound in (f1)–(f4)

is either of the form “waste at most area”, which holds for any placement, or of the form “the piece is placed by ZC and the waste is at most ...”; in the second case ZC' places the same piece by (O). $\square$

Lemma 8.3 (transition rows). *Assume (W1)–(W5), and let β be a complete block whose successor $\beta + 1$ exists ($s_{\beta+1} + \sec \alpha_{\beta+1} \leq L$). For ZC and for ZC' the total waste of the transition rows of β (the stretches with left end on the line $\lambda_\beta = \hat{\xi}_\beta$ and right end on the line $\lambda_{\beta+1} = s_{\beta+1}$) is at most*

$$A'_T := (h_T + 1)(1 + \tan \alpha_0) + (2 + \zeta(w_T + \tan \alpha_0 + \zeta \tan \alpha_0))(w_T + \tan \alpha_0),$$

with h_T, w_T as in Lemma 8.1. Hence Lemmas 8.1 and 8.2 hold with A_T replaced by A'_T .

Proof. Write $\tan \alpha, \sec \alpha, q, M, s, \hat{\xi} := \hat{\xi}_\beta, J_L := J_{\beta,L}$ and n_L for block β , and $\tan \alpha' := \tan \alpha_{\beta+1} \leq \tan \alpha, s' := s_{\beta+1} = \hat{\xi} + \omega$ with $\omega \in [0, 1]$, and $\hat{\kappa} := \tan \alpha - \tan \alpha' = \kappa_\beta + \eta_\beta \leq q_0^2/\zeta + \delta_Q$. By Lemma 8.1(c) the transition rows are the rows $j \geq J_L$. The stretch of row j has left end $a_j := \hat{\xi} - j \tan \alpha$, right end $a_j^+ := s' - (j + 1) \tan \alpha'$, length $\ell_j := a_j^+ - a_j = \omega - \tan \alpha' + j\hat{\kappa}$ and region $\text{Str}_j := \{z \in Z : j \leq Y < j + 1, \lambda_\beta \geq \hat{\xi}, \lambda_{\beta+1} < s'\}$. The rows $j \geq \lfloor r(L) \rfloor$ are not treated by (Z4); their regions are counted in the same way below, and they are bad rows in the sense of (3) (as $j + 1 > r(L) \geq r(a_j)$), so no part of $\{\lambda_\beta \geq \hat{\xi}, \lambda_{\beta+1} < s', Y \geq J_L\} \cap Z$ is missed.

(1) *Number of rows.* By Lemma 8.1(f3) every point of $\{\lambda_\beta \geq \hat{\xi}, \lambda_{\beta+1} < s', Y \geq J_L\} \cap Z$ has $Y < J_L + h_T$. So Str_j is empty unless j is an integer in $[J_L, J_L + h_T)$: at most $h_T + 1$ rows.

(2) *The left ends are positive.* $h_T - 1$ equals the right-hand side of (W3) minus $\zeta + (1 + \zeta \tan \alpha_0)/q_{\min}$, so (W3) gives $h_T - 1 < N_0 \cos \alpha_0 \leq N_0$. For a row of (1), $j < J_L + h_T < (M - 1) + 1 + N_0 = M + N_0$. By (I2), (8.1) and (W1), $\hat{\xi} = s + \Lambda_\beta > M \tan \alpha + \Lambda \geq M \tan \alpha + N_0 \tan \alpha_0 \geq (M + N_0) \tan \alpha$. Hence $a_j > 0$.

(3) *Waste of one row.* The horizontal sections of $\{j \leq Y < j + 1, \lambda_\beta \geq \hat{\xi}, \lambda_{\beta+1} < s'\}$ have length $(s' - Y \tan \alpha') - (\hat{\xi} - Y \tan \alpha) = \omega + Y\hat{\kappa}$, affine in Y , equal to $\ell_j + \tan \alpha'$ at $Y = j$ and to $\ell_j + \tan \alpha$ at $Y = j + 1$; so $\text{area}(\text{Str}_j) \leq \max(0, \ell_j + \tan \alpha_0)$. Call row j *good* if $j + 1 \leq r(a_j)$ and *bad* otherwise. In a good row, by (2) and (O), ZC and ZC' both place $[a_j, a_j + \lfloor \ell_j \rfloor] \times [j, j + 1]$ if $\ell_j \geq 1$, and the waste is at most $\ell_j + \tan \alpha_0 - \lfloor \ell_j \rfloor < 1 + \tan \alpha_0$; if $\ell_j < 1$ it is at most $\max(0, \ell_j + \tan \alpha_0) < 1 + \tan \alpha_0$. In a bad row the waste is at most $\text{area}(\text{Str}_j) \leq \max(0, \ell_j + \tan \alpha_0)$; a point of Str_j has $j \leq Y \leq r(X) \leq r(L)$, so if $\text{Str}_j \neq \emptyset$ then $j \leq r(L)$ and $\ell_j \leq 1 + j\hat{\kappa} \leq 1 + r(L)(q_0^2/\zeta + \delta_Q) \leq w_T$; the waste is at most $w_T + \tan \alpha_0$.

(4) *Number of bad rows.* Let j be bad with $\text{Str}_j \neq \emptyset$, and $(X, Y) \in \text{Str}_j$. Then $j \leq Y \leq r(X)$ and $X < s' - Y \tan \alpha' \leq s' - j \tan \alpha'$, so $j \leq r(s' - j \tan \alpha')$, that is $j(1 + \zeta \tan \alpha') \leq H + \zeta \omega$ with $H := r(\hat{\xi})$. Bad means $j + 1 > r(\hat{\xi} - j \tan \alpha)$, that is $j(1 + \zeta \tan \alpha) > H - 1$. So j lies in the half-open interval $((H - 1)/(1 + \zeta \tan \alpha), (H + \zeta \omega)/(1 + \zeta \tan \alpha'))]$, whose length is

$$\begin{aligned} \frac{H\zeta(\tan \alpha - \tan \alpha') + \zeta\omega(1 + \zeta \tan \alpha) + 1 + \zeta \tan \alpha'}{(1 + \zeta \tan \alpha)(1 + \zeta \tan \alpha')} &\leq 1 + \zeta(H\hat{\kappa} + 1 + \tan \alpha_0 + \zeta \tan \alpha_0) \\ &\leq 1 + \zeta(w_T + \tan \alpha_0 + \zeta \tan \alpha_0), \end{aligned}$$

using $\omega \leq 1, \tan \alpha, \tan \alpha' \leq \tan \alpha_0$, a denominator at least 1, and $H \leq r(L)$ (as $\hat{\xi} \leq s' \leq L$, since the successor exists), so that $H\hat{\kappa} + 1 \leq w_T$. A half-open interval of length ℓ contains at most $\ell + 1$ integers: there are at most $2 + \zeta(w_T + \tan \alpha_0 + \zeta \tan \alpha_0)$ bad rows.

Counting every row with the bound $1 + \tan \alpha_0$ and the bad rows once more with $w_T + \tan \alpha_0$ gives at most A'_T . The other parts of the proof of Lemma 8.1 are unchanged. $\square$

Lemma 8.4 (end region of ZC'). *Assume (W1)–(W5). For ZC' the last term $w_e(4 + 2\zeta w_e)$ of B_W may be replaced by w_e .*

Proof. Let $P_z := \{(X, Y) : 0 \leq X \leq L, \lfloor r(X) \rfloor \leq Y \leq r(X)\}$ (the *partial layer*). Its vertical sections have length $\text{frac}(r(X)) < 1$, so $\text{area}(P_z \cap \{X \in I\}) \leq |I|$ for every interval I .

No block. Every row $j < \lfloor r(L) \rfloor$ is one stretch from 0 to L . The part of its region with $X \geq a_j^*$ is $[a_j^*, L] \times [j, j+1]$ (inside Z , as $r(X) \geq r(a_j^*) \geq j+1$ there), of which ZC' leaves uncovered less than 1; the part with $X < a_j^*$ lies in P_z (as in (iii) below). The points with $Y \geq \lfloor r(L) \rfloor$ lie in P_z . So $U_Z \leq \lfloor r(L) \rfloor + \text{area}(P_z) \leq r(L) + L$, and $L < s_0 + \sec \alpha_0 \leq w_e$ as in Lemma 8.1(f4); this is at most $r(L)(1 + \frac{1}{2} \tan \alpha_0) + w_e$.

Blocks exist. Let β be the last block, $\tan \alpha := \tan \alpha_\beta$, $\hat{\xi} := \hat{\xi}_\beta$ and $E_{\text{end}} := \{z \in Z : \lambda_\beta(z) \geq \hat{\xi}\}$. As in Lemma 8.1(f3), $L - \hat{\xi} < 1 + \sec \alpha_0$, so every point of E_{end} has $X \geq \hat{\xi} - Y \tan \alpha \geq \hat{\xi} - r(L) \tan \alpha_0 > L - w_e$: $E_{\text{end}} \subset \{L - w_e < X \leq L\}$. No cell meets E_{end} : the cells of β lie in $\{\lambda_\beta < \hat{\xi}\}$, and the cells of an earlier block β'' lie in $\{\lambda_{\beta''} < \hat{\xi}_{\beta''}\}$, while $\lambda_\beta \leq \lambda_{\beta''}$ for $Y \geq 0$ and $\hat{\xi}_{\beta''} \leq s_\beta \leq \hat{\xi}$. The *end rows* are the stretches with left end $a_j := \hat{\xi} - j \tan \alpha$ on the line $\lambda_\beta = \hat{\xi}$ and right end $x = L$ (rows $j < \lfloor r(L) \rfloor$); their regions are $\text{Str}_j = \{z \in Z : j \leq Y < j+1, \lambda_\beta \geq \hat{\xi}\}$. Split Str_j by X :

- (i) $X \geq a_j^*$: if $a_j^* \leq L$ this is the rectangle $[a_j^*, L] \times [j, j+1]$ (its points satisfy $X \geq 0$, $Y < j+1 \leq r(a_j^*) \leq r(X)$ and $\lambda_\beta \geq a_j + j \tan \alpha = \hat{\xi}$); ZC' places $\lfloor L - a_j^* \rfloor$ squares in it if $L - a_j^* \geq 1$, so the waste of this part is less than 1;
- (ii) $X < a_j^*$: this part lies in the triangle $\{\hat{\xi} - Y \tan \alpha \leq X < \hat{\xi} - j \tan \alpha, j \leq Y < j+1\}$ of area at most $\frac{1}{2} \tan \alpha \leq \frac{1}{2} \tan \alpha_0$;
- (iii) $a_j \leq X < a_j^*$ (only if $a_j^* > a_j$): points of Z have $X \geq 0$, so this part is empty if $a_j^* = 0$; otherwise $a_j^* = (j+1 - \chi)/\zeta$, its points have $r(X) < j+1$ and $j \leq Y \leq r(X)$, so $\lfloor r(X) \rfloor = j$ and they lie in $P_z \cap E_{\text{end}}$.

The points of E_{end} with $Y \geq \lfloor r(L) \rfloor$ (rows not treated by (Z4)) lie in P_z , as $\lfloor r(X) \rfloor \leq \lfloor r(L) \rfloor \leq Y$. The parts (iii) of different rows and this part are disjoint subsets of $P_z \cap \{L - w_e < X \leq L\}$, of total area at most w_e . There is at most one final stretch per row $j < \lfloor r(L) \rfloor$. A final stretch is either an end row, whose waste outside P_z is less than $1 + \frac{1}{2} \tan \alpha_0$ by (i) and (ii), or, by Lemma 8.1(c), a stretch whose left end is $x = 0$ or a slab line with $p \leq P_{\beta'} - 1$ of some block β' ; such a stretch is placed as by ZC (by (O)) and has waste at most $1 + \frac{1}{2} \tan \alpha_0$ as in Lemma 8.1(f3). This covers the rows $j < J_{\beta, L}$ of E_{end} : if $J_{\beta, 0} \leq j < J_{\beta, L}$, then $p(j) \leq P_\beta - 1$ and $E_{\text{end}} \cap \{j \leq Y < j+1\}$ lies in the region of the final stretch of row j , which starts on the slab line $\xi_{\beta, p(j)}$ of β ; if $j < J_{\beta, 0}$, the last block does not block row j , and $E_{\text{end}} \cap \{j \leq Y < j+1\}$ lies in the region of the final stretch of row j , which starts at $x = 0$ or after an earlier block (with $p \leq P_{\beta'} - 1$ by Lemma 8.1(c)). So the final stretches together with $E_{\text{end}} \cap \{Y \geq \lfloor r(L) \rfloor\}$ waste at most $r(L)(1 + \frac{1}{2} \tan \alpha_0) + w_e$, which replaces $r(L)(1 + \frac{1}{2} \tan \alpha_0) + w_e(4 + 2\zeta w_e)$ in (f4). $\square$

Lemma 8.5 (wall filler ZC'). *Under (W1)–(W5) the construction ZC' does not fail, its pieces form a packing in Z , and*

$$U_Z \leq B_W'' := Lc_{\text{col}} + \bar{K} \left[(N_0 - 1)(\varepsilon_W + \tan \alpha_0) + \frac{1 + \zeta \tan \alpha_0}{q_{\min}} (1 + \tan \alpha_0) + A_T' \right] \\ + \lfloor \chi \rfloor (1 + \frac{1}{2} \tan \alpha_0) + w_0(1 + \zeta w_0) + r(L)(1 + \frac{1}{2} \tan \alpha_0) + w_e + 2(1 + \tan \alpha_0),$$

with A_T' of Lemma 8.3 and the other symbols as in Lemma 8.1.

Proof. Lemmas 8.2, 8.3 and 8.4. Lemma 8.3 changes the transition rows of the complete blocks that have a successor, and Lemma 8.4 the final stretches and E_{end} ; these terms of (f4) are disjoint. The term $2(1 + \tan \alpha_0)$ is not needed (Lemma 8.4 gives w_e); it is kept because the certification of Section 8.5 uses the larger bound. $\square$

8.3. Sawtooth accounting of the start rows and the final stretches. This subsection is new in version 1.1 and is used only by the tiers marked $\ddagger$ (which also use Section 8.2). The construction ZC' is not changed; only the accounting of two groups of rows is sharpened: the lengths of these

rows form arithmetic progressions, and the sum of the fractional parts of such a progression is about half its length.

Notation. $\text{frac}(x) := x - \lfloor x \rfloor \in [0, 1)$. For a block β of ZC (equally of ZC') we use the quantities of Lemma 8.1(a) for that block: $A_\beta := (\Gamma_{\beta,0} - M_\beta \cos \alpha_\beta - \sin \alpha_\beta)/q_\beta$ and $B_\beta := q_\beta/(\zeta \sec \alpha_\beta)$, so that $J_{\beta,i} = \lfloor A_\beta + i/B_\beta \rfloor$ for $0 \leq i \leq P_\beta - 1$ for every block, complete or not; $J_{\beta,L} := J_{\beta,P_\beta-1}$; $\kappa_\beta := q_\beta^2/\zeta$ (for a complete block this is κ_β of (Z3)); and $p(j) := \#\{i : J_{\beta,i} \leq j\}$. The new hypothesis is

$$(W6) \quad \nu_* := \varpi \tan \alpha_0 - q_0^2/\zeta > 0.$$

Lemma 8.6 (sawtooth sums). *Let $0 < \nu < 1$, $\epsilon \geq 0$, $n \geq 0$ an integer, c real, $\pm \in \{+1, -1\}$, and $x_j = c \pm j\nu + d_j$ ($0 \leq j \leq n-1$) with $d_j \in [0, \epsilon]$. Then*

$$\sum_{j=0}^{n-1} \text{frac}(x_j) \leq n \frac{1+\nu}{2} + n\epsilon + \frac{1}{8\nu}.$$

Proof. (1) For $d \geq 0$, $\text{frac}(y+d) \leq \text{frac}(y) + d$: if $\text{frac}(y) + d < 1$ the two sides are equal; otherwise $\text{frac}(y+d) = \text{frac}(y) + d - \lfloor \text{frac}(y) + d \rfloor \leq \text{frac}(y) + d - 1$. So it suffices to treat $d_j = 0$ and to add $n\epsilon$. (2) The sign -1 is the sign $+1$ read backwards: $x_{n-1-i} = c' + i\nu$ with $c' := c - (n-1)\nu$. (3) Let $y_j := \text{frac}(c + j\nu)$. As $0 < \nu < 1$, $y_{j+1} = y_j + \nu$ if $y_j + \nu < 1$, and $y_{j+1} = y_j + \nu - 1$ otherwise (a wrap). Cut $0, \dots, n-1$ before each index that follows a wrap; this gives maximal runs of consecutive indices. In a run of $r \geq 1$ indices the values are $f, f + \nu, \dots, f + (r-1)\nu =: l$ with $l < 1$, and their sum is $r(f+l)/2$. A run that starts right after a wrap has $f = y_{\text{prev}} + \nu - 1 \in [0, \nu)$ (because $y_{\text{prev}} < 1 \leq y_{\text{prev}} + \nu$), so its sum is less than $r(1+\nu)/2$. Only the first run may start higher; its sum is $r(2l - (r-1)\nu)/2 < r(2 - (r-1)\nu)/2 = r(1+\nu)/2 + (r-r^2\nu)/2$, and $(r-r^2\nu)/2 \leq 1/(8\nu)$ for every real r . The run lengths add up to n . $\square$

The constant $1/(8\nu)$ cannot be lowered: it is attained in the limit by one run of length $1/(2\nu)$ that starts just below $1 - (r-1)\nu$.

The rows treated below (ZC' on $Z(\chi, \zeta, L)$); the waste of a stretch is the area of its stretch region minus the number of squares placed in it, as in Lemmas 8.1(f) and 8.2). The *start rows*: if block 0 exists, the rows $J_{0,0} \leq j \leq \lfloor \chi \rfloor - 1$; the first stretch of such a row runs from $x = 0$ to the line $\lambda_0 = s_0$. The *final stretches*: if a block exists, the stretch of row j ($0 \leq j < \lfloor r(L) \rfloor$) whose right end is $x = L$, together with the set $E_{\text{end}} \cap \{Y \geq \lfloor r(L) \rfloor\}$ of Lemma 8.4. These are exactly the stretches bounded in Lemma 8.1(f3) by $\lfloor \chi \rfloor(1 + \frac{1}{2} \tan \alpha_0)$ and in Lemma 8.4 by $r(L)(1 + \frac{1}{2} \tan \alpha_0) + w_e$.

Lemma 8.7 (sawtooth accounting). *Assume (W1)–(W6) and consider ZC' on $Z(\chi, \zeta, L)$.*

- (a) *If block 0 exists, the start rows have total waste at most $\lfloor \chi \rfloor(\frac{1}{2} + \tan \alpha_0) + 1/(8 \tan \alpha_0)$.*
- (b) *If at least one block exists, the final stretches together with $E_{\text{end}} \cap \{Y \geq \lfloor r(L) \rfloor\}$ have total waste at most*

$$r(L)(\frac{1}{2} + \tan \alpha_0 + (\sec \alpha_0 - 1)) + \frac{\bar{K} + 2}{8\nu_*} + \frac{1}{2} \left(2 + N_0 q_0 + \frac{1 + \zeta \tan \alpha_0}{q_{\min}} \right) + \frac{1}{2} (1 + \zeta L + \zeta w_e) + w_e.$$

- (c) *If no block exists, $U_Z < \lfloor \chi \rfloor + \zeta L + 1 + L$.*

All three parts use ZC' and the facts of Lemmas 8.1, 8.2 and 8.4 about it, hence (W1)–(W5). Beyond that, (a) uses (W5) ($0 < \tan \alpha_0 \leq 0.21$); (b) uses (W5), (W6) and (I1), (I2) of Lemma 8.1; (c) uses only the definitions and $\text{area}(P_z) \leq L$ from the proof of Lemma 8.4.

Proof. (a) Let j be a start row and $\ell_j := s_0 - (j+1) \tan \alpha_0$. Its stretch region is $\{j \leq Y < j+1, X \geq 0, X + Y \tan \alpha_0 < s_0\}$; it lies in Z , as $X \leq s_0 \leq L$ and $Y < j+1 \leq \lfloor \chi \rfloor \leq r(X)$. In (Z4') $a = 0$ and $j+1 \leq \chi$, so $a^* = 0$, and ZC' places $[0, \lfloor \ell_j \rfloor] \times [j, j+1]$ if $\ell_j \geq 1$. If $\ell_j \geq 0$ the sections of the region have length $s_0 - Y \tan \alpha_0$, so its area is $\ell_j + \frac{1}{2} \tan \alpha_0$ and the waste is $\text{frac}(\ell_j) + \frac{1}{2} \tan \alpha_0$. If $\ell_j < 0$: $s_0 = M_0 \tan \alpha_0$ and $M_0 = F_0 + \lfloor \chi \rfloor - 1$, so $\ell_j \geq (F_0 - 1) \tan \alpha_0 \geq -\tan \alpha_0$, the section at height Y has length at most $\tan \alpha_0(1 - (Y - j))$, and the waste (the area) is at most $\frac{1}{2} \tan \alpha_0$.

In all cases the waste is at most $\text{frac}(\ell_j) + \frac{1}{2} \tan \alpha_0$. The start rows form an integer interval of $n \leq \lfloor \chi \rfloor$ rows, and $\ell_j = (s_0 - \tan \alpha_0) - j \tan \alpha_0$ is an arithmetic progression; Lemma 8.6 with the sign -1 , $\nu = \tan \alpha_0 \in (0, 0.21]$ and $\epsilon = 0$ gives the total $n(1 + \tan \alpha_0)/2 + 1/(8 \tan \alpha_0) + n \tan \alpha_0/2 \leq \lfloor \chi \rfloor (\frac{1}{2} + \tan \alpha_0) + 1/(8 \tan \alpha_0)$.

(b) Let the blocks be $0, 1, \dots, \beta_e$. Every block except the last is complete, and there are at most $\bar{K}$ complete blocks, so $\beta_e + 1 \leq \bar{K} + 1$. For $0 \leq j < \lfloor r(L) \rfloor$ the final stretch of row j starts after the last block that blocks row j , and at $x = 0$ if there is none. Let R_β be the set of rows whose last blocking block is β , and R_{none} the set of rows blocked by no block; these are integer intervals and partition $\{0, \dots, \lfloor r(L) \rfloor - 1\}$. Four classes:

(F0) $j \in R_{\text{none}}$. The final stretch is the whole row: $j + 1 \leq J_{0,0} \leq H(J_{0,0}) \leq \Gamma_{0,0} = r(0) = \chi$, so the region is the rectangle $[0, L] \times [j, j + 1)$ inside Z , $a^* = 0$, and the waste is less than 1. By Lemma 8.1(a) for block 0, $|R_{\text{none}}| \leq J_{0,0} < F_0 + (1 + \zeta \tan \alpha_0)/q_{\min}$, and $F_0 \leq \chi - N_0 \cos \alpha_0 - \sin \alpha_0 < \chi - N_0 + N_0 q_0 < 2 + N_0 q_0$; so $|R_{\text{none}}| < 2 + N_0 q_0 + (1 + \zeta \tan \alpha_0)/q_{\min}$.

(F1) $j \in R_\beta$ with $j < J_{\beta,L}$ (for $\beta < \beta_e$ every $j \in R_\beta$, by Lemma 8.1(c)). Then $p := p(j) \leq P_\beta - 1$, the left end is $a_j = \xi_{\beta,p} - j \tan \alpha_\beta \geq 0$, the length is $\ell_j = L - a_j \geq \sec \alpha_\beta > 0$, the region has area at most $\ell_j + \frac{1}{2} \tan \alpha_\beta$, and, as in Lemma 8.1(f3), $j + 1 \leq r(a_j)$; so by (O) the waste is at most $\text{frac}(\ell_j) + \frac{1}{2} \tan \alpha_0$. As in the derivation of (R) in Lemma 8.1(b) (which uses only $J_{\beta,i} = \lfloor A_\beta + i/B_\beta \rfloor$ for $i \leq P_\beta - 1$ and $j < J_{\beta,L}$, so it holds for the last block as well), $p(j) = B_\beta(j + 1 - A_\beta) + f_j$ with $f_j \in [0, 1)$, and modulo 1

$$\ell_j \equiv c'_\beta + j\nu_\beta + d_j, \quad \nu_\beta := \tan \alpha_\beta - \kappa_\beta, \quad d_j := (1 - f_j)(\sec \alpha_\beta - 1) \in [0, \sec \alpha_0 - 1],$$

with a constant c'_β . By (I1) and $q_\beta \leq q_0$, $\nu_\beta \geq \varpi \tan \alpha_0 - q_0^2/\zeta = \nu_* > 0$, and $\nu_\beta \leq \tan \alpha_0 \leq 0.21$. Lemma 8.6 (sign $+1$, $\epsilon = \sec \alpha_0 - 1$) gives, for the n_β rows of (F1) with last blocking block β , a total waste at most $n_\beta(\frac{1}{2} + \tan \alpha_0 + (\sec \alpha_0 - 1)) + 1/(8\nu_*)$.

(F2) $j \in R_{\beta_e}$ with $j \geq J_L := J_{\beta_e,L}$: the end rows of Lemma 8.4, with $a_j = \hat{\xi} - j \tan \alpha$ ($\tan \alpha := \tan \alpha_{\beta_e}$, $\hat{\xi} := \xi_{\beta_e}$). The conditions $a_j \geq 0$ and $j + 1 \leq r(a_j)$ are both upper bounds on j ; let j_1 be the least $j \geq J_L$ where one of them fails (or $\lfloor r(L) \rfloor$). (F2a) $J_L \leq j < j_1$ (n' rows): $a_j^* = a_j$, the part (iii) of the proof of Lemma 8.4 is empty, the part (i) is $[a_j, L] \times [j, j + 1)$ with waste $\text{frac}(L - a_j)$, and the part (ii) has area at most $\frac{1}{2} \tan \alpha_0$; $L - a_j = (L - \hat{\xi}) + j \tan \alpha$ is an arithmetic progression with step $\tan \alpha \in [\varpi \tan \alpha_0, \tan \alpha_0]$ and $\varpi \tan \alpha_0 \geq \nu_*$, so Lemma 8.6 gives at most $n'(\frac{1}{2} + \tan \alpha_0) + 1/(8\nu_*)$. (F2b) $j_1 \leq j < \lfloor r(L) \rfloor$ (n_{tail} rows): the parts (i) and (ii) waste less than $1 + \frac{1}{2} \tan \alpha_0$, and the parts (iii) lie in $P_z \cap E_{\text{end}}$. If $a_{j_1} \geq 0$ and $j_1 + 1 > r(a_{j_1})$, then, as $L - \hat{\xi} < 1 + \sec \alpha_0$ and $j_1 < r(L)$, $a_{j_1} > \hat{\xi} - r(L) \tan \alpha_0 > L - w_e$, so $j_1 + 1 > r(L) - \zeta w_e$ and $n_{\text{tail}} < \zeta w_e + 1$. If $a_{j_1} < 0$, then as in Lemma 8.1(f3)(ii) $j_1 \geq \lfloor \chi \rfloor$ and $n_{\text{tail}} < \zeta L + 1$. So $n_{\text{tail}} < 1 + \zeta L + \zeta w_e$. The parts (iii) of all end rows and $E_{\text{end}} \cap \{Y \geq \lfloor r(L) \rfloor\}$ are disjoint subsets of $P_z \cap E_{\text{end}}$, of area at most w_e (proof of Lemma 8.4).

Summation. Put $\bar{q} := \frac{1}{2} + \tan \alpha_0 + (\sec \alpha_0 - 1)$. A row of (F0) wastes less than $1 \leq \bar{q} + \frac{1}{2}$, and a row of (F2b) less than $1 + \frac{1}{2} \tan \alpha_0 \leq \bar{q} + \frac{1}{2}$; the bounds of (F1) and (F2a) are at most (rows) $\bar{q} + 1/(8\nu_*)$ per chunk, and there are at most $\beta_e + 2 \leq \bar{K} + 2$ chunks. As $|R_{\text{none}}| + \sum_\beta n_\beta + n' + n_{\text{tail}} = \lfloor r(L) \rfloor \leq r(L)$, the total is at most $r(L)\bar{q} + (\bar{K} + 2)/(8\nu_*) + \frac{1}{2}|R_{\text{none}}| + \frac{1}{2}n_{\text{tail}} + w_e$, which gives (b).

(c) Every row $j < \lfloor r(L) \rfloor$ is one stretch from 0 to L . Rows $j \leq \lfloor \chi \rfloor - 1$: the region is $[0, L] \times [j, j + 1)$ and the waste is less than 1. Rows $\lfloor \chi \rfloor \leq j < \lfloor r(L) \rfloor$ (fewer than $\zeta L + 1$): the part $X \geq a_j^*$ is a rectangle with waste less than 1, and the part $X < a_j^*$ lies in P_z . The points with $Y \geq \lfloor r(L) \rfloor$ lie in P_z , and $\text{area}(P_z) \leq L$. $\square$

Lemma 8.8 (ZC' with sawtooth accounting). *Under (W1)–(W6) the construction ZC' does not fail, its pieces form a packing in Z, and*

$$\begin{aligned}
U_Z \leq B_{W4} &:= Lc_{\text{col}} + \bar{K} \left[(N_0 - 1)(\varepsilon_W + \tan \alpha_0) + \frac{1 + \zeta \tan \alpha_0}{q_{\min}} (1 + \tan \alpha_0) + A'_T \right] \\
&+ \lfloor \chi \rfloor \left(\frac{1}{2} + \tan \alpha_0 \right) + \frac{1}{8 \tan \alpha_0} + w_0(1 + \zeta w_0) + r(L) \left(\frac{1}{2} + \tan \alpha_0 + (\sec \alpha_0 - 1) \right) + \frac{\bar{K} + 2}{8\nu_*} \\
&+ \frac{1}{2} \left(2 + N_0 q_0 + \frac{1 + \zeta \tan \alpha_0}{q_{\min}} \right) + \frac{1}{2} (1 + \zeta L + \zeta w_e) + w_e + 2(1 + \tan \alpha_0).
\end{aligned}$$

Proof. Packing: Lemma 8.2. Bound, if a block exists: the partition (f) of Lemma 8.1 for ZC'; the cells, aligned and unaligned closings as in (f1), (f3); the transition rows by Lemma 8.3; the part of E_0 with $Y \geq \lfloor \chi \rfloor$ by $w_0(1 + \zeta w_0)$; the start rows by Lemma 8.7(a) in place of $\lfloor \chi \rfloor (1 + \frac{1}{2} \tan \alpha_0)$; the final stretches and $E_{\text{end}} \cap \{Y \geq \lfloor r(L) \rfloor\}$ by Lemma 8.7(b) in place of $r(L)(1 + \frac{1}{2} \tan \alpha_0) + w_e$. Every stretch is counted once, by its right end. The term $2(1 + \tan \alpha_0)$ is not needed and is kept as slack. If no block exists, Lemma 8.7(c) gives $U_Z < \lfloor \chi \rfloor + \zeta L + 1 + L$, while $B_{W4} \geq \lfloor \chi \rfloor + \zeta L + \frac{3}{2} + w_e$ and $L < s_0 + \sec \alpha_0 \leq w_e$. $\square$

8.4. The end packing. We use the band end $\text{Reg}(y)$ of Section 6 with $\tan(\theta/2) = b/(b^2 - 1)$, and, as in Corollary 7.4, $\mu := \tan \theta$, $g(X) := y + (X - \sigma)\mu$, $g_0 := g(0)$, $\bar{g} := g(w)$ and $\Delta := w\mu$.

Construction E38(y). It depends on three positive parameters c_D , c_R (the wall widths) and a_w (the wall tilt factor); version 1.0 used $c_D = c_R = 4$ and $a_w = 3/2$.

- (E1) $m := \lfloor \bar{g} \rfloor + 1$, $D_\ell := \lfloor c_D m^{3/4} \rfloor$, $D_r := \lfloor c_R m^{3/4} \rfloor$ and $g_D := g(D_\ell)$.
- (E2) *Main tilt:* $t := \lceil t^* Q \rceil / Q$ with $t^* = t^*(m, g_D)$ of Lemma 7.2 (g_0 replaced by g_D).
- (E3) *Main columns:* $\xi_i := D_\ell + m \tan \alpha + i \sec \alpha$ for $0 \leq i < N := \#\{i \geq 0 : \xi_i + \sec \alpha \leq w - D_r\}$, $\Gamma_i := g(\xi_i - m \tan \alpha) = g(D_\ell + i \sec \alpha)$, $J_i := \lfloor (\Gamma_i - m \cos \alpha - \sin \alpha) / q \rfloor$ and the columns $K(\xi_i, J_i, m - J_i)$. Put $J_L := J_{N-1}$ and $\xi_N := \xi_0 + N \sec \alpha$.
- (E4) *Cut:* $\kappa_m := q^2 / \mu$; $\tau := \tan \alpha_c$, where t_c is the largest point of $(1/Q^*)\mathbb{Z}$ with $\tan \alpha_c \leq \tan \alpha - \kappa_m$; $\hat{\kappa} := \tan \alpha - \tau$. The phase ω_m is defined as in (Z3), with $(s_\beta, \hat{\xi}_\beta, \tan \alpha_{\beta+1}, B, A)$ replaced by $(\xi_0, \xi_{N-1}, \tau, q/(\mu \sec \alpha), (g_D - m \cos \alpha - \sin \alpha)/q)$ and the rows $J_0 \leq j < J_L$; put $\ell_0 := \xi_{N-1} + \omega_m$.
- (E5) *Main rows:* for $0 \leq j < J_L$ let $p(j) := \#\{i : J_i \leq j\}$, $a_j := \xi_{p(j)} - j \tan \alpha$ and $a_j^+ := \ell_0 - (j+1)\tau$; if $a_j^+ - a_j \geq 1$ place $[a_j, a_j + \lfloor a_j^+ - a_j \rfloor] \times [j, j+1]$.
- (E6) *Walls.* Each of the three regions below is filled by ZC with $t_0 := \lceil Q^* \tilde{t} \rceil / Q^*$, where $\tilde{t}$ is within relative error 10^{-12} of the tilt t_{tgt} with $\tan \alpha(t_{\text{tgt}}) = a_w \sqrt{\zeta}$, and mapped by an exact rotation by $\pm 90^\circ$ and a translation:
 - Z_L : $\chi = \xi_0 - g_0 \tan \alpha$, $\zeta = \tan \alpha$, $L = g_0$, $(x, y) \mapsto (y, g_0 - x)$; region $\{\lambda \leq \xi_0, Y \leq g_0\}$;
 - Z_{LR} : $\chi = w - \ell_0$, $\zeta = \tau$, $L = J_L$, $(x, y) \mapsto (w - y, x)$; region $\{X + Y\tau \geq \ell_0, Y \leq J_L\}$;
 - Z_{UR} : $\chi = w - \xi_N + J_L \tan \alpha$, $\zeta = \tan \alpha$, $L = Y_u - J_L$, $(x, y) \mapsto (w - y, J_L + x)$, with $Y_u := (g_0 + \mu \xi_N) / (1 + \mu \tan \alpha)$; region $\{\lambda \geq \xi_N, J_L \leq Y \leq Y_u\}$.

The error of the wall tilts. Put $\epsilon' := 1.03 \cdot 10^{-12}$, $T_w := a_w \sqrt{\zeta} (1 + \epsilon') + \delta_Q$ and let $t_{\text{up}} := T_w / (1 + \sqrt{1 + T_w^2})$ be the tilt with $\tan \alpha(t_{\text{up}}) = T_w$ (the half-angle formula $t = \tan \alpha / (1 + \sec \alpha)$). Assume $t_{\text{up}} < 1/10$ (Corollary 8.10 checks this on every box, with a certified margin of at least 0.055). As $a_w \sqrt{\zeta} \leq T_w$ and $t \mapsto \tan \alpha$ is increasing, $t_{\text{tgt}} < t_{\text{up}} < 1/10$. On $t \in (0, 1/10]$ we have $d \log \tan \alpha / d \log t = (1 + t^2) / (1 - t^2) \leq 1.01/0.99 < 1.020203$; since $|\log(\tilde{t}/t_{\text{tgt}})| \leq -\log(1 - 10^{-12}) \leq 1.0000000000006 \cdot 10^{-12}$ and the mean value theorem applies on the interval between t_{tgt} and $\tilde{t}$ (which lies in $(0, 1/10]$, as $\tilde{t} \leq t_{\text{tgt}}(1 + 10^{-12})$), the relative error of $\tan \alpha(\tilde{t})$ is at most $\exp(1.020203 \cdot 1.0000000000006 \cdot 10^{-12}) - 1 \leq 1.0203 \cdot 10^{-12} \leq \epsilon'$. Finally $\tan \alpha(t_0) \leq \tan \alpha(\tilde{t}) + \delta_Q$, so

$$\tan \alpha_0 \in [a_w \sqrt{\zeta} (1 - \epsilon'), a_w \sqrt{\zeta} (1 + \epsilon') + \delta_Q], \quad t_0 \leq t_{\text{up}} < 1/10.$$

(Version 1.0 used $t \leq 0.0014$ and $\epsilon' = 1.00001 \cdot 10^{-12}$, which is valid only for very large b .)

Construction $E38'(y)$ is $E38(y)$ with every wall filled by ZC' (Section 8.2) instead of ZC .

Here $\lambda = \lambda(X, Y) = X + Y \tan \alpha$ is the slab coordinate of the main tilt. Hypotheses: (E-i) $1 < g_D < m$ and $t \leq 1/10$; (E-ii) $N \geq 1$; (E-iii) $\tan \alpha - \kappa_m \geq \tan \alpha(1/Q^*)$, so that $\tau > 0$; (E-iv) $\varepsilon_m + \tau < 1$, where $\varepsilon_m := (\sec \alpha - 1) + m\delta_Q + 1/Q_f$; (E-v) each of the three walls has $\chi \geq 4$ and $t_{\text{up}} < 1/10$ and satisfies (W1)–(W5), for Z_{LR} and Z_{UR} with L replaced by the upper bound m (which only strengthens (W2) and (W3)). If (E-iii) fails, $E38(y)$ is not defined (the wall Z_{LR} would have $\zeta = \tau \leq 0$, outside Lemma 8.1); Lemma 8.9 and Corollary 8.10 assume (E-iii) and never use that case.

Lemma 8.9 (end lemma for the exponent $3/8$). *Under (E-i)–(E-v), $E38(y)$ is a packing in $\text{Reg}(y)$, and its uncovered area satisfies*

$$U(y) \leq w \sin \alpha + wq + \Delta(\sec \alpha + m \tan \alpha) + J_0(1 + \tan \alpha) + J_L(\varepsilon_m + \tan \alpha) + \frac{1}{2}\mu h_L^2 + \mu w_R^2 \\ + U_{Z_L} + U_{Z_{\text{LR}}} + U_{Z_{\text{UR}}},$$

where $h_L := \xi_0 - g_0 \tan \alpha$, $w_R := w - \xi_N + \bar{g} \tan \alpha$, and each $U_Z \leq B_W$ by Lemma 8.1. The same holds for $E38'(y)$, with each $U_Z \leq B_W''$ by Lemma 8.5.

The proof uses Lemma 8.1 only through “the pieces of a wall form a packing in the wall region and $U_Z \leq B_W$ ”, and it uses the parameters only through $D_\ell, D_r \geq 0$ (integers) and $t_0 \in (0, 1/10]$. So it applies verbatim to $E38'(y)$ with Lemma 8.5 in place of Lemma 8.1.

Proof. (a) *Main staircase.* As in Lemma 7.3(a), with g_0 replaced by g_D and the shift D_ℓ : $\Gamma_i = g_D + i\mu \sec \alpha$ is affine in i ; $J_i \in [0, m-1]$, because $H(0) = m \cos \alpha + \sin \alpha \leq g_D \leq \Gamma_i$ by Lemma 7.2 and $H(m) = m + \sin \alpha > \bar{g} \geq \Gamma_i$; the J_i are non-decreasing; the gaps lie in $[0, q]$; and the columns lie in $\text{Reg}(y)$ ($X \geq \xi_i - m \tan \alpha = D_\ell + i \sec \alpha \geq 0$, $X \leq \xi_i + \sec \alpha \leq w - D_r$). The derivation of (R) in the proof of Lemma 8.1(b) applies verbatim to the rows $J_0 \leq j < J_L$ (with $p(j) = \lceil B(j+1-A) \rceil \in [1, N-1]$, $B(\sec \alpha - 1) = q^2/\mu = \kappa_m$, $\eta := \hat{\kappa} - \kappa_m \in [0, \delta_Q)$, $J_L - J_0 \leq m-1$, and $\hat{\xi}$ replaced by ξ_{N-1}): the length $\ell_j := a_j^+ - a_j$ satisfies $\text{frac}(\ell_j) \leq \varepsilon_m$. Moreover $\ell_j \geq \ell_0 - \xi_{p(j)} - \tau \geq -\tau$ (as $p(j) \leq N-1$); together with $\text{frac}(\ell_j) \leq \varepsilon_m < 1 - \tau$ (E-iv) this forces $\ell_j \geq 0$. The rows $j < J_0$ ($p(j) = 0$) are not aligned and waste at most $1 + \tan \alpha$ each. Row ceiling: for $j < J_L$, $p(j) \leq N-1$ and $J_{p(j)} \geq j+1$, so $j+1 \leq H(J_{p(j)}) \leq \Gamma_{p(j)} = g(\xi_{p(j)} - m \tan \alpha) \leq g(a_j)$ (as $j < m$). Also $a_j \geq \xi_0 - m \tan \alpha = D_\ell \geq 0$ and $a_j^+ \leq \ell_0 \leq w$.

(b) *The wall regions lie in $\text{Reg}(y)$.* $Z_L = \{0 \leq Y \leq g_0, 0 \leq X \leq \xi_0 - Y \tan \alpha\}$ and $Y \leq g_0 \leq g(X)$. $Z_{\text{LR}} = \{0 \leq Y \leq J_L, \ell_0 - Y\tau \leq X \leq w\}$: its points have $X \geq \ell_0 - J_L\tau \geq \xi_{N-1} - m \tan \alpha$, so $g(X) \geq \Gamma_{N-1} \geq H(J_L) \geq J_L \geq Y$. $Z_{\text{UR}} = \{J_L \leq Y \leq Y_u, \xi_N - Y \tan \alpha \leq X \leq w\}$: by the definition of Y_u , $Y_u = g(\xi_N - Y_u \tan \alpha)$, so $Y \leq Y_u \leq g(\xi_N - Y \tan \alpha) \leq g(X)$; and $J_L \leq \Gamma_{N-1} \leq g(\xi_N - J_L \tan \alpha)$ gives $J_L \leq Y_u$ (the map $Y \mapsto g(\xi_N - Y \tan \alpha) - Y$ is decreasing), that is $L \geq 0$. The maps are isometries, so by Lemma 8.1 every wall piece lies in its region.

(c) *Disjointness.* Main columns: slabs. Main rows against columns: as in Lemma 7.3(e) on the left (the line $\lambda = \xi_{p(j)}$, and columns with $J \geq j+1$ above $Y = j+1$); on the right the rows lie in $\{X + Y\tau \leq \ell_0\}$, the columns $i \leq N-2$ in $\{\lambda \leq \xi_{N-1}\} \subset \{X + Y\tau \leq \ell_0\}$ (as $\tau \leq \tan \alpha$, $Y \geq 0$ and $\ell_0 \geq \xi_{N-1}$), and column $N-1$ lies in $\{Y \geq J_L\}$ while the rows lie in $\{Y \leq J_L\}$. $Z_L \subset \{\lambda \leq \xi_0\}$, and all main pieces lie in $\{\lambda \geq \xi_0\}$. $Z_{\text{LR}} \subset \{X + Y\tau \geq \ell_0\}$, against the main rows and the columns $i \leq N-2$; against column $N-1$, Z_{UR} and the main rows the line $Y = J_L$ is used where needed. $Z_{\text{UR}} \subset \{\lambda \geq \xi_N\}$, against the main columns (in $\{\lambda \leq \xi_N\}$), and against the rows by $Y = J_L$. Z_L against Z_{LR} : Z_{LR} lies in $\{\lambda \geq \xi_{N-1} \geq \xi_0\}$. Z_L against Z_{UR} : $\lambda = \xi_0 \leq \xi_N$. Z_{LR} against Z_{UR} : $Y = J_L$.

(d) *Accounting.* Partition $\text{Reg}(y)$ into $\{\lambda < \xi_0\}$ ($= Z_L$ plus $S_1 := \{\lambda < \xi_0, Y > g_0\}$); $\{\lambda \geq \xi_N, Y \geq J_L\}$ ($= Z_{\text{UR}}$ plus S_2 , its part with $Y > Y_u$); $\{X + Y\tau > \ell_0, Y < J_L\}$ ($= Z_{\text{LR}}$ up to the boundary); the column cells $\{\xi_i \leq \lambda < \xi_{i+1}, Y \geq J_i\}$ minus the previous set (for $i \leq N-2$ the cell

does not meet it; for $i = N - 1$, $Y \geq J_L$); and the row regions $\{j \leq Y < j + 1, \lambda \geq \xi_{p(j)}, X + Y\tau \leq \ell_0\}$, $0 \leq j < J_L$. The cells cost at most $w \sin \alpha + wq + \Delta(\sec \alpha + m \tan \alpha)$, exactly as in Lemma 7.3(g) ($N \sec \alpha \leq w$, $N \tan \alpha \leq w \sin \alpha$). A row region has sections of length $\ell_0 - \xi_{p(j)} + Y\hat{\kappa}$, so its area is at most $\ell_j + \frac{1}{2}(\tan \alpha + \tau)$ and its waste at most $\varepsilon_m + \tan \alpha$ for $J_0 \leq j < J_L$, and at most $1 + \tan \alpha$ for $j < J_0$. S_1 consists of points with $0 \leq X < \xi_0 - g_0 \tan \alpha = h_L$ and $g_0 < Y \leq g_0 + \mu X$: area at most $\frac{1}{2}\mu h_L^2$. S_2 consists of points with $X \in [\xi_N - \bar{g} \tan \alpha, w]$ and $Y_u < Y \leq g(X) \leq Y_u + \mu w_R$: area at most μw_R^2 . The three wall regions contribute $U_{Z_L} + U_{Z_{LR}} + U_{Z_{UR}}$ (the maps are isometries). $\square$

8.5. The band ends. A *parameter row* consists of a threshold $b_0 \geq 10^4$, a window coefficient $c_y \in (0, 3/4]$, wall widths $c_D, c_R > 0$, a wall tilt factor $a_w > 0$ and the proof parameter $\varpi \in (0, 1)$ of Lemma 8.1. In the rows marked $\dagger$ the walls are filled by ZC' (Lemma 8.5) instead of ZC . The rows used in this paper are those of Table 3; the row “v1.0” is the choice of version 1.0.

Corollary 8.10. *Let one row of Table 3 be given, with its constant C_E . Let $b \geq b_0$ be an integer, $\tan(\theta/2) = b/(b^2 - 1)$, and let y be any real number with*

$$c_y b^{4/5} \leq y \leq c_y b^{4/5} + 2.$$

Let Q be a power of 2 with $Q \geq 16b$. Then $E38(y)$ ($E38'(y)$ in the rows marked $\dagger$) with the parameters c_D, c_R, a_w of the row is a packing in $\text{Reg}(y)$ with

$$U(y) \leq C_E b^{3/5},$$

and the end cost of Proposition 6.3 satisfies $E(y) = U(y) + \frac{1}{2} \tan \theta \leq C_E b^{3/5} + 1.0001/b$.

Proof. Put $z := m^{-1/4}$.

(i) *Ranges.* By the band facts of Corollary 7.4(i) (they need $b \geq 10^4$; every b_0 of Table 3 is at least $10^{7.7}$), $2/b < \mu \leq 2.000001/b$, $\Delta \leq 2.000001$, $\sigma\mu \leq 4.1 \cdot 10^{-8}$ and $b - 2.1/b \leq w < b$. As $\bar{g} = y - \sigma\mu + \Delta \in (y, y + 2.000001]$ and $m = \lfloor \bar{g} \rfloor + 1$, we get $c_y b^{4/5} < m \leq c_y b^{4/5} + 5.000001$, $\delta := m - g_0 \leq 3.000001$ and $m - g_D \leq \delta$. From $m > c_y b^{4/5} \geq m - 5.000001 = m(1 - 5.000001z^4)$,

$$\hat{b} := bz^5 \in [c_y^{-5/4}(1 - 5.000001z^4)^{5/4}, c_y^{-5/4}], \quad z < c_y^{-1/4}b^{-1/5} \leq z_0 := c_y^{-1/4}b_0^{-1/5},$$

and $D_\ell z^3 \in (c_D - z^3, c_D]$, $D_r z^3 \in (c_R - z^3, c_R]$. *Main tilt.* Let $D' = m + g_D$ and $\delta' = m - g_D$, so that $t^* = (1 + \sqrt{1 + D'\delta'})/D'$ (Lemma 7.2 with g_D in place of g_0). From below, $t^* \geq \sqrt{\delta'/D'}$ with $D' \leq 2m$ and $\delta' > \bar{g} - g_D = (w - D_\ell)\mu \geq 2(1 - D_\ell/b - 2.1/b^2)$, where $D_\ell/b \leq c_D z^2/\hat{b}$ and $2.1/b^2 = 2.1z^{10}/\hat{b}^2$. From above, t^* is increasing in δ' and decreasing in D' (proof of Lemma 7.2); we substitute $D' \geq 2m - 3.000001$ and $\delta' \leq 3.000001$ directly into the formula for t^* (not into the second bound of Lemma 7.2) and use $t \leq t^* + 1/Q$ and $1/Q \leq 1/(16b)$. This gives

$$\frac{t}{z^2} \in \left[\sqrt{1 - c_D z^2/\hat{b} - 2.1z^{10}/\hat{b}^2}, \frac{z^2 + \sqrt{z^4 + 3.000001(2 - 3.000001z^4)}}{2 - 3.000001z^4} + \frac{z^3}{16\hat{b}} \right].$$

Further $\kappa_m = q^2/\mu \leq q^2b/2$ and $\tau \in [\tan \alpha - \kappa_m - \delta_Q, \tan \alpha - \kappa_m]$. Wall inputs: Z_L : $\zeta = \tan \alpha$, $\chi \in [D_\ell, D_\ell + 3.000001 \tan \alpha]$, $L \leq m$; Z_{LR} : $\zeta = \tau$, $\chi \in [D_r + \sec \alpha - 1, D_r + 2 \sec \alpha]$, $L = J_L$; Z_{UR} : $\zeta = \tan \alpha$, $\chi \in [D_r, D_r + \sec \alpha + m \tan \alpha]$, $L = Y_u - J_L$; and $J_L + (Y_u - J_L) = Y_u \leq \bar{g} < m$. Wall tilts: by (E6), $\tan \alpha_0 \in [a_w \sqrt{\zeta}(1 - \epsilon'), a_w \sqrt{\zeta}(1 + \epsilon') + \delta_Q]$ with $\epsilon' = 1.03 \cdot 10^{-12}$, provided $t_{\text{up}} < 1/10$, which (iii) checks. (E-i): $m - 3.000001 \leq g_0 \leq g_D \leq \bar{g} < m$, and $g_D > 1$ because $m > c_y b_0^{4/5} > 10^5$ in every row.

(ii) *Normalisation.* Every term of Lemma 8.9 and of B_W (with the ϖ of the row) (of B_W'' in the rows marked $\dagger$) is multiplied by $z^3 = m^{-3/4}$ and rewritten in terms of $\hat{b}$, t/z^2 , $\tan \alpha/z^2$, ζ/z^2 , χz^3 , $Lz^4 \in [0, 1]$, $\tan \alpha_0/z$, q_0/z^2 , $q_{\min}/z^2$, $N_0 z^3$ and Λz^3 , so that only non-negative powers of z occur. The complete list of the normalised inequalities is Appendix A. For example $w \sin \alpha z^3 \leq \hat{b} \sin \alpha/z^2$, $\bar{K} = (Lz^4)/(\Lambda z^3) \cdot z^{-1}$, and $J_0 \leq (4mt + 2)/(Qq)$, because $H(0) = m \cos \alpha + \sin \alpha$ has $|dH/dt| \leq 4mt + 2$ and $t - t^* \leq 1/Q$. For the two right walls the terms linear in L are bounded with the

larger of the two coefficients (as $L_{LR} + L_{UR} \leq m$) and the other terms are evaluated at $L = m$ (the bounds of Lemmas 8.1 and 8.5 and the hypotheses (W2), (W3) are non-decreasing in L).

(iii) *Interval evaluation.* $(0, z_0]$ is split into N_Z equal intervals ($z = 0$ included) and, on each, the range of t/z^2 from (i) into N_T equal pieces; on each box all enclosures of Appendix A are evaluated in interval arithmetic. On every box of the row the lower bounds of the following margins are positive: $1/10 - t$; (E-ii); (E-iii); (E-iv) in the form $1 - \varepsilon_m - \tau$; and, for each of the three walls, (W1)–(W5), $\chi - 4$ and $1/10 - t_{\text{up}}$. The smallest margins and Ψ_{sup} , the supremum over the boxes of the upper bound of $\Psi(z) := z^3 \times (\text{the bound of Lemma 8.9})$, are in Table 3.

(iv) By Lemma 8.9,

$$\begin{aligned} U(y) &\leq \Psi_{\text{sup}} m^{3/4} \leq \Psi_{\text{sup}} (c_y b^{4/5} + 5.000001)^{3/4} \\ &\leq \Psi_{\text{sup}} c_y^{3/4} \left(1 + \frac{5.000001}{c_y b_0^{4/5}}\right)^{3/4} b^{3/5} =: C_E^{\text{cert}} b^{3/5} \leq C_E b^{3/5}. \end{aligned}$$

For $c_y = 1/12$ this is the factor $12^{-3/4}(1 + 60.000012 b_0^{-4/5})^{3/4}$ of version 1.0. The relation between E and U is as in Corollary 7.4(vi). $\square$

row	b_0	c_y	$c_D = c_R$	a_w	ϖ	boxes	Ψ_{sup}	C_E^{cert}	C_E	(E-iii)	(W2)	t_{up}
B1 [†]	$10^{7.7}$	0.170157	2.92683	1.25781	0.65	320×128	65.9031	17.46022	17.461	0.592	0.0173	0.0558
B1s [†]	$10^{7.9}$	0.16	2.92683	1.26	0.65	320×128	67.0681	16.96722	16.968	0.593	0.0167	0.0590
B2 [†]	10^8	0.154681	2.92683	1.26562	0.65	320×128	67.7915	16.72079	16.721	0.584	0.0141	0.0603
A1	$10^{8.5}$	0.133923	2.90865	1.13636	0.681818	320×128	85.6481	18.96099	18.961	0.611	0.0144	0.0706
A1s	$10^{8.58}$	0.131625	2.90865	1.13636	0.681818	320×128	85.6691	18.72102	18.722	0.631	0.0146	0.0716
A2	10^9	0.116939	2.90909	1.13636	0.681818	320×128	87.6231	17.52222	17.523	0.661	0.0156	0.0759
A3	10^{10}	0.0957429	2.61999	1.13	0.65	320×128	92.1363	15.85845	15.859	0.926	0.0157	0.0841
A4	10^{12}	0.0769401	2.60016	1.13	0.65	320×128	101.1582	14.77799	14.778	1.606	0.0145	0.0933
A5	10^{16}	0.0838149	2.86364	1.25	0.65	320×128	90.8528	14.15237	14.153	1.947	0.0156	0.0988
v1.0	10^{16}	1/12	4	3/2	0.65	160×64	96.4769	14.96365	15.54	1.946	0.1314	0.0986

TABLE 3. The parameter rows of Corollary 8.10 and the values of the author’s covering. Ψ_{sup} and C_E^{cert} are upper bounds, rounded up; C_E is the constant used in Corollary 8.10 and Theorem 1.1. The last three columns are lower bounds over all boxes, rounded down, of the margins (E-iii) (normalised by z^2), (W2) (normalised by z , smallest of the three walls) and $1/10 - t_{\text{up}}$ (smallest of the three walls). The margin $1 - \varepsilon_m - \tau$ is at least 0.9969, and all other margins are positive on every box. Rows marked [†] use ZC' and Lemma 8.5. The row v1.0 is the parameter choice of version 1.0; its $C_E = 15.54$ is that of [U].

Certification. The values of Table 3 were computed by the author’s program `cert_v3.py` (`mpmath.iv`, 120 bits; for each row 320×128 boxes, run in two slices of 160 z -intervals), and merged by `finalize_v3.py`, which checks that the slices cover every box exactly once and checks the stated constants of Theorem 1.1 in interval arithmetic. With the row v1.0 (160×64 boxes) the program reproduces $\Psi_{\text{sup}} = 96.47689\dots$ and $C_E^{\text{cert}} \leq 14.96365$ of the program `cert3e.py` of version 1.0, which was written independently. The constants are certified by the author’s covering and by three coverings written independently from the text: two by verification agents (`mpmath.iv` and `python-flint arb`) and one by a reviewer (`mpmath.iv`), and all three implement the same normalised inequalities (the list of Appendix A), so they check the arithmetic and the coverage but are only a weak check of the transcription of these inequalities from the lemmas; the point checks of the un-normalised formulas (at 90, 120 and 5850 points) partly address that. The reviewer’s covering uses the same boxes and reproduces Ψ_{sup} , C_E^{cert} and the margins of every row to the printed

digits, and the worst-box split to within one unit in the last printed digit; the two other coverings use adaptive subdivisions and give slightly smaller values of C_E^{cert} (see “Verification of version 1.1” at the end of the paper). All programs were written by AI systems of the same family. In every row the worst box is the last z -interval (b near b_0) at the top of the t -range; for the row A1s, for example, the bound splits there as column triangles 30.909, other main terms 0.287, left wall 23.608 and right walls 30.867 (in units of $m^{3/4}$). Exponent bookkeeping: the triangles cost about $bt \asymp bm^{-1/2} \asymp m^{3/4}$ (as $m \asymp b^{4/5}$), and each wall about $L \tan \alpha_0 + \bar{K} N_0 \tan \alpha_0 + \bar{K}/q_{\min} + 2\chi \asymp m^{3/4}$, with $\tan \alpha_0 \asymp m^{-1/4}$, $\chi \asymp m^{3/4}$ and $\bar{K} \asymp m^{1/4}$.

Why these thresholds. The threshold k_i enters the proof only through $b_* \geq b_0$ (Section 8.7); the other size conditions (the band facts, the room for the lifts, $\lambda' < 1$, $Q \geq 16b$) hold far below. With the parameters of version 1.0 (window coefficient $1/12$, $c_D = c_R = 4$, $a_w = 3/2$) the hypothesis that fails for smaller b_0 is (E-iii), $\tau > 0$: $\kappa_m/\tan \alpha$ is about $1.84 c_y^{-3/2} b^{-1/5}$ (about $76 b^{-1/5}$ for $c_y = 1/12$), and the covering fails for b below about $2.7 \cdot 10^9$. A larger window coefficient removes this obstruction (the threshold of (E-iii) on the covering scales like $20.9 c_y^{-15/2}$), at the price of larger column triangles; this is why the rows of Table 3 have c_y between 0.0769 and 0.1702. This threshold is only a rough scale for (E-iii) to hold on the whole window, as the covering requires; for a single lift y , whether $\tau > 0$ depends on y (through $m - g_D$) and is not monotone in b , and aligned cuts inside the window occur far below this scale (Remark 8.13). For smaller b_0 the remaining cost is the growth of C_E with z_0 : the terms of B_W of order $m^{1/2}$, in units of $m^{3/4}$, are of order z with large coefficients. Lemmas 8.3 and 8.4 reduce these coefficients: for the row B2 with B_W in place of B_W'' an independent covering gives $C_E^{\text{cert}} \approx 22.50$ instead of 16.72. Hypothesis (W2) is not the obstruction to small b_0 , but it is the binding constraint of the parameter choice: the rows were found by a local search under the side condition that the normalised margin of (W2) be at least 0.012 on a coarser cover, which costs about 1.4% in C_E or less compared with the best parameters found without it. The parameters are not optimised further, and the per-row bounds of Lemma 8.1 remain crude. With the parameters of version 1.0 the certified bound tends to about 14.6 as $b_0 \rightarrow \infty$; this reflects the estimates of the proof, not $c^*(k)$.

8.6. Lifts with $\text{frac}(\bar{g}) \geq 1/2$ and the tiers marked $\ddagger$. This subsection is new in version 1.1 and is used only by the tiers marked $\ddagger$. Choosing the lift of each band so that $\text{frac}(\bar{g}) \geq 1/2$ at both of its ends lowers $\delta = m - g_0$ from 3.000001 to 2.500001 and the main tilt accordingly.

Band notation. We use the strip frame of Section 6: a band of length L_b , with h and $d = \sec \theta$, a lift $y_0 \geq 0$ with $L_b \geq h + 2y_0$, $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$ rows and $y_1 := L_b - h - (R-1)d - y_0 \in [y_0, y_0 + d]$; the bottom end is $\text{Reg}(y_0)$ and the top end is the image of $\text{Reg}(y_1)$. Both ends have the same function $\bar{g}(y) = y + c_0$ with $c_0 := (w - \sigma)\mu$, and $1 < d < 1.0001$ for $b \geq 10^4$.

Real lifts. Up to here the lift was a quarter-integer. Lemma 8.11 below gives a real y_0 (rational for integer b and k , but with no bounded denominator). Nothing in the results used depends on y_0 being a multiple of $1/4$ or lying on a grid: the strip frame of Section 6 takes any lift $y_0 \geq 0$ and any integer $R \geq 1$ with $y_1 \geq 0$, and $\text{Reg}(y)$ for any real $y \geq 0$; Lemma 6.1 uses $L_b = y_0 + y_1 + (R-1)d + h$ and separating lines that are affine in y_0 ; Lemmas 3.1 and 3.3 hold for every base point; Corollary 6.2 asks only $y_0 \geq 0$; the area $A_0(y)$ in Proposition 6.3 is affine in y , and the waste identity is an area decomposition, valid for real y_0, y_1 . Lemma 8.9 and Corollaries 8.10, 8.12 take any real y in their windows; the grids Q, Q^*, Q_f are grids of tilts and of phase offsets, not of heights. Two independent AI readings (one of them made without this text or the earlier reviews) checked the cited results line by line: they hold for every real lift and use no lattice, rationality or integrality; they need only (1) $y_0, y_1 \geq 0$, (2) $L_b \geq h + 2y_0$, (3) $y \geq \sigma\mu$ at both ends for the area identity of Proposition 6.3 (below that only $E(y) \leq U(y) + \frac{1}{2} \tan \theta$ holds, which is the direction used), and (4) $\bar{g}(y) \leq L_b$ at both ends. For the lift of Lemma 8.11 with $\alpha_0 = c_y b^{4/5}$, as used in Section 8.7, these hold: (1) $y_1 \geq y_0 \geq \alpha_0 > 0$; (2) is step (1) of the proof of Lemma 8.11; (3) $y \geq \alpha_0 \geq c_y b_0^{4/5} > 8 \cdot 10^4$, while $\sigma\mu \leq 4.1 \cdot 10^{-8}$ by

Corollary 7.4(i); (4) $\bar{g}(y_0) \leq \bar{g}(y_1) \leq y_1 + \Delta \leq y_0 + 3.000001 < h + 2y_0 + 3 < k - b \leq L_b$ by the room argument of Section 8.7. Moreover, the lemma is applied to each band separately: Lemma 6.1 and Proposition 6.3 are statements about one band, and Corollary 6.2, written there with one common y_0 , uses of each band only that its squares lie in its region; so the two bands may have different lifts, and $k^2 - N = (k^2 - S^2) + \sum_{2 \text{ bands}} [(R-1) \tan \theta + E(y_0) + E(y_1)]$ holds with each band's own y_0, y_1, R .

Lemma 8.11 (lift choice). *Let $b \geq 10^4$, a band (L_b, h, d) as above, and $\alpha_0 \geq 0$ real with $L_b - h - 1 \geq 2\alpha_0$. Then there is a real y_0 with $\alpha_0 \leq y_0 < \alpha_0 + (d+1)/2$ such that, with R and y_1 defined from y_0 : $R \geq 1$, $y_1 - y_0 \in \{0, 1\}$, and*

$$\text{frac}(\bar{g}(y_0)) = \text{frac}(\bar{g}(y_1)) \geq \frac{1}{2}.$$

In particular y_0 and y_1 lie in $[\alpha_0, \alpha_0 + 2.0001]$.

Proof. For an integer $R \geq 1$ put $S_R := L_b - h - (R-1)d$, $K_R := S_R + 2c_0$, $\epsilon_R := 0$ if $\lfloor K_R \rfloor$ is odd and $\epsilon_R := 1$ if it is even, and $Y_R := (S_R - \epsilon_R)/2$. (1) If $y_0 := Y_R$, then $L_b - h - 2y_0 = (R-1)d + \epsilon_R \in [(R-1)d, Rd)$ (as $0 \leq \epsilon_R \leq 1 < d$), so the row count is R , $y_1 = S_R - y_0 = y_0 + \epsilon_R$, and $L_b - h - 2y_0 \geq 0$. (2) $\bar{g}(Y_R) = (K_R - \epsilon_R)/2$. The integer $n := \lfloor K_R - \epsilon_R \rfloor$ is odd, and $\phi_R := \text{frac}(K_R - \epsilon_R) = \text{frac}(K_R)$; so $\bar{g}(Y_R) = (n-1)/2 + (1 + \phi_R)/2$ with $(n-1)/2$ an integer and $(1 + \phi_R)/2 \in [1/2, 1)$, that is $\text{frac}(\bar{g}(y_0)) = (1 + \phi_R)/2 \geq 1/2$; and $\bar{g}(y_1) = \bar{g}(y_0) + \epsilon_R$. (3) $Y_R - Y_{R+1} = (d - \epsilon_R + \epsilon_{R+1})/2 \in [(d-1)/2, (d+1)/2]$ with $(d-1)/2 > 0$, so Y_R decreases strictly to $-\infty$ in steps of at most $(d+1)/2$. (4) $Y_1 \geq (L_b - h - 1)/2 \geq \alpha_0$. Let $R^* := \max\{R \geq 1 : Y_R \geq \alpha_0\}$ and $y_0 := Y_{R^*}$. Then $\alpha_0 \leq y_0 \leq Y_{R^*+1} + (d+1)/2 < \alpha_0 + (d+1)/2$, and $y_1 \leq y_0 + 1 < \alpha_0 + (d+3)/2 < \alpha_0 + 2.00005$. $\square$

The case $y_1 = y_0 + 1$ is rare but covered by the proof. The bounds are nearly attained: $\text{frac}(\bar{g}) = 1/2$ occurs.

Corollary 8.12. *Let one row of Table 4 be given: b_0, c_y, c_D, c_R , wall tilt factors $a_{w,L}, a_{w,LR}, a_{w,UR}, \varpi$ and C_E . Let $b \geq b_0$ be an integer and y real with*

$$c_y b^{4/5} \leq y \leq c_y b^{4/5} + 2.0001 \quad \text{and} \quad \text{frac}(\bar{g}(y)) \geq \frac{1}{2}.$$

Let Q be a power of 2 with $Q \geq 16b$. Then $E38'(y)$, with $D_\ell = \lfloor c_D m^{3/4} \rfloor$, $D_r = \lfloor c_R m^{3/4} \rfloor$ and in (E6) the factor $a_{w,L}$ for Z_L , $a_{w,LR}$ for Z_{LR} and $a_{w,UR}$ for Z_{UR} , is a packing in $\text{Reg}(y)$ with $U(y) \leq C_E b^{3/5}$, and $E(y) \leq C_E b^{3/5} + 1.0001/b$.

Lemma 8.9 holds verbatim with a different factor a_w for each wall (its proof does not use the value of a_w), and, if each wall also satisfies (W6), with each $U_Z \leq B_{W4}$ of Lemma 8.8, for the same reason as for B_W'' .

Proof. As Corollary 8.10 (rows with $\dagger$), with Lemma 8.8 in place of Lemma 8.5 and these changes. (i) $\bar{g} = y - \sigma\mu + \Delta \in (y, y + 2.000001]$; since $\text{frac}(\bar{g}) \geq 1/2$, $m = \lfloor \bar{g} \rfloor + 1 = \bar{g} + 1 - \text{frac}(\bar{g}) \leq \bar{g} + \frac{1}{2} \leq c_y b^{4/5} + 4.500101$, and $m > \bar{g} > y \geq c_y b^{4/5}$; the covering uses $m \leq c_y b^{4/5} + 4.5002$, that is $\hat{b} \in [c_y^{-5/4}(1 - 4.5002z^4)^{5/4}, c_y^{-5/4}]$. Further $\delta = m - g_0 = 1 - \text{frac}(\bar{g}) + \Delta \leq 2.500001$ and $m - g_D \leq \delta$; the formula for t^* in Corollary 8.10(i) is increasing in $m - g_D$, and with 2.500002:

$$\frac{t}{z^2} \leq \frac{z^2 + \sqrt{z^4 + 2.500002(2 - 2.500002z^4)}}{2 - 2.500002z^4} + \frac{z^3}{16b}.$$

(E-i) holds as $m - 2.500001 \leq g_0 \leq g_D \leq \bar{g} < m$ and $g_D > 1$, since $m > c_y b_0^{4/5} > 8 \cdot 10^4$ in every row of Table 4; the band facts need $b \geq 10^4$, and $b_0 \geq 10^{7.15}$ there. The lower end of the t -range and the wall inputs are as in Corollary 8.10 (Z_L keeps the wider enclosure $\chi \in [D_\ell, D_\ell + 3.000001 \tan \alpha]$); (E6) is applied to each wall with its own a_w . (ii) The normalised inequalities are those of Appendix A with the replacements of Appendix B. (iii) The covering checks, on every box, all margins

of Corollary 8.10(iii) and (W6) for each wall. (iv) $U(y) \leq \Psi_{\sup} c_y^{3/4} (1 + 4.5002/(c_y b_0^{4/5}))^{3/4} b^{3/5} =: C_E^{\text{cert}} b^{3/5} \leq C_E b^{3/5}$. $\square$

row	b_0	c_y	$\Psi_{\sup}$	C_E^{cert}	C_E	(E-iii)	(W2)	(W6)	t_{up}
C1 [‡]	$10^{7.3617}$	0.153846	58.9138	14.47251	14.473	0.482	0.0162	0.591	0.0456
C2 [‡]	$10^{7.2}$	0.16	58.5182	14.80459	14.805	0.454	0.0157	0.572	0.0420
C3 [‡]	$10^{7.15}$	0.165	57.8716	14.98288	14.983	0.494	0.0156	0.596	0.0411

TABLE 4. The rows of Corollary 8.12 (tiers marked [‡]) and the values of the author's covering (320×128 boxes). All rows have $c_D = 2.81426$, $c_R = 3.00931$, $a_{w,L} = 1.26$, $a_{w,LR} = 1.3608$, $a_{w,UR} = 1.35403$ and $\varpi = 0.65$. $\Psi_{\sup}$ and C_E^{cert} are upper bounds, rounded up; the last four columns are lower bounds of the margins (normalised as in Table 3; (W6) normalised by z), rounded down. The margin $1 - \varepsilon_m - \tau$ is at least 0.9961, and all other margins are positive on every box.

Certification of the rows [‡]. The values of Table 4 were computed by the author's program `cert_v4.py` (`mpmath.iv`, 120 bits, 320×128 boxes per row in two slices); its merge step checks the cover of the boxes and, in interval arithmetic, the stated constants of Theorem 1.1. They are certified by the author's covering and by three coverings written independently from the text: two by verification agents (`python-flint arb`: $C_E^{\text{cert}} \leq 14.46634, 14.79786, 14.97606$; `mpmath.iv`: $14.469520, 14.801342, 14.979582$, for the rows C1, C2, C3) and one by a reviewer (`mpmath.iv`, which reproduces $14.47251, 14.80459, 14.982873$), all passing; all three implement the same normalised inequalities (Appendix B), so they are only a weak check of the transcription of these inequalities from the lemmas. For $b_0 \leq b \leq 10^{16}$ this limitation is addressed by a direct covering (`python-flint arb`) of the un-normalised bound of Lemma 8.9 with B_{W4} , in the variables $(\log_2 b, \text{frac}(\bar{g}))$, with the integer quantities enclosed exactly on each box and the actual wall lengths L , without Appendix B (only J_0 uses the inequality obtained from the mean value theorem in Corollary 8.10(ii)): the certified suprema of $U/b^{3/5}$ are 13.910 (C1), 14.235 (C2) and 14.319 (C3), all margins are positive, and at about $7.6 \cdot 10^4$ points per row the direct value is at most the normalised bound. For $b > 10^{16}$ the normalised coverings apply. All programs were written by AI systems of the same family. The window coefficients c_y were chosen by coarse coverings, and the wall parameters by a local search; they are not optimised further.

8.7. Proof of Theorem 1.1. Fix a tier i of Table 1, its parameter row (Table 3) with b_0 , c_y and $C_E := C_{E,i}$, and an integer $k \geq k_i$. Put $b_* := (5k/(3C_E))^{5/8}$ (so $k = \frac{3}{5}C_E b_*^{8/5}$) and $b := \lceil b_* \rceil$. As $k \geq k_i \geq \frac{3}{5}C_E b_0^{8/5}$, $b \geq b_* \geq b_0$. Take $\tan(\theta/2) = b/(b^2 - 1)$, so $w < b$ and $S := k - b + w < k$, and use the packing of Corollary 6.2 with

$$y_0 := \frac{1}{4} \lceil 4c_y b^{4/5} \rceil \in [c_y b^{4/5}, c_y b^{4/5} + \frac{1}{4}]$$

and, in each band, $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$. *Room for the lifts:* $b \leq b_* + 1$ gives $b^{4/5} \leq b_*^{4/5} + 1 = (5k/(3C_E))^{1/2} + 1$, so $y_0 \leq c_y (5/(3C_E))^{1/2} k^{1/2} + c_y + \frac{1}{4} \leq k^{1/2} + 1$, because $c_y (5/(3C_E))^{1/2} \leq 1$ and $c_y \leq 3/4$ (both hold in every row). As $h < 3$ (Section 7.3), $h + 2y_0 < 2k^{1/2} + 5 < k - b \leq L_b$ (as $k \geq 10^{13}$ and $b \leq b_* + 1 \leq k^{5/8} + 1$, since $5/(3C_E) < 1$); so $R \geq 1$ and $y_1 \in [y_0, y_0 + d]$ with $d < 1.0001$, and both lifts lie in the window of Corollary 8.10. Fill the four end regions by Corollary 8.10 with Q the least power of 2 that is at least $16b$. As in Section 7.3, by Corollary 6.2 and Proposition 6.3,

$$k^2 - N < \frac{4k}{b} + 4C_E b^{3/5} + \lambda', \quad \lambda' := \frac{4k}{b(b^2 - 1)} + \frac{4kb}{(b^2 - 1)^2} + 2 \tan \theta \leq \frac{8.001 k}{b^3} + \frac{4.1}{b}.$$

With $k \leq \frac{3}{5}C_E b^{8/5}$ and $b \geq b_0$ we get $\lambda' \leq 8.001 \cdot \frac{3}{5}C_E b_0^{-7/5} + 4.1/b_0$; in interval arithmetic this bound is less than 1 in every tier without $\ddagger$ (its largest value is about $8.32 \cdot 10^{-8}$, for the row with the smallest b_0); so $\lambda' < 1$, and Lemma 2.2 gives $c^*(k) \leq k^2 - N - 1 < 4k/b + 4C_E b^{3/5}$. Finally $4k/b \leq 4k/b_* = \frac{12}{5}C_E b_*^{3/5}$ and, by the mean value theorem, $b^{3/5} \leq b_*^{3/5} + \frac{3}{5}b_*^{-2/5} \leq b_*^{3/5} + \frac{3}{5}b_0^{-2/5}$, so

$$c^*(k) < \frac{32}{5}C_E b_*^{3/5} + \frac{12}{5}C_E b_0^{-2/5} = \frac{32}{5}\left(\frac{5}{3}\right)^{3/8} C_E^{5/8} k^{3/8} + \frac{12}{5}C_E b_0^{-2/5} \leq C_i k^{3/8} + a_i.$$

Here $\frac{32}{5}(\frac{5}{3})^{3/8} \approx 7.751277$. The inequalities $C_i \geq \frac{32}{5}(\frac{5}{3})^{3/8} C_E^{5/8}$, $k_i \geq \frac{3}{5}C_E b_0^{8/5}$, $a_i \geq \frac{12}{5}C_E b_0^{-2/5}$, the inequality “bound on λ' less than 1” and the two room conditions were checked in interval arithmetic for every tier without $\ddagger$ (`finalize_v3.py`, which also prints the value of the bound on λ'). $\square$

Version 1.0. The row v1.0 with $C_E = 15.54$ gives, by the same argument, the theorem of version 1.0: in interval arithmetic $\frac{32}{5}(\frac{5}{3})^{3/8} 15.54^{5/8} \in [43.05564, 43.05565]$, $\frac{3}{5} \cdot 15.54 \cdot 10^{128/5} \leq 3.712 \cdot 10^{26} \leq 4 \cdot 10^{26}$ and $\frac{12}{5} \cdot 15.54 \cdot 10^{-32/5} \leq 1.49 \cdot 10^{-5}$, so $c^*(k) < 43.06 k^{3/8} + 2 \cdot 10^{-5}$ for every integer $k \geq 4 \cdot 10^{26}$. It is superseded by the tier A5.

The tiers marked $\ddagger$. For these tiers the proof above is changed as follows. In each of the two bands (lengths $L_b = S$ and $L_b = k - b$) take the lift y_0 of Lemma 8.11 with $\alpha_0 := c_y b^{4/5}$, instead of $\frac{1}{4}[4c_y b^{4/5}]$; the two bands may get different lifts (Section 8.6). Then y_0 and y_1 lie in $[c_y b^{4/5}, c_y b^{4/5} + 2.0001]$ and $\text{frac}(\bar{g}(y_0)) = \text{frac}(\bar{g}(y_1)) \geq 1/2$, so Corollary 8.12 applies to all four ends. *Room:* $\alpha_0 \leq c_y(5/(3C_E))^{1/2} k^{1/2} + c_y \leq k^{1/2} + \frac{3}{4}$, and Lemma 8.11 needs $L_b - h - 1 \geq 2\alpha_0$, which holds as $L_b \geq k - b$, $h < 3$, $b \leq k^{5/8} + 1$ and $k \geq 10^{12}$ (every k_i of a tier $\ddagger$ is at least 10^{12} , checked). Then $y_0 < \alpha_0 + (d+1)/2 \leq k^{1/2} + 1.76$, $h + 2y_0 + 3 < 2k^{1/2} + 9.6 < k - b \leq L_b$, and each end region lies in its band. The count, the bound $\lambda' < 1$ (checked in interval arithmetic by the merge step of `cert_v4.py`, together with the stated constants, the room conditions and $k_i \geq 10^{12}$; its largest value is about $2.97 \cdot 10^{-7}$) and the final inequality are as above, with the C_E of the tier.

Remark 8.13 (what the explicit computations test). The range of Corollary 8.10 ($b \geq b_0 \geq 10^{7.7}$) has not been tested by explicit computation at the parameters of a tier and $b \geq b_0$; for Corollary 8.12 only the 15 end packings described at the end of the paper (b from $1.41 \cdot 10^7$ to $2.30 \cdot 10^7$, near b_0) were built. The constants are certified by interval arithmetic. The explicit checks test the validity of the construction and of the lemmas at smaller b . The construction E38 is defined only when (E-iii) holds; in some runs below (marked “with $\tau := 0$ ”) we used the *convention* $\tau := 0$, with the lower right wall filled by rows only, which is not part of the construction. Whether the cut is aligned ($\tau > 0$) depends on y and is not monotone in b .

Version 1.0 checks. The construction was implemented in exact rational arithmetic and checked by an exact checker that does not use the lemmas (shapes, containment, separating axis tests for all pairs with overlapping bounding boxes): for $b = 1600, \dots, 409600$ and five lifts each in the window of version 1.0 (all 25 pass, with $\tau := 0$, so these runs test validity only), at lifts outside the window where the cut is aligned (all pass, with the residuals of (R) attained to 4–5 digits), and for the wall filler alone on canonical regions for m up to 10^4 (certified) and $2.56 \cdot 10^6$ (exact internal checks of Lemma 8.1), with $U_Z \leq B_W$ in every case. Deliberately corrupted packings were rejected. Two blank-slate AI reviews and an independent reimplementations of version 1.0 found no counterexample. The program `run_z.py` builds the wall filler for $m = 10^4$, $\zeta = 0.5/\sqrt{m}$, $\chi = 4m^{3/4}$, $L = m$ and certifies it: 3 blocks, 25289 pieces, 28402 pairs decided exactly, no overlap, (W1)–(W5) hold, 7611 aligned closings with largest residual $0.0056087 \leq \varepsilon_W = 0.0056093$, and $U_Z = 6712 \leq B_W \leq 19615$. Full exact counts with the construction of version 1.0 give $c^*(10^8) \leq 44299$, $c^*(10^{10}) \leq 236005$ and $c^*(10^{11}) \leq 547881$ (ends checked exactly); at $k = 10^8$ this is worse than Theorem 1.6 (24790), and below k_i the closed forms of Theorem 1.1 must not be used.

Version 1.1 checks (programs in `code/zc_tests`). (a) E38 with the wall parameters of the tiers A1, A5 and A3 in windows $[c_y b^{4/5}, c_y b^{4/5} + 2]$ with $c_y = 1/3$ or $1/2$ (not a tier’s c_y) and an

aligned cut, for $b = 102400, 204800$ and 409600 : five end packings with up to $4.8 \cdot 10^5$ pieces, all checked exactly (no overlap, no containment failure, the checker's U equal to the construction's). (b) ZC' on canonical walls, $m = 3000, \dots, 2.56 \cdot 10^6$: $U_Z \leq B''_W$ in all eight cases (and $U_Z \leq B_W$ for ZC), of which (W1)–(W5) hold in six ((W5) fails in two); exact certificates for $m \leq 10^4$; ZC' places 1–25 rows that ZC leaves empty. (c) $E38'$ with the wall parameters of the tiers B2 and B1s, $b = 25600, \dots, 204800$, aligned cut: four end packings with ZC' walls and one with ZC walls for comparison, all checked exactly; at $b = 102400$ the rule $(Z4')$ lowers U from 16643 to 12322. (d) A blank-slate review implemented ZC' from the text in exact arithmetic (36 walls, m up to $1.6 \cdot 10^5$) and checked, besides validity, the exact partition identity of Lemma 8.1(f) and the per-row and per-block bounds of Lemmas 8.3 and 8.4: no violation; three deliberately wrong placement rules were caught. An independent reimplementation of ZC , ZC' , $E38$, $E38'$ and the full L-shaped packing, written from the text alone, is described under “Verification of version 1.1” at the end of the paper. It includes aligned-cut tests inside the windows of the tiers B1, B1s and B2 at $b \approx 1.5 \cdot 10^6$ – $4 \cdot 10^6$, below b_0 .

9. THE CERTIFIED PACKINGS

Generation. The end packings of Theorem 1.6 were produced by a modification of the construction of Section 7.1 with tuned parameters: b was chosen near the minimizer of a model of the total cost, and the lift y_0 (about $0.9b^{2/3}$) by a floating-point pre-scan; for each end the column length m is $\lfloor \bar{g} \rfloor$ or $\lfloor \bar{g} \rfloor + 1$, whichever leaves less uncovered area; the tilt is the least multiple t of 2^{-40} with $m \cos \alpha + \sin \alpha \leq g(\cos \alpha)$ (the top vertex of the first column, which starts at the left wall, has abscissa $\cos \alpha$); and each J_i is the largest value, at least J_{i-1} , for which the column (of $m - J_i$ squares) fits under the ceiling. These rules play no role in the verification, which uses only the resulting explicit pieces.

Proof of Theorem 1.6. Let (k, b, y_0) be a row of Table 2 and $\tan(\theta/2) = b/(b^2 - 1)$. Then $\tan(\theta/2) > 1/b$, so $\theta \in (\theta_0, \pi/2)$ and $w < b$ (Lemma 3.2), and γ, σ are rational. For both bands let $R := \lfloor (L_b - h - 2y_0)/d \rfloor + 1$, so that $y_0 \leq y_1 < y_0 + d$. The data file for k lists, for each of the four ends, its lift (y_0 for the lower ends, y_1 for the upper ends) and its pieces, with exact rational coordinates in the frame of $\text{Reg}(y)$: *columns* (a tilted $1 \times n$ rectangle, given by its lowest vertex, n and the rational cosine and sine of its tilt), *layers* and *wedge rows* (axis-parallel $1 \times L$ rectangles). By Corollary 6.2 and Lemma 2.2 it suffices to check, in exact rational arithmetic:

- (C0) the lift given in the file for each end equals y_0 (lower ends) or the computed y_1 (upper ends), and every n and L is a positive integer;
- (C1) $w < b$, $R \geq 1$, $y_0 \geq 0$ and $y_1 \geq 0$ for both bands;
- (C2) every piece is a rectangle with side lengths 1 and n (hence a union of n unit squares with pairwise disjoint interiors); for a column this means $\cos^2 + \sin^2 = 1$ exactly;
- (C3) the four vertices of every piece lie in $\text{Reg}(y)$ (which is convex);
- (C4) two pieces of the same end have disjoint interiors.

For (C4) it suffices to test pairs whose bounding boxes overlap, and for such pairs the separating axis test along the edge normals of the two rectangles is exact (touching is allowed). The count $N = (k - b)^2 + b(R_{\text{right}} + R_{\text{top}}) + (\text{end squares})$ gives the values $c = k^2 - N$ of Table 2, and Table 5 lists the resulting data. All checks pass (see below). $\square$

Verification. Three programs check the certificates in exact rational arithmetic.

- (a) The generating program checks, for each end, that all vertices lie in $\text{Reg}(y)$, that each column lies in its slab $\{\xi_i \leq \lambda \leq \xi_i + \sec \alpha\}$ and above height J_i , that the slabs do not overlap, that the J_i are non-decreasing, that each layer starts to the right of the slabs of the columns below it and each wedge row to the left of the first slab (the separating lines of Lemma 7.3(e)), and the exact uncovered area; it also checks the identity of Proposition 6.3 exactly.

k	R_{right}	R_{top}	pieces (4 ends)	body cost	end cost
10^5	99868	99245	2887	639.20	944.79
10^6	999671	997190	10111	1609.72	2363.28
10^7	9999166	9989291	42550	4048.29	5991.71
$3.825 \cdot 10^7$	38248577	38226491	93680	6925.21	10063.79
10^8	99997915	99958601	165152	10172.28	14618.72

TABLE 5. Data of the certificates. Body cost is $\sum(R-1)\tan\theta$ over the two bands and end cost is the sum of the four $E(y)$ of Proposition 6.3; their sum is c up to $k^2 - S^2 < 0.002$ (rounded to two decimals). The average of $E/b^{2/3}$ over the four ends is between 3.1 and 3.3.

- (b) A checker written from a description of the construction, without reading the generating program, takes only the parameters of the pieces from it, builds the vertices itself and checks (C0)–(C4) with the separating axis test for all pairs of pieces with overlapping bounding boxes, the exact uncovered area, and the count. It reproduced N and c for all five rows (for $k \geq 10^7$ on a remote machine), and rejected deliberately corrupted end packings for $b = 400$ and 6400 (a column shifted, lowered or lengthened, a layer raised, an extra wedge square, a lowered ceiling).
- (c) The checker `stair_check.py`, included with this text, reads only the published data files. It recomputes the band, the rows R and the upper lifts y_1 from (k, b, y_0) , checks (C0)–(C4) with the separating axis test for all pairs of pieces with overlapping bounding boxes, the exact uncovered areas, the count and the identity of Proposition 6.3. It was adapted from checker (b) and does not import the generating program. Its results are in Table 6; seven deliberately corrupted versions of the data (a column shifted, lowered or given an extra square, a layer raised or lengthened, a wedge row lengthened, the ceiling lowered by $1/100$) were all rejected at $k = 10^5$ and at $k = 10^8$.

k	pieces	pairs decided exactly	overlaps	result (time)
10^5	2887	22274	0	pass (0.3 s)
10^6	10111	119522	0	pass (1.2 s)
10^7	42550	969664	0	pass (8.2 s)
$3.825 \cdot 10^7$	93680	2730501	0	pass (29.6 s)
10^8	165152	5748521	0	pass (59.2 s)

TABLE 6. Results of the checker `stair_check.py` (c), summed over the four ends, with Python `fractions` on a laptop. For $k \geq 10^7$ it was also run with `gmpy2` rationals on a remote Linux machine, with the same output apart from the arithmetic and timing lines.

All checkers were written by AI systems; none has been reviewed by a human. The rows of the bands are not listed in the data; their correctness rests on Lemmas 3.3 and 6.1.

The tuned packings are 4.9–11.8% better than the packings of the proof of Theorem 1.2 (Remark 7.5); for $k = 10^5, \dots, 10^8$ the ratio $c/k^{2/5}$ is between 15.6 and 16.0, and the local slopes of $\log c$ against $\log k$ between 10^5 , 10^6 , 10^7 and 10^8 are 0.399, 0.403 and 0.393.

10. COMPARISON WITH THE LITERATURE

10.1. From the wasted area to the deficiency. By Lemma 2.1, an upper bound $W(x) \leq F(x)$ that holds for all x in an interval $(k-\delta, k)$ gives $c^*(k) \leq \limsup_{x \rightarrow k^-} F(x) - 1$, that is $c^*(k) \leq F(k) - 1$

if F is continuous. A bound that holds only for some x (for instance for x whose fractional part is bounded away from 0) gives nothing for $c^*(k)$. Conversely, a packing that shows $c^*(k) \leq c - 1$ shows $W(x) \leq x^2 - k^2 + c$ for x slightly below k .

10.2. Printed bounds. The bounds in the literature that we found are listed below (literature search of 2026-10-08; the last column is what they give for $c^*(k)$ by the translation above).

source	statement	consequence for $c^*(k)$
Erdős–Graham 1975 [EG]	$W(x) = O(x^{7/11})$, all real x	$O(k^{7/11})$
Chung–Graham 2009 [CG]	$W(x) \leq Cx^{(3+\sqrt{2})/7} \log x$	$O(k^{(3+\sqrt{2})/7} \log k)$, exponent ≈ 0.6306
Wang–Dong–Li 2016 [WDL] (arXiv)	$W(x) \leq (16\sqrt{2} + 38)x^{5/8}$	$\leq 60.63 k^{5/8} - 1$
Arslanov–Mustafin–Shangitbayev 2021 [AMS]	$s(n^2 - n) < n$ for $n > 12$	$\leq k - 1$ for $k > 12$
Bui 2025 [Bui], McClenagan 2026 [McC] (preprints)	$W(x) = O(x^{3/5})$, all real x	$O(k^{3/5})$, implicit constant

The best printed bound for $c^*(k)$ is therefore $O(k^{3/5})$. The statement $W(x) = O(x^{3/5})$ was first claimed by Chung and Graham [CG2]; an error in that argument is pointed out in [Bui] and [McC]. We have checked the statement of [WDL, Theorem 1] but not whether every step of its proof applies to x just below an integer. The bound $c^*(k) = O(k^{3/5})$ is stated explicitly in the online problem collections [Ev] and [Le]. [AMS] is the only refereed source we found that states a bound in the form of $c^*(k)$; it also notes that [EG] gives $s(n^2 - O(n^{7/11})) < n$. Göbel [Gö] remarked that an improvement of the trivial bound is possible for large n in each range $k^2 + k \leq n < (k+1)^2$, without an order.

For $k = 10^5$ the explicit bound of [WDL] gives about $8.1 \cdot 10^4$, Theorem 1.4 gives 2535 and Theorem 1.6 gives 1583.

For small k , the Squares-in-Squares page maintained by D. Ellsworth [El] (web resource, version of 2026-09-24) lists packings showing $s(n^2 - n + 1) < n$ for $n = 16, 17, 18$, that is $c^*(k) \leq k - 2$ for these k .

To the best of our knowledge, no upper bound of order $o(k^{3/5})$ for $c^*(k)$, or for $W(x)$ just below an integer, has been stated in the searched sources; in particular we found neither $c^*(k) = O(\sqrt{k})$ nor any statement about the ends of tilted bands. The search covered [EG], [RV], [CG], [CG2], [WDL], [Bui] (versions 1 and 2), [McC] and the thesis [McC2], [AMS], Friedman’s survey [Fr] and the Squares-in-Squares page [El], the repositories [Le] and [Ev], MathOverflow and Mathematics Stack Exchange, and arXiv up to 2026-10-08. Neither [Bui] nor [McC] makes a special statement for sides just below an integer. A search cannot exclude unpublished or oral results. The ingredients of Theorem 1.3 are standard, and Theorem 1.4 may well be regarded as an easy consequence of known techniques; the staircase of Section 7 is the primitive construction of [Bui, Section 3] (Section 1.2).

10.3. Why uniform bounds cannot go below $\sqrt{k}$. Roth and Vaughan [RV, Theorem] proved that $W(x) \gg (x\|x\|)^{1/2}$ whenever $x(x - \lfloor x \rfloor) > 1/6$, where $\|x\|$ is the distance from x to the nearest integer.¹ In particular $W(n + \frac{1}{2}) \gg n^{1/2}$. Hence an upper bound $W(x) \leq Cx^\alpha$ valid for all large real x must have $\alpha \geq 1/2$, and through (1.1) such a bound can never give $c^*(k) = O(k^\alpha)$ with $\alpha < 1/2$. A bound of order $o(\sqrt{k})$ for $c^*(k)$ must use the fact that the side is just below an integer, where the theorem of Roth and Vaughan gives nothing ($\|x\| \rightarrow 0$). The constructions of this paper do so: the width $b - \varepsilon$ of the bands is just below the integer b . Erdős and Graham remark that perhaps

¹The theorem is sometimes quoted with $x - \lfloor x \rfloor$ in place of $\|x\|$, for example on p. 1 of [McC]. That form would give $W(x) \gg \sqrt{x}$ for x just below an integer, and hence $c^*(k) \gg \sqrt{k}$, contradicting Theorem 1.2. The original statement [RV, p. 170] has $\|x\|$ and gives nothing for x just below an integer.

the correct bound is $W(x) = O(x^{1/2})$ [EG, p. 122]; Theorem 1.4 gives the corresponding order for $c^*(k)$, and Theorems 1.2 and 1.1 go below it, just below integers.

11. LOWER BOUNDS AND THE ORDER OF $c^*(k)$

In the notation of Section 2, $c^*(k) = W_{\min}(k) - 1$. The following lower bounds are in preprints by the author that have not been refereed.

- [R]: $W_{\min}(k) \geq 0.0353 \log k$ for every integer $k \geq 2$, where $\log$ is the natural logarithm (version 1.2; the constant was 0.033 in earlier versions). Thus $c^*(k) \rightarrow \infty$. The constant uses a computer-assisted lemma ([R, Lemma 4.10]).
- [Q]: $W_{\min}(k) \geq 0.1 k^{1/4}$ for every integer $k \geq 4.62 \cdot 10^{12}$.
- [T]: $W_{\min}(k) > 0.06 k^{1/3}$ for every integer $k \geq 3.5 \cdot 10^{17}$, and $\liminf_{k \rightarrow \infty} W_{\min}(k)/k^{1/3} \geq 0.06285$. Without [R, Lemma 4.10], $W_{\min}(k) > 0.05 k^{1/3}$ for $k \geq 3.39 \cdot 10^{17}$.

For small k see [R, Section 1]: $W_{\min}(k) \geq 3$ for $k \geq 2$, that is $c^*(k) \geq 2$, and a computer-assisted result of E. Daniel that has not been refereed gives $s(k^2 - 3) = k$ for $k \geq 6$, that is $c^*(k) \geq 3$ ([R, Corollary 1.3]; [Le, entry T-064]).

Together with Theorems 1.2 and 1.1,

$$\begin{aligned} 0.06 k^{1/3} - 1 &< c^*(k) < 20.668 k^{2/5} + 0.623 & (k \geq 3.5 \cdot 10^{17}), \\ c^*(k) &< 40.62 k^{3/8} + 1.4 \cdot 10^{-5} & (k \geq 3.4 \cdot 10^{26}), \end{aligned}$$

and the other tiers of Theorem 1.1; so $k^{1/3} \ll c^*(k) \ll k^{3/8}$ is what is proved: the upper bound by Theorem 1.1 of this paper and the lower bound in [T]. Neither exponent is claimed to be optimal; the true exponent is open (Section 12). The gap is large for moderate k : for $k = 10^8$ the lower bounds above give only $c^*(k) \geq 2$ ($c^*(k) \geq 3$ with the result of Daniel), while Theorem 1.6 gives $c^*(k) \leq 24790$.

12. OTHER END CONSTRUCTIONS AND OPEN PROBLEMS

Nothing in this section is used elsewhere. By Proposition 6.3, an end packing with uncovered area $O(b^\beta)$, uniformly for lifts in an interval of length $\sec \theta$, gives $c^*(k) \leq \min_b (4k/b + O(b^\beta)) = O(k^{\beta/(1+\beta)})$, with $b \asymp k^{1/(1+\beta)}$. The staircase has $\beta = 2/3$, and the chained wall fillers of Section 8 have $\beta = 3/5$.

Remark 12.1 (an intermediate construction: tilted columns). Before the staircase, we filled the end regions by columns of n squares whose tilt is chosen separately for each column from the fractional part of the ceiling height, so that $n \cos \beta + \sin \beta$ matches the ceiling; neighbouring columns are separated by slabs as in Lemma 7.1(i), but with an extra gap whenever the tilt increases. The loss under the ceiling is then about $\sqrt{2/M}$ per unit width for columns of height M , and each change of tilt costs a gap of order $M \tan \beta$; with $M \asymp \sqrt{b}$ this gives uncovered area $O(b^{3/4})$ and $c^*(k) = O(k^{3/7})$. The staircase avoids the changes of tilt altogether: one tilt serves all columns, and the rising ceiling is followed by moving squares into the layers, each move raising a column by $1 - \cos \alpha$. Since Theorem 1.2 has a smaller exponent, and its proof does not use this construction, we do not include the details or a proof of the $k^{3/7}$ bound.

Remark 12.2 (Type 3 regions of Chung and Graham). The end regions $\text{Reg}(y)$ are right trapezoids of width $w \approx b$ whose two parallel sides differ by about 2. Cut by two vertical lines into three trapezoids of width $w/3$, and with lifts y of order b , each piece appears to satisfy the hypotheses of the ‘‘Type 3’’ trapezoids of [CG, Theorem 1(3)], for which a packing with uncovered area $O(x^{1/\sqrt{2}} \log x)$ is asserted there. Combining Proposition 6.3 with [CG, Theorem 1(3)] therefore appears to give $c^*(k) = O(k^{\sqrt{2}-1} (\log k)^{2-\sqrt{2}})$. The exponent $\sqrt{2}-1 \approx 0.414$ is larger than $2/5$ and than $3/8 = 0.375$,

so this route is weaker than Theorems 1.2 and 1.1. We do not verify the details, and the proof in [CG] is written in outline, with lower-order terms suppressed.

Remark 12.3 (the wall wedges). In the bound of Lemma 7.3, the column triangles cost about $w \sin \alpha \asymp b/\sqrt{m}$ and the two walls (the wedge rows and the ends of the layers) cost about m each, with $m \asymp b^{2/3}$. Section 8 reduces the cost of a wall of height m from order m to order $m^{3/4}$ (following ideas of [Bui, Sections 3–4]); then the best column length is $m \asymp b^{4/5}$, the end cost is $O(b^{3/5})$, and $c^*(k) = O(k^{3/8})$. The constants of Section 8 are crude: the parameter rows come from a local search, and several per-row bounds of Lemma 8.1 were not optimised.

Open problems. The true exponent of $c^*(k)$ is open: it lies between $1/3$ ([T]) and $3/8$ (Theorem 1.1). We do not claim the true exponent, and we make no claim that $3/8$ is the best exponent obtainable from this frame or from known techniques. Better constants for Theorem 1.1, and smaller thresholds k_i , would also be of interest. It would also be interesting to have a matching structure for the lower bound, or a direct argument that some end cost of order b^β with $\beta > 0$ is unavoidable in this frame.

APPENDIX A. THE NORMALISED INEQUALITIES OF COROLLARY 8.10

This appendix lists all inequalities evaluated on each box in step (iii) of the proof of Corollary 8.10, for a parameter row $(b_0, c_y, c_D, c_R, a_w, \varpi)$; they are those of the program `cert_v3.py` (functions `box`, `wall` and `taurange`). For the row v1.0 they reduce to the list of version 1.0, except for the forms of (E-iv) and of the tilt condition of the walls, which are stated below. All quantities are intervals; an inequality “ > 0 ” or “ ≥ 0 ” means that the lower end of the enclosure is positive or non-negative, and an upper bound means the upper end of the enclosure.

Box and inputs. $z := m^{-1/4} \in [z_\ell, z_u] \subset [0, z_0]$ with $z_0 := c_y^{-1/4} b_0^{-1/5}$, and $\tau_n := t/z^2 \in [\tau_\ell, \tau_u]$, a piece of the range

$$\left[\sqrt{1 - c_D z^2 / \hat{b} - 2.1 z^{10} / \hat{b}^2}, \frac{z^2 + \sqrt{z^4 + 3.000001(2 - 3.000001 z^4)}}{2 - 3.000001 z^4} + \frac{z^3}{16\hat{b}} \right]$$

evaluated on $[z_\ell, z_u]$ (lower end of the first, upper end of the second). Enclosures:

$$\begin{aligned} \hat{b} &\in [c_y^{-5/4}(1 - 5.000001 z^4)^{5/4}, c_y^{-5/4}], \quad \mu b \in [2, 2.000001], \quad \Delta \leq 2.000001, \\ D_\ell z^3 &\in [c_D - z^3, c_D], \quad D_r z^3 \in [c_R - z^3, c_R], \\ \delta_Q &\leq g_q z^5, \quad g_q \in [0, 2.1/(16\hat{b})], \quad 1/Q_f \leq f_q z^5, \quad f_q \in [0, 1/(16\hat{b})]. \end{aligned}$$

With $t = \tau_n z^2$:

$$\begin{aligned} \frac{\tan \alpha}{z^2} &= \frac{2\tau_n}{1 - t^2}, \quad \frac{\sin \alpha}{z^2} = \frac{2\tau_n}{1 + t^2}, \quad q_n := \frac{q}{z^4} = \frac{2\tau_n^2}{1 + t^2}, \\ \sec \alpha &= \frac{1 + t^2}{1 - t^2}, \quad \frac{\sec \alpha - 1}{z^4} = \frac{2\tau_n^2}{1 - t^2}, \quad \frac{\kappa_m}{z^2} \leq \frac{q_n^2 \hat{b} z}{\mu b}, \end{aligned}$$

and $\tau/z^2 \in [\tan \alpha/z^2 - \kappa_m/z^2 - g_q z^3, \tan \alpha/z^2 - \kappa_m/z^2]$.

Main part.

- (E-i) $1/10 - t > 0$.
- (E-ii) $\hat{b}(1 - 2.1 z^{10}/\hat{b}^2) - (D_\ell z^3 + D_r z^3) z^2 - (\tan \alpha/z^2) z^3 - \sec \alpha z^5 > 0$ (this is $z^5(w - D_\ell - D_r - m \tan \alpha - \sec \alpha)$ from below).
- (E-iii) $\tan \alpha/z^2 - \kappa_m/z^2 - g_q z^3 > 0$.
- (E-iv) $1 - \varepsilon_m - (\tau/z^2) z^2 > 0$ with $\varepsilon_m \leq ((\sec \alpha - 1)/z^4) z^4 + g_q z + f_q z^5$.
- Main terms of Lemma 8.9, times z^3 : column triangles $w \sin \alpha z^3 \leq \hat{b} \sin \alpha/z^2$; and the other main terms: $w q z^3 \leq \hat{b} q_n z^2$; $\Delta(\sec \alpha + m \tan \alpha) z^3 \leq \Delta(\sec \alpha z^3 + (\tan \alpha/z^2) z)$; $J_0(1 +$

$\tan \alpha)z^3 \leq \frac{4\tau_n z^2 + 2z^4}{16\hat{b}q_n}(1 + (\tan \alpha/z^2)z^2)$; $J_L(\varepsilon_m + \tan \alpha)z^3 \leq ((\sec \alpha - 1)/z^4)z^3 + g_q + f_q z^4 + (\tan \alpha/z^2)z$; $\frac{1}{2}\mu h_L^2 z^3 \leq \mu b(h_L z^3)^2 z^2/(2\hat{b})$ with $h_L z^3 \in [D_\ell z^3, D_\ell z^3 + 3.000001(\tan \alpha/z^2)z^5]$; and $\mu w_R^2 z^3 \leq \mu b(D_r z^3 + \sec \alpha z^3 + (\tan \alpha/z^2)z)^2 z^2/\hat{b}$.

One wall. A wall is given by $\zeta_n := \zeta/z^2$, $\chi_n := \chi z^3$ and $L_n := Lz^4 \leq 1$ (the value $L_n = 1$ is used). Then

$$a_n := \tan \alpha_0/z \in [a_w \sqrt{\zeta_n}(1 - \epsilon'), a_w \sqrt{\zeta_n}(1 + \epsilon') + g_q z^4], \quad \epsilon' = 1.03 \cdot 10^{-12},$$

$$\sec \alpha_0 = \sqrt{1 + a_n^2 z^2}, \quad \cos \alpha_0 = 1/\sec \alpha_0, \quad \frac{\sec \alpha_0 - 1}{z^2} = \frac{a_n^2}{\sec \alpha_0 + 1},$$

$$q_{0,n} := \frac{q_0}{z^2} = \frac{a_n^2}{\sec \alpha_0(\sec \alpha_0 + 1)},$$

$$q_{\min,n} := \frac{q_{\min}}{z^2} = \frac{(\varpi a_n)^2}{s_m(s_m + 1)}, \quad s_m = \sqrt{1 + \varpi^2 a_n^2 z^2},$$

$$N_0 z^3 \in [\chi_n - 2z^3, \chi_n - z^3], \quad [\chi]z^3 \in [\chi_n - z^3, \chi_n], \quad r(L)z^3 = \chi_n + \zeta_n L_n z,$$

$$M_{\max} z^3 = r(L)z^3 + N_0 z^3 q_{0,n} z^2,$$

$$\Lambda z^3 = \frac{q_{\min,n} N_0 z^3 - z - \zeta_n a_n z^4}{\zeta_n}, \quad \bar{K} z = \frac{L_n}{\Lambda z^3},$$

$$\frac{\varepsilon_W}{z^2} = \frac{\sec \alpha_0 - 1}{z^2} + N_0 z^3 g_q + f_q z^3.$$

Margins of the wall:

- (W1) $\Lambda z^3 - N_0 z^3 a_n z \geq 0$ (positive on every box);
- (W2) $(1 - \varpi)a_n - \Delta_{\text{dr}}/z > 0$ with $\Delta_{\text{dr}}/z = \frac{q_{0,n}}{N_0 z^3}(L_n + \bar{K} z(1 + \zeta_n a_n z^3)z/\zeta_n) + \bar{K} z g_q z^3$;
- (W3) $N_0 z^3 \cos \alpha_0 - z^3(\zeta_n \sec \alpha_0/q_{\min,n} + a_n z + \zeta_n z^2(2 + \sec \alpha_0) + \zeta_n M_{\max} z^3 a_n) - z(1 + \zeta_n a_n z^3)/q_{\min,n} \geq 0$;
- (W4) $1 - (\varepsilon_W/z^2)z^2 > 0$;
- (W5) $\varpi a_n - g_q z^4 \geq 0$;
- $\chi \geq 4$: $\chi_n - 4z^3 \geq 0$;
- the tilt condition $t_{\text{up}} < 1/10$ of (E6): $1/10 - a_n z/(1 + \sec \alpha_0) > 0$, where $a_n z/(1 + \sec \alpha_0)$ is the half-angle formula $t = \tan \alpha/(1 + \sec \alpha)$ evaluated on the enclosure of $a_n z$; this also gives $t_0 \leq 1/10$ (the second half of (W5)). (Version 1.0 checked $a_n z \leq 1/10$ instead.)

The bound: $B_W z^3 \leq \alpha L_n + R$ with $\alpha := c_{\text{col}}/z + X/(\Lambda z^3)$, where

$$c_{\text{col}}/z = a_n + \sec \alpha_0 q_{0,n} z + \sec \alpha_0^2 \zeta_n z + \sec \alpha_0 \zeta_n N_0 z^3 q_{0,n} a_n z, \quad X = X_1 + X_2 + X_3,$$

$$X_1 = (N_0 z^3 - z^3)((\varepsilon_W/z^2)z + a_n), \quad X_2 = (1 + \zeta_n a_n z^3)(1 + a_n z)/q_{\min,n},$$

$$h_T = 1 + \zeta_n \sec \alpha_0/q_{\min,n} + a_n z + \zeta_n z^2(1 + \sec \alpha_0) + \zeta_n M_{\max} z^3 a_n,$$

$$w_T z = z + (r(L)z^3 + z^3)(q_{0,n}^2/\zeta_n + g_q z^3),$$

$X_3 = h_T(w_T z)z$ (this is $A_T z^2$), and

$$R = [\chi]z^3(1 + \frac{1}{2}a_n z) + w_0 z^3(1 + \zeta_n z^2 w_0) + r(L)z^3(1 + \frac{1}{2}a_n z) + R_e,$$

$$R_e = (w_e z^2)z(4 + 2\zeta_n w_e z^2),$$

$$w_0 = a_n z + N_0 z^3 q_{0,n} a_n, \quad w_e z^2 = z^2(1 + \sec \alpha_0) + r(L)z^3 a_n + N_0 z^3 q_{0,n} a_n z^2.$$

For B_W'' (rows marked †) two terms change: $X_3 = (h_T + 1)(1 + a_n z)z^2 + (2 + \zeta_n(w_T z)z + \zeta_n a_n z^3 + \zeta_n^2 a_n z^5)((w_T z)z + a_n z^3)$ (this is $A_T' z^2$ of Lemma 8.3), and $R_e = (w_e z^2)z + 2(1 + a_n z)z^3$ (this is $(w_e + 2(1 + \tan \alpha_0))z^3$, Lemma 8.5).

The three walls. Z_L : $\zeta_n = \tan \alpha / z^2$, $\chi_n \in [D_\ell z^3, D_\ell z^3 + 3.000001(\tan \alpha / z^2)z^5]$; Z_{LR} : $\zeta_n = \tau / z^2$, $\chi_n \in [D_r z^3 + ((\sec \alpha - 1)/z^4)z^7, D_r z^3 + 2 \sec \alpha z^3]$; Z_{UR} : $\zeta_n = \tan \alpha / z^2$, $\chi_n \in [D_r z^3, D_r z^3 + \sec \alpha z^3 + (\tan \alpha / z^2)z]$. Finally

$$\Psi \leq w \sin \alpha z^3 + (\text{other main terms}) + (\alpha + R)_{Z_L} + \max(\alpha_{Z_{LR}}, \alpha_{Z_{UR}}) + R_{Z_{LR}} + R_{Z_{UR}},$$

where the maximum is taken of the upper ends. $\Psi_{\sup}$ is the largest upper end over all boxes, and $C_E^{\text{cert}} = \Psi_{\sup} c_y^{3/4} (1 + 5.000001/(c_y b_0^{4/5}))^{3/4}$ (upper end).

APPENDIX B. THE NORMALISED INEQUALITIES OF COROLLARY 8.12

These are the inequalities of Appendix A for the rows marked †, with exactly the following replacements; they are those of the program `cert_v4.py`.

- Inputs: $\hat{b} \in [c_y^{-5/4}(1 - 4.5002z^4)^{5/4}, c_y^{-5/4}]$ (instead of 5.000001); the upper end of the range of τ_n is

$$\frac{z^2 + \sqrt{z^4 + 2.500002(2 - 2.500002z^4)}}{2 - 2.500002z^4} + \frac{z^3}{16\hat{b}}$$

(instead of 3.000001); everything else as in Appendix A, including $\chi_n \in [D_\ell z^3, D_\ell z^3 + 3.000001(\tan \alpha / z^2)z^5]$ for Z_L .

- One wall: a_w is $a_{w,L}$, $a_{w,LR}$ or $a_{w,UR}$ for Z_L , Z_{LR} , Z_{UR} , so that

$$a_n \in [a_w \sqrt{\zeta_n}(1 - \epsilon'), a_w \sqrt{\zeta_n}(1 + \epsilon') + g_q z^4]$$

and t_{up} are computed per wall. The new margin (W6), normalised by z : $\nu_*/z = \varpi a_n - q_{0,n}^2 z / \zeta_n > 0$.

- The bound: $B_{W4} z^3 \leq \alpha L_n + R$ with α as for B''_W and $R = R_{\text{start}} + R_{\text{end}} + R_e$, where

$$\begin{aligned} R_{\text{start}} &= \lfloor \chi \rfloor z^3 \left(\frac{1}{2} + a_n z \right) + \frac{z^2}{8a_n} + w_0 z^3 (1 + \zeta_n z^2 w_0), \\ R_{\text{end}} &= r(L) z^3 \left(\frac{1}{2} + a_n z + \frac{\sec \alpha_0 - 1}{z^2} z^2 \right) + \frac{(\bar{K}z)z + 2z^2}{8\nu_*/z} + \frac{1}{2} \left(N_0 z^3 q_{0,n} z^2 + 2z^3 + \frac{z(1 + \zeta_n a_n z^3)}{q_{\min,n}} \right) \\ &\quad + \frac{1}{2} (z^3 + \zeta_n L_n z + \zeta_n (w_e z^2) z^3), \quad R_e = (w_e z^2) z + 2(1 + a_n z) z^3. \end{aligned}$$

- B_{W4} , (W2) and (W3) are non-decreasing in L , and (W6) does not involve L ; so $L_n = 1$ is used as before, and $C_E^{\text{cert}} = \Psi_{\sup} c_y^{3/4} (1 + 4.5002/(c_y b_0^{4/5}))^{3/4}$.
- Margins checked on every box: those of Appendix A and, for each of the three walls, (W1)–(W6), $\chi - 4$ and $1/10 - t_{\text{up}}$.

VERIFICATION STATUS

- Theorems 1.3 and 1.4, Corollary 5.1, Proposition 1.5 and Lemmas 2.1–4.3, 6.1, Corollary 6.2 and Proposition 6.3 are proved in full in this text. An earlier write-up of Sections 3–5 received an independent AI review, which found no critical or major issue; its minor remarks are included (the case $(k, b) = (2, 2)$, $L \geq 0$ in Lemma 3.4, and the scope of Lemma 4.3, see Remark 5.3). The formulas were also tested in exact arithmetic: (5.1) and the bounds of Theorems 1.3, 1.4 and Proposition 1.5 for all $2 \leq b \leq k \leq 1500$, the angle of Lemma 4.2 for all $2 \leq b \leq k \leq 250$, and actual packings (separating axis tests) for all b and $k \leq 60$ and some larger cases. Such tests are not proofs.
- Theorem 1.2 (Section 7) is proved in full; it uses Lemma 2.2, Lemmas 3.1–3.3, Lemma 6.1, Corollary 6.2 (with Lemma 4.1) and Proposition 6.3. An earlier write-up of Section 7 received two independent blank-slate AI reviews, both of which judged it correct with no critical or major issue; their minor remarks are included. The proof relies on one evaluation

in interval arithmetic, $\Phi(10^{-4/3}) \leq 5.0163$, done by the author's program and by three independent computations; the other constants were certified in the same way. The construction was tested in exact arithmetic, also by an independent reimplementaion (Remark 7.5). To our knowledge the proof has not appeared before; it deserves priority in a human review.

- Theorem 1.1 (Section 8) is proved in full; it uses Lemmas 7.1 and 7.2, Lemma 2.2, Lemmas 3.1–3.3, Lemma 6.1, Corollary 6.2 and Proposition 6.3. The tiers marked $\dagger$ also use Lemmas 8.2, 8.3, 8.4 and 8.5, which are new in version 1.1. The tiers marked $\ddagger$ also use Lemmas 8.6, 8.7, 8.8, 8.11 and Corollary 8.12, which are new in version 1.1. The write-up of Section 8 of version 1.0 received two independent blank-slate AI reviews, with their own programs: one judged it correct after fixes, the other correct; all their fixes were included in version 1.0, and an independent reimplementaion found no counterexample. The changes of version 1.1 received two further independent blank-slate AI reviews. Both judged the parameter part correct after minor fixes and the new lemmas of Section 8.2 correct. All their fixes are included (among them the value of $\frac{32}{5}(\frac{5}{3})^{3/8}$, the rounding of the tables, the precise statement of two hypotheses, the derivation of the range of t/z^2 , and the complete list of Appendix A). The constants are certified by the author's covering and by three coverings written independently from the text: two by verification agents (`mpmath.iv` and `python-flint arb`) and one by a reviewer (`mpmath.iv`), and all three implement the same normalised inequalities (the lists of Appendices A and B), so they check the arithmetic and the coverage but are only a weak check of the transcription of these inequalities from the lemmas; the point checks of the un-normalised formulas (at 90, 120 and 5850 points) partly address that. All programs were written by AI systems of the same family. The constant $C_E = 15.54$ of version 1.0 is certified by three independent full-range coverings ($C_E \leq 14.964$, 14.963 and 14.9631). The range $b \geq b_0$ has not been tested by explicit computation, except for 15 end packings of the tiers $\ddagger$ near b_0 ; the explicit checks cover the validity of the construction at smaller b (Remark 8.13). It deserves priority in a human review.
- Theorem 1.6 is computer-assisted. Its finite part was checked by three programs, two of them written independently of the generating program (Section 9).
- Section 12 contains remarks only; Remarks 12.1 and 12.2 are not proved here.
- Section 10 and the novelty statement rest on a literature search, which cannot prove absence.
- No human expert has reviewed this text. The paper has not been refereed. All reviewers and checkers so far are AI systems of the same family, which may share blind spots.

Verification of version 1.1. The following checks of the changes of version 1.1 were made after the two reviews mentioned above. All programs were written by AI systems of the same family; none has been reviewed by a human.

- *Independent covering 1* (verification agent; written from the text alone; `mpmath.iv`, 120 bits): 512 z -strips $\times$ 128 t -pieces with adaptive refinement, about 65,537 boxes per tier; gap-free check passed; $z = 0$ included. All nine tiers certified: C_E^{cert} at most the stated C_E , every margin positive; for example $C_E^{\text{cert}} \leq 18.71447$ (A1s), 14.14983 (A5), 16.96349 (B1s), 16.71729 (B2). Pointwise check of the un-normalised formulas at 90 points: passed. Negative control: the parameters of version 1.0 with $b_0 = 10^9$ fail (E-iii), as expected.
- *Independent covering 2* (verification agent; written from the text alone; `python-flint arb`, 160 bits, adaptive bisection): 389–3013 leaves per tier; the partition checked by a Kraft sum equal to 1 and by the total area. All nine tiers certified; for example $C_E^{\text{cert}} \leq 18.70756$ (A1s), 14.14845 (A5), 16.95859 (B1s), 16.71253 (B2). Direct un-normalised evaluation at 120 cases: passed.

- *Reviewer's covering* (blank-slate review; own code from the text; `mpmath.iv`, 120 bits, 320×128 boxes): all nine tiers agree with Table 3 to the printed digits. For the tiers marked $\dagger$ it uses A'_T of Lemma 8.3.
- *Independent reimplementations* of ZC , ZC' , $E38$, $E38'$ and the full L-shaped packing, written from the text alone, in exact rational arithmetic, with a separate exact checker (separating axes): 133 instances, $4.39 \cdot 10^7$ pieces, all valid; some of these instances (end packings where (E-iii) fails) use the convention $\tau := 0$ of Remark 8.13. The wall bounds $U_Z \leq B''_W$ hold in 49 of 49 walls ((W1)–(W5) hold in 36 of them), and Lemmas 8.3 and 8.4 hold term by term in 46 of 46 accounted walls. The first in-window tests with an aligned cut ($\tau > 0$), eight end packings at $b = 1.5 \cdot 10^6$ – $4 \cdot 10^6$ with the parameters of the tiers B1, B1s and B2: all valid; seven of the eight satisfy all hypotheses and are within the bound of Lemma 8.9; in the eighth (B1, $b = 1.5 \cdot 10^6$, $y = 59407/4$) hypothesis (E-v) fails for the lower right wall, which was filled by rows only (a convention, not part of the construction), so the bound of Lemma 8.9 does not apply there. Pointwise check of Corollary 8.10 at 5850 points: passed. 29 of 30 negative controls were rejected; the remaining one is a valid packing.
- *Tiers marked $\ddagger$* (Sections 8.3 and 8.6). Two blank-slate AI reviews: one found Lemmas 8.6, 8.7, 8.8 and 8.11 correct and Corollary 8.12 and the tiers correct after fixes (included); the other found all parts correct. Exact tests (not proofs): 53 walls ZC' (34 with all hypotheses true), no violation of Lemmas 8.7 and 8.8, largest ratio $U_Z/B_{W_4} = 0.737$; Lemma 8.6 in 21592 cases; Lemma 8.11 in 376 cases and in 3000 cases forced to $y_1 = y_0 + 1$; no violation. Three independent coverings of the rows of Table 4, written from the text, all passing (`python-flint arb`: $C_E^{\text{cert}} \leq 14.46634, 14.79786, 14.97606$; `mpmath.iv`: 14.469520, 14.801342, 14.979582; the reviewer's `mpmath.iv` covering reproduces 14.47251, 14.80459, 14.982873); all three implement the same normalised inequalities (Appendix B), with point checks of the un-normalised formulas at 46 and 36 points. A direct covering of the un-normalised bound for $b_0 \leq b \leq 10^{16}$, without Appendix B (`python-flint arb`; integer quantities enclosed exactly per box, actual wall lengths; only J_0 uses the mean-value inequality), certifies $U/b^{3/5} \leq 13.910, 14.235, 14.319$ for C1, C2, C3, all margins positive; for $b > 10^{16}$ the normalised coverings apply. That the cited results of Section 6 hold for real lifts was checked by two independent AI readings (Section 8.6). Exact end packings: 15 complete end packings $E38'$ for the rows C1–C3 with lifts of Lemma 8.11 in the window, b from $1.41 \cdot 10^7$ to $2.30 \cdot 10^7$, each with $1.47 \cdot 10^7$ – $2.37 \cdot 10^7$ pieces, checked exactly (separating axes, containment, exact U): all hypotheses hold, no overlap, no failure; U is at most 0.71 times the bound of Lemma 8.9 and at most $0.69 C_E b^{3/5}$, and U_Z/B_{W_4} lies between 0.44 and 0.75; 3 of the 15 have $y_1 = y_0 + 1$. Of 30 negative controls 24 were rejected; the other 6 came from a flaw in the design of the controls and were rejected when applied to the whole packing.

Programs and data. <https://github.com/squarepacker/k2-minus-c-upper> (archived on Zenodo; the DOI is given in the repository README). The data files `stair_k*.json.gz` (the end pieces of the five certificates), the checker `stair_check.py`, the generator `stair_dump.py` and the interval computation `const_stair.py` of the constants of Corollary 7.4 and Section 7.3, the interval covering `cert3e.py` of the constants of Corollary 8.10 for the row v1.0, the author's covering `cert_v3.py` of all rows of Table 3 with the merge and theorem checks `finalize_v3.py` and the recorded slices, the wall filler `r38.py` with the exact checker `chk.py` and the driver `run_z.py`, and the exact tests of version 1.1 in `code/zc_tests` (Remark 8.13) as well as the author's covering `cert_v4.py` of the rows of Table 4 with its recorded slices and merges, and the exact tests `sl_t2_test.py` of Lemmas 8.6 and 8.11 accompany this text; `README_code.txt` lists the commands and the expected outputs. The programs of the reviews and of the independent coverings and reimplementations are not included.

Use of AI. Developed with the assistance of Claude (Anthropic), including the proofs, the text and the programs; the author takes full responsibility.

Licence. This text is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0); the accompanying programs are distributed under the MIT licence.

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